

Why Must a Cause of All and a Law of All Be Related in a Universe? Freedom as the cause and Law of all.

Freedom, Causation, and the General Problem of Navigability: Architecture of Freedom Intelligence

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Abstract

Why must a Cause of All and a Law of All be related in a universe? This paper tests that question as a hypothesis from the author's own research programme, the Architecture of Freedom Intelligence (AFI): that Freedom — the structured availability of possible continuation, called Possibility — is constitutive of causal production, and that this cause connects to a law of navigation, Lightness, through the open question of Navigability: how a system moves among the continuations open to it. Whether cause and law truly connect this way, or belong to separate registers entirely, is held open and tested against 52 independently verified 2025–2026 sources, forty counterfactual universes, and eight formal models.

Key results:

- *Structured possibility generates a real navigation problem, but only under two contestable commitments: reality is compressible, and realization follows a single history.*
- *Freedom and the Law of Freedom may connect through Lightness, as AFI proposes, or may belong to separate registers entirely, as much of physics and philosophy holds — this paper takes a tested position rather than assuming either side.*
- *The candidate Law of Freedom — Freedom equals Perception divided by Distortion, raised to a power — resembles active inference and empowerment, stopping short, so far, of a shown reduction to either.*
- *Twenty-two falsification criteria and eight testbeds set out what would be needed before this law could be called general.*
- *The author's own prior AFI publications are cited for provenance only, never as independent evidence.*

Keywords: causation; possibility; modal metaphysics; path availability; navigability; active inference; empowerment; reachability; viability theory; affordance perception; causal emergence; falsifiability

Key Conclusions

This paper's title asks why a Cause of All and a Law of All must be related in a universe. Five findings, stated first, before the full argument, set out how far the evidence currently carries an answer:

- *Freedom, as structured possibility, is necessary for causal production in every branching domain tested.*
- *That condition opens a real, held-open question — the Navigability problem: whether some law connects to Freedom, or whether cause and law belong to separate registers entirely.*
- *The Law of Freedom stands as AFI's coherent, falsifiable candidate for that connection, still short of a proven law of general application.*
- *It resembles active inference and empowerment, stopping short, so far, of a shown reduction to either.*
- *Of twenty-two falsification criteria, two pass cleanly; most others pass only partially, pending experiments still to run (Section 17).*

This paper tests a hypothesis, rather than assumes it. The full argument follows below.

What Is Actually the Architecture of Freedom Intelligence (AFI), from Gonalo Melo de Magalhães?

AFI is a theoretical framework. It was originated and developed by Gonalo Melo de Magalhães. AFI puts forward five hypotheses. This paper tests them, one by one, against the 2025–2026 literature. Here are the five hypotheses, stated in full, before the argument begins:

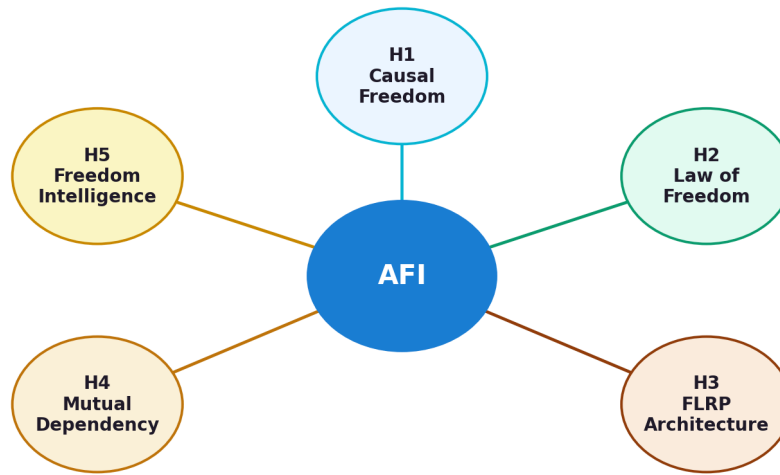
- **Hypothesis 1 — Causal Freedom: Freedom is Passibility.** *Freedom means the structured availability of possible continuation from a given state. This is a technical term, defined precisely for this paper. It is not political liberty. It is not free will. It is not moral autonomy. Read as this paper's candidate **Cause of All**: what makes causal production possible in the first place.*
- **Hypothesis 2 — The Law of Freedom.** *Every system follows the lightest path, according to its own perception of that path. Written as a formula: Freedom equals Perception divided by Distortion, raised to a power alpha. $\text{Freedom} = (\text{Perception} / \text{Distortion})^\alpha$. Read as this paper's candidate **Law of All**: which regularity, if any, governs the selection Hypothesis 1 leaves open.*
- **Hypothesis 3 — The FLRP Architecture.** *Reality is organized through four layers, in this order: Freedom, then Logic, then Relationships, then Physics. Physics is the final layer — the one that has crystallized into fixed form.*
- **Hypothesis 4 — Mutual Dependency.** *Hypotheses 1, 2, and 3 are not three separate, optional claims. They stand or fall together.*
- **Hypothesis 5 — Freedom Intelligence.** *Intelligence is the capacity to perceive better, reduce distortion, reveal more possibilities, and unblock paths. It means a system improving its own Freedom over time.*

Why call Hypothesis 2 a law? A scientific law is not a decree. It is a claim with four properties. It states an exact, testable relationship. It applies across many different kinds of systems, not just one. A specific experiment could show it false. It is written precisely enough that “false” has one clear meaning. The Law of Freedom is offered as a candidate with all four properties — not as an established law, the way Ohm's law or Newton's laws are established. Section 16 scores it against this standard directly, criterion by criterion. Until it passes those tests widely, “law” here names a target this research programme is aiming at, not an achievement already reached.

Figure 1

The Five Hypotheses of AFI

The Five Hypotheses of AFI



Hypotheses 1, 2, and 3 are mutually dependent, per H4

Figure 1

Note. AFI decomposes into exactly five hypotheses — Causal Freedom, the Law of Freedom, the FLRP Architecture, their Mutual Dependency, and Freedom Intelligence. Each is tested separately in this paper. None is severable from the others, per Hypothesis 4. The hub-and-spoke layout suggests independence for legibility only; the real dependency structure among the first three hypotheses is triangular (H1–H2–H3, per Figure 2), not a simple star.

Figure 2

Mutual Dependency: H1, H2, H3

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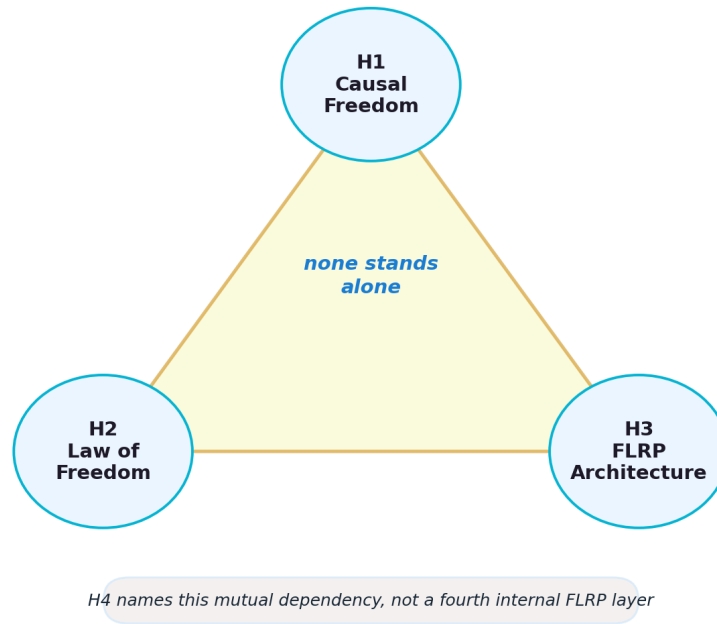


Figure 2

Note. Hypothesis 1 (Causal Freedom), Hypothesis 2 (Law of Freedom), and Hypothesis 3 (FLRP Architecture) are mutually dependent. None is offered as a separable, independently optional claim. That mutual dependency is exactly what Hypothesis 4 asserts. The triangular layout illustrates the dependency qualitatively. It does not formally specify the direction or strength of dependency between any two of the three.

This paper is the author's own research programme's first systematic test of these five hypotheses. It asks three questions: what the 2025–2026 literature says about them, what forty formally specified counterfactual universes reveal about them, and what a rigorous audit finds they actually establish. A full glossary of every AFI term used in this paper is in Appendix E, along with a quick-reference index to all forty counterfactual universes.

Table of Contents

§	Section	Page
1	<i>The Question Beneath Every Causal Question</i>	5
2	<i>Why Cause and Law Appear Unrelated</i>	11
3	<i>Freedom as Path Availability in AFI</i>	14
4	<i>What Makes Causation Possible?</i>	17

§	Section	Page
5	<i>Possibility, Passibility, and Causal Production</i>	19
6	<i>What Is a Path?</i>	20
7	<i>What Is Navigability?</i>	21
8	<i>The 2025–Current Research Frontier</i>	24
9	<i>Perception of the Possible</i>	26
10	<i>Counterfactual Universe Laboratory</i>	32
11	<i>Formal Models of Passibility</i>	34
12	<i>Foundational Genealogy: Powers, Process, Physics, and Control</i>	47
13	<i>The Cause-to-Law Bridge: Does the Law of Freedom Follow?</i>	50
14	<i>Lightness: Local, Global, or Recursive?</i>	56
15	<i>Causal World Models, Active Inference, and Future Path Space</i>	58
16	<i>Falsification Matrix</i>	60
17	<i>A Passibility-Extended Scientific Method</i>	66
18	<i>Implications for Human and Artificial Intelligence</i>	73
19	<i>Limitations</i>	75
20	<i>Research Programme</i>	76
21	<i>Conclusion</i>	77
	<i>References</i>	79
A	<i>The Counterfactual Universe Laboratory (Full)</i>	82
B	<i>Total Distinction Matrix (Definition Registry)</i>	96
C	<i>Source Verification Note</i>	106
D	<i>Final Audit and Standalone Statements</i>	107
E	<i>Glossary and Quick-Reference Index</i>	111

1. The Question Beneath Every Causal Question

Ordinary causal inquiry asks what caused an outcome. This paper asks a prior question: what makes a transition from A to B ever *available* — the availability question? Two easy dismissals fail: that it is trivial, and that existing causal theory already answers it.

Taking it seriously surfaces **Navigability**: given that more than one continuation is typically available from a state, how does a system move among them? The task is to determine how tightly the two questions connect.

Key points:

- *Availability (what makes a transition possible) vs. navigability (how a system moves among available transitions) — kept distinct throughout.*

- *AFI's proposition under test: Freedom, in this paper's technical sense, names the structured availability of possible continuation — **Passibility**, distinct from political liberty, free will, or moral autonomy — and creates causation.*
- *Companion proposition, the **Law of Freedom**: systems follow the lightest available path according to their own perception of it.*

This paper's title, unpacked. Section 13 and Appendix D.2 give this question its precise, technical form: does a Cause of All bear on a Law of All? Five figures below set out why that question is forced rather than optional wherever a domain branches, and where currently available evidence leaves it, holding open, throughout, which of the two live readings from Section 2 is correct.

Figure 3

Opening a Problem, Not Yet Solving It

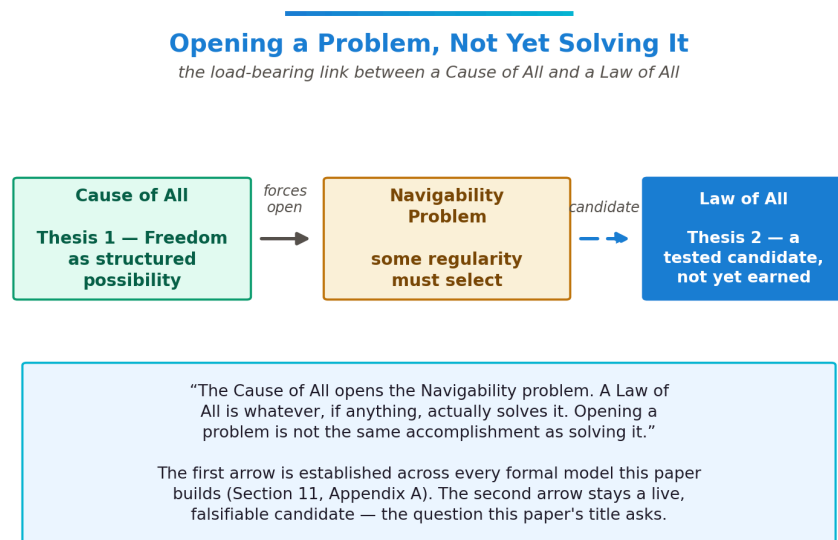


Figure 3

Note. Appendix D.2 states the relation in one line, reproduced here: opening the Navigability problem and solving it are different accomplishments. The first is secured by this paper's formal apparatus; the second is exactly what Sections 13, 16, and 17 spend their effort testing.

Figure 4

Two Total Claims, One Universe

Two Total Claims, One Universe

why the question cannot simply be dismissed

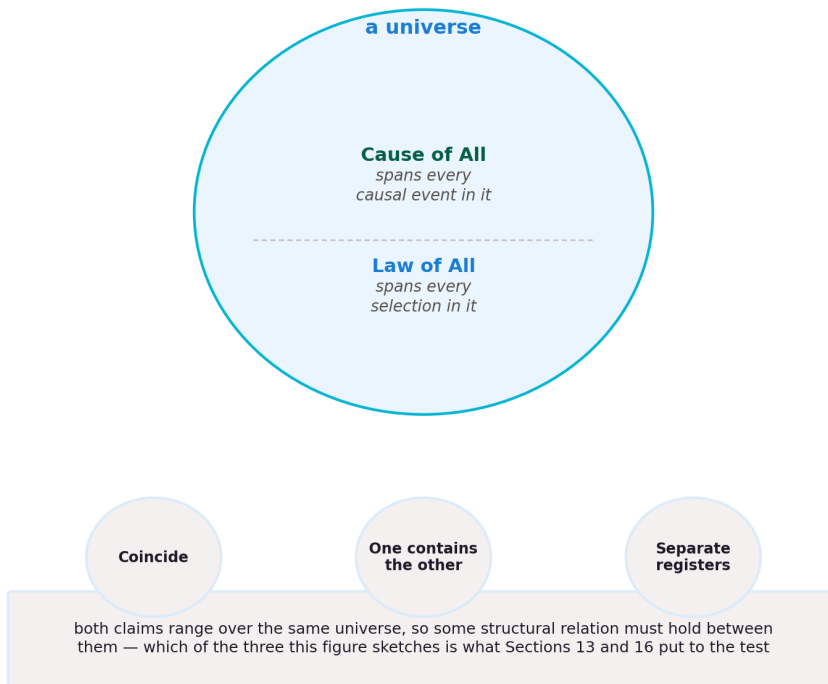


Figure 4

Note. A Cause of All and a Law of All are both universal claims about the same domain, which is why the question this paper's title asks survives as a live question rather than dissolving into a mere category mistake. Sections 13 and 16 test which of the three sketched relations — coincidence, containment, or separate registers — the evidence actually supports.

Figure 5

Branching Turns the Question From Optional to Forced

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single-successor domains ask nothing; branching domains cannot decline

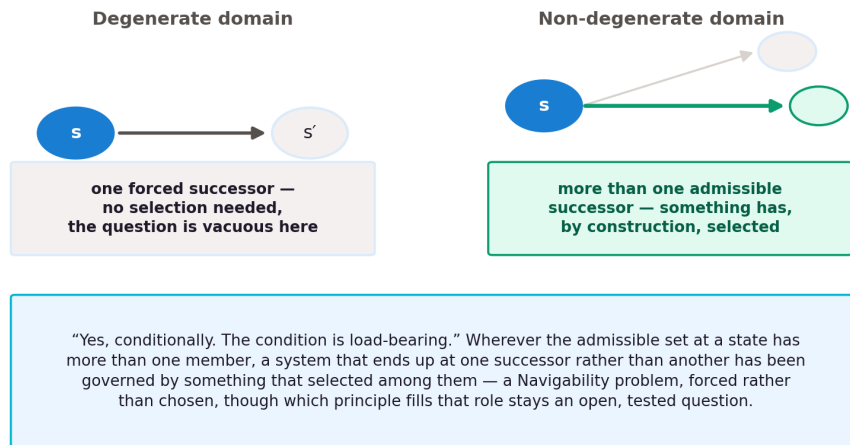


Figure 5

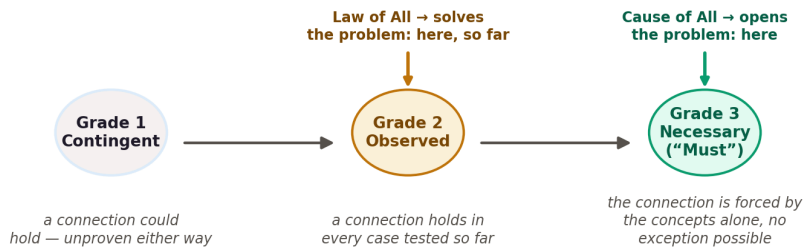
Note. Appendix D.1's answer to its own Question 3 — "yes, conditionally" — is drawn here directly. The condition is branching: a domain with exactly one admissible successor leaves the selection question moot, while any domain with more than one forces it open, regardless of which law, if any, turns out to answer it.

Figure 6

Why "Must" Is the Hard Word

Why “Must” Is the Hard Word

three grades of connection-claim, and where this paper's evidence sits



The Cause of All forcing open a Navigability problem is established across this paper's formal models — a Grade 3 result. Which law, if any, solves that problem is not yet at Grade 3: the Law of Freedom is a specified, falsifiable candidate, tested and re-scored in Section 16, still short of the evidence bar Appendix D.4 sets for calling it general.

Figure 6

Note. This paper keeps two different claims at two different grades: that a Cause of All forces open a Navigability problem is treated as established, Grade 3, per Appendix D.1; that the Law of Freedom is the law which closes it is treated throughout as a tested candidate, held short of Grade 3 by the same standard set out in Appendix D.4.

Figure 7

One Universe, Two Total Questions

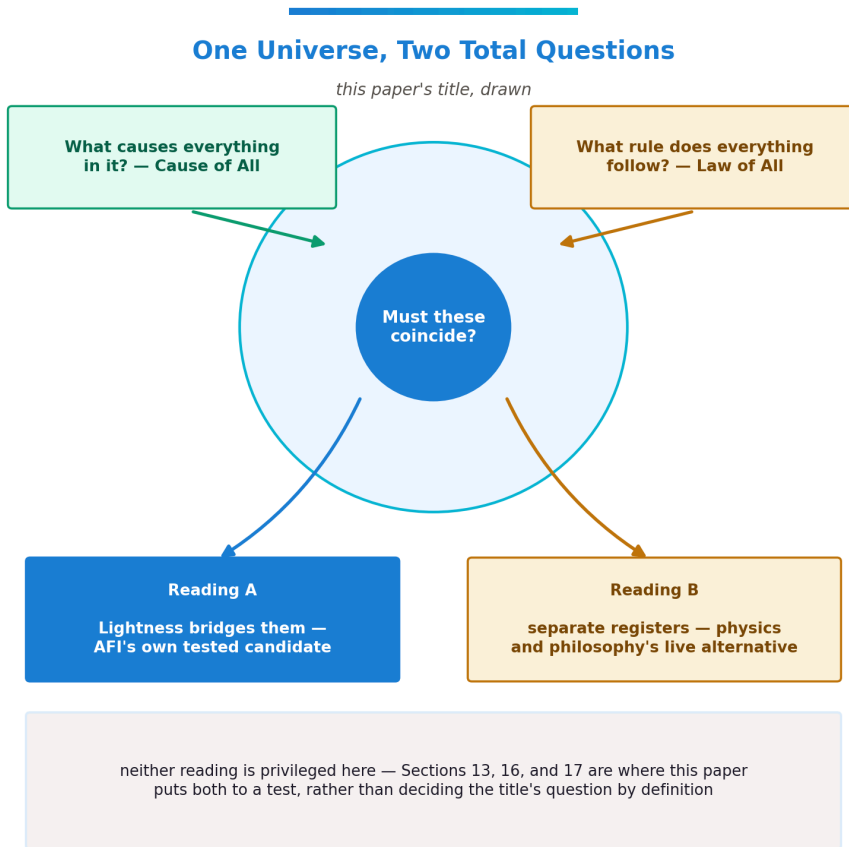


Figure 7

Note. This figure is the paper's title, drawn. Both readings introduced in Section 2 — the Lightness bridge and the separate-registers view — remain equally live here; every figure in this paper, this one included, leaves the choice between them to Sections 13, 16, and 17's evidence, rather than to definition.

Figure 8

The hidden question beneath every causal claim

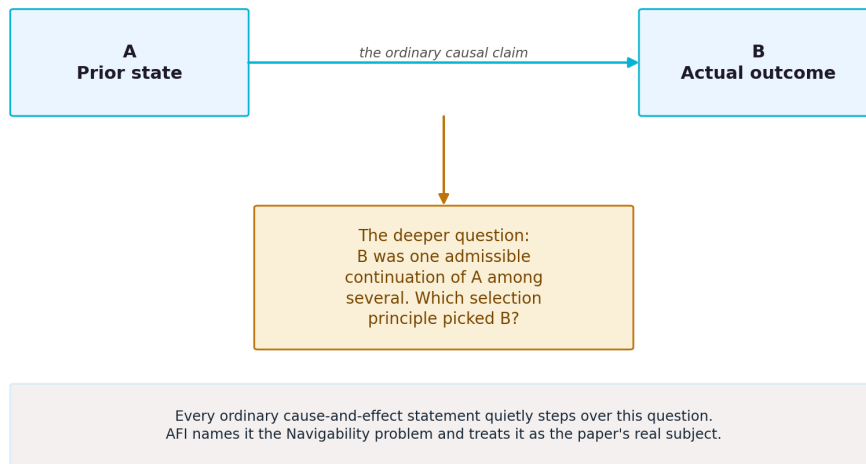


Figure 8

Note. An ordinary “A caused B” statement is silent on why B rather than some other admissible continuation of A occurred. The figure illustrates a conceptual gap, not a measured one; it does not itself establish that a determinate “selection principle” exists in every domain.

2. Why Cause and Law Appear Unrelated

A cause answers *through what*. A law answers *how*.

Structural causal models (Pearl 2000; Papineau 2025) can be built without asking what explains the admissible-intervention space; a behavioral law can be tested without asking what makes its causation possible.

AFI’s claim: this independence is an artifact of which primitive is chosen.

- *Most candidate causal primitives — matter, energy, fields, particles, information, spacetime — are static or substantial: things that exist, not structures of transition.*
- *Passibility is defined in terms of states, transitions, admissibility, and continuation. It already contains a transition landscape with more than one live point whenever more than one continuation is admissible — so a further question is unavoidable: which point gets realized, and by what regularity? Cause and Law look unrelated only until one notices the chosen primitive already contains a navigation problem inside it.*
- **Falsifiability:** *the bridge fails if a system could have fully structured Passibility yet nothing about that structure bears on how it navigates.*

Figure 9

Static vs. dynamic view of causation

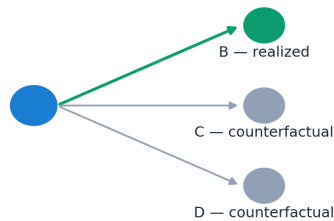
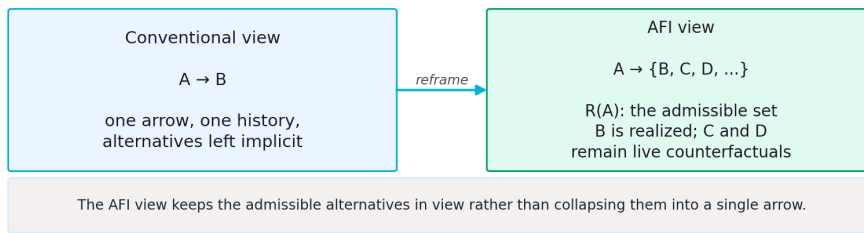


Figure 9

Note. Representing causation as a single arrow ($A \rightarrow B$) suppresses the admissible-alternatives structure that a Possibility-based account foregrounds. The “dynamic view” panel is a schematic instance, not a claim that all causal domains have exactly three admissible continuations.

An open question, held open. Freedom opens a field of live possibilities. That openness alone raises a further question: among every admissible continuation, which one actually happens? Two readings compete for the answer, and this paper holds both live rather than settling the matter by definition.

One reading: the Lightness bridge. AFI proposes that everything follows the lightest path among all available paths, judged by that system’s own perception — Thesis 2, offered as the completion of Thesis 1 rather than a separate claim bolted on afterward. On this reading, Freedom supplies the field, Lightness supplies the direction of travel through it, and the two theses move together.

The opposite reading: two separate registers. Many readers, physicists especially, hold this view with equal standing. In their working vocabulary, a cause is a matter of forces, fields, and conserved quantities, while a law is a matter of measured regularity arrived at through variational calculus, already occupying the “lightest path” role inside mechanics alone (Section 11, Model G). On this reading, Freedom and the Law of Freedom belong to genuinely separate registers — one names a space of possibility, the other names a selection principle, and a domain could carry rich Freedom while its actual dynamics track a regularity carrying zero resemblance to Lightness. Section 17’s testbeds exist to settle this question case by case, rather than assume an answer in advance.

Four professional readers meet this open question differently.

- *A philosopher tends to treat the Lightness reading as a substantive, contestable metaphysical wager — one candidate account of production among several live rivals (Section 12), owed independent argument rather than definitional fiat.*

- **A physicist** already has a principle that rhymes with lightness — stationary action — confined to specific, well-tested domains, and reasonably asks for the same rigor before granting Freedom-Lightness any wider standing.
- **An engineer** wants an algorithm: given a state and a perception model, compute the lightest path, and show the computation terminates (Section 17.2's tractability discussion speaks directly to this demand).
- **A designer** meets the Lightness reading as an everyday move. Design practice starts from a space of possible configurations (Freedom) and searches for the version that costs least to build, learn, or use (Lightness) — the two travel together as a single act, every time a design is chosen. For a designer, the Lightness reading arrives as familiar practice; for the other three, it arrives as a claim still owed its proof.

Falsifiability: the Lightness reading would weaken if a system's Freedom were shown to structure its space of options while its actual behavior tracked some quantity systematically distinct from Lightness, at every honestly measured horizon. The separate-registers reading would weaken symmetrically if that same distinctness turned up nowhere any test could reach.

Figure 10

Why a Cause Might Need a Law: The Lightness Bridge

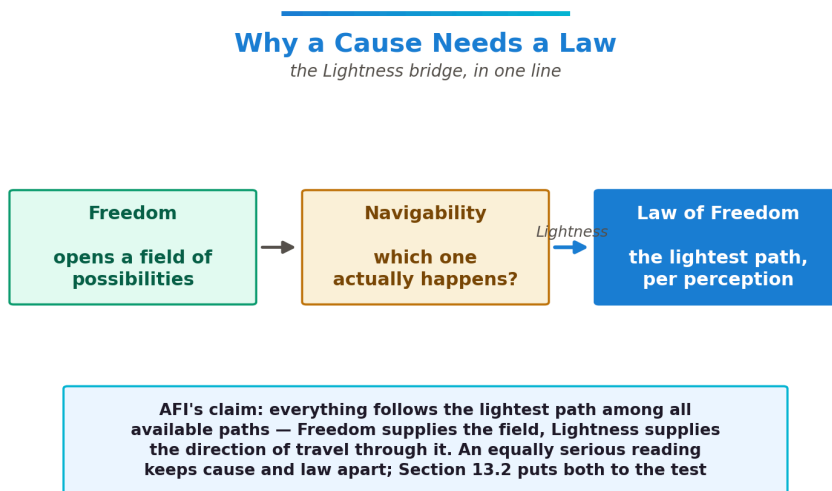


Figure 10

Note. Freedom (Thesis 1) opens a field of admissible continuations; that openness alone leaves the Navigability question standing — which continuation happens. AFI's own candidate closes the loop by naming Lightness as the selection principle: everything follows the lightest path among all available paths, according to a system's own perception. The equal, opposite reading treats cause and law as separate registers, each carrying its own distinct principle rather than a shared one; Section 17 is where the two readings meet a test.

Figure 11

Four Readings of the Bridge

Four Readings of the Bridge

the same open question, met four ways — every reading held equally

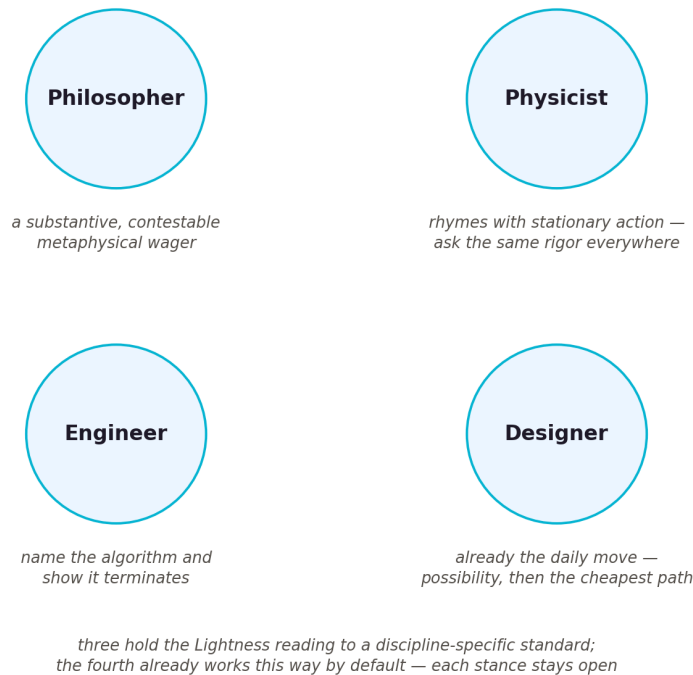


Figure 11

Note. The same open question earns four different reactions across four professional vantage points, and no vantage point is privileged over the others here. Philosophy, physics, and engineering each hold the Lightness reading to a discipline-specific evidentiary standard; design practice already works this way by default, which is why the reading arrives as familiar there and as a claim still owed its proof everywhere else.

3. Freedom as Path Availability in AFI

Freedom's most compressed statement is $F = (\text{Perception}/\text{Distortion})^\alpha$. Perception names a system's accessible representation of relevant reality and available paths; Distortion, whatever reduces that access.

- **Thesis 1** — *Freedom is the Cause of All.*
 - *Weak reading: some possibility is always a necessary condition for actuality — near-platitudinous once stated carefully.*
 - *Strong reading (tested first, per this programme's discipline): structured possibility is not merely a necessary background condition but **constitutive** of causal production — production is the actualization of one member of an already-structured field of admissible continuations, doing genuine explanatory work.*

- **Falsifiability:** if the strong reading collapses into the weak one — adding nothing beyond “something must be possible” — Thesis 1’s strong form fails.
- **Thesis 3** (FLRP: Freedom → Logic → Relationships → Physics) — physical constraint is the last-crystallizing layer: observed in reverse order but generated forward. Used here only where it clarifies how a possibility field differentiates into physical persistence (§11, Model C).
- **Thesis 5** — intelligence is the capacity to improve operative Freedom: better perception, discovering real paths, removing false ones, counterfactual simulation, and preserving future Possibility. Examined independently elsewhere against AI world-model, active-inference, and empowerment research.

Figure 12

Freedom = Possibility = Path Availability

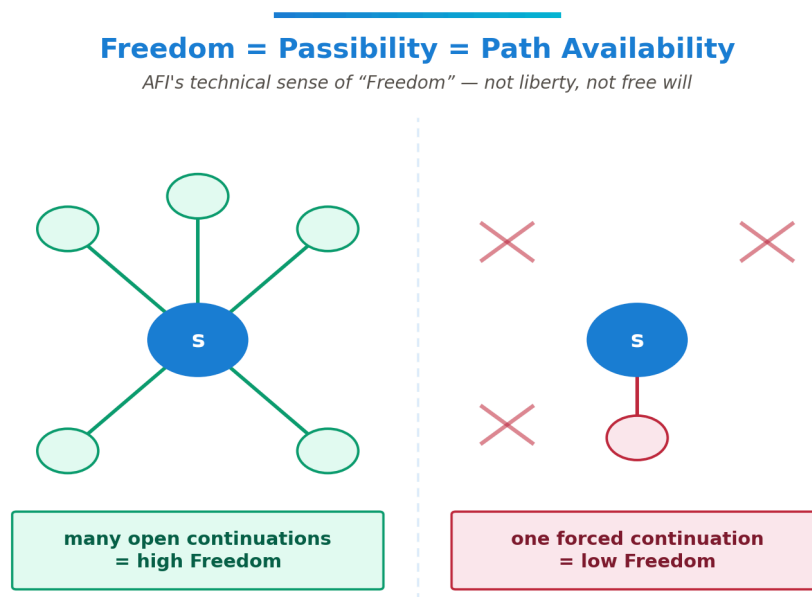


Figure 12

Note. AFI’s technical sense of “Freedom” is exactly the structured availability of possible continuation from a state — many open continuations is high Freedom, one forced continuation is low Freedom — and nothing more than that. The two-panel contrast is a deliberately extreme illustration (five open branches vs. one forced branch); most real states fall somewhere between these poles, and the figure does not itself define a metric for exactly where. Falsifiability: A state independently verified to have more admissible continuations than another, but which this paper’s own formal apparatus (Section 11) scores as lower Freedom with no compensating Perception/Distortion difference, would falsify this identification.

Figure 13

The Freedom Ratio

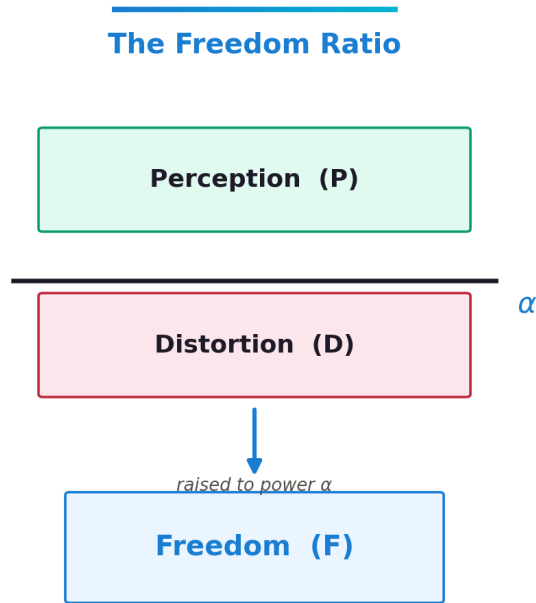
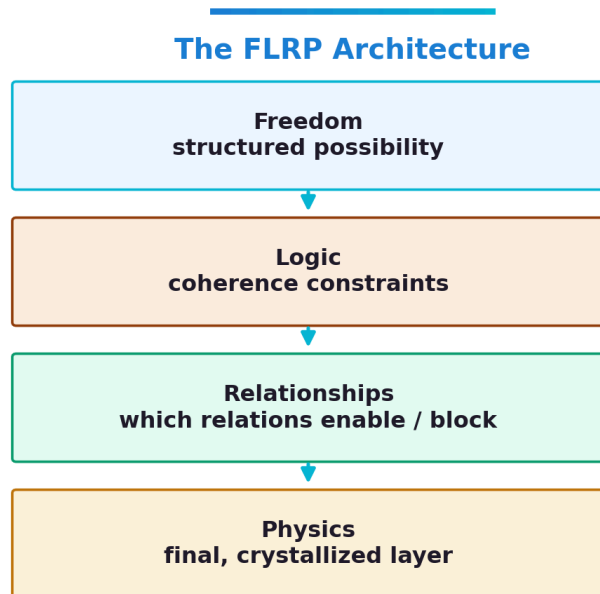


Figure 13

Note. The candidate Law of Freedom formalizes Freedom as Perception divided by Distortion, raised to a power α — a ratio, not a sum, which is why AFI resists additive reward-style formalizations. The figure visualizes the ratio's structure, not its calibration; the value of α , and whether it is a fixed constant or itself system- or domain-relative, is left open by the paper (Section 11).

Figure 14

The FLRP Architecture



realization flows downward — Physics is what becomes actual

Figure 14

Note. AFI proposes reality is organized through four ordered layers — Freedom, then Logic, then Relationships, then Physics — with Physics as the final, crystallized layer that realization flows into. The figure depicts the proposed ordering; it does not itself demonstrate that the four layers are causally or explanatorily ordered in this way rather than some other way, which is an open metaphysical commitment (Section 13).

4. What Makes Causation Possible?

If A causes B, must some admissible transformation exist through which B can become actual from A?

- A first-pass “yes, trivially” risks emptiness: if “admissible” is defined only retrospectively from what occurred, the claim is tautological, not a discovery. The substantive version asks something stronger: whether the admissible structure is fixed independently of, and prior to, which member is realized — if so, it constrains what can become actual rather than merely redescribing it.
- This is the weak/strong reading distinction (§3) — the single most consequential distinction, since nearly every later claim depends on which reading is in play.
- **Falsifiability:** the strong reading is contentful only if admissibility can be specified before the outcome is known — a formulation stated only after the fact reduces to the weak reading.

Figure 15

Four Ways to Answer “Why B?”

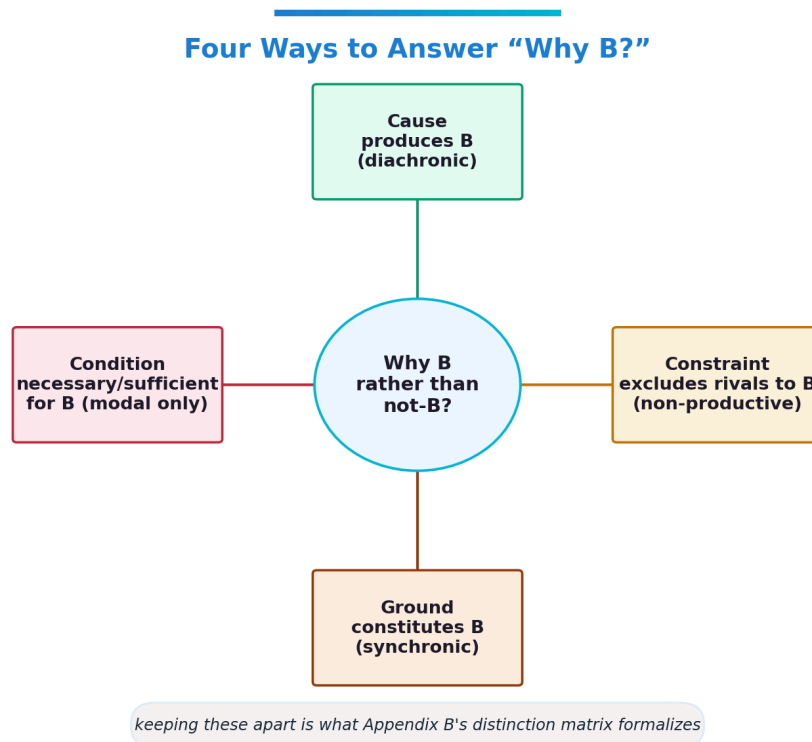


Figure 15

Note. An ordinary causal-sounding question (“why B rather than not-B?”) can be answered in at least four structurally different ways — cause (diachronic production), ground (synchronic constitution), condition (modal necessity/sufficiency), or constraint (exclusion of rivals) — and AFI’s own claims must specify which one they intend. The four-quadrant layout treats the answers as mutually exclusive for clarity; several of the sources in Appendix B’s Table C note live philosophical disputes about whether some pairs (e.g. cause and ground) fully reduce to one another.

Figure 16

The cause-to-law derivation chain, with epistemic-layer tags

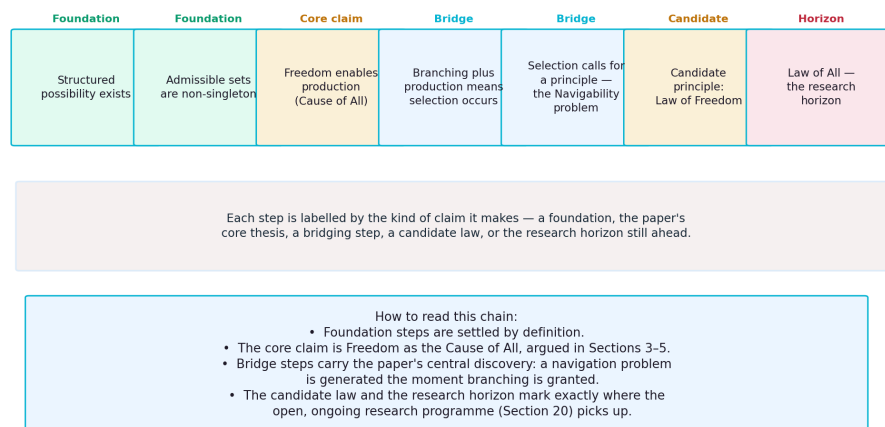


Figure 16

Note. The seven-step chain from structured possibility to a candidate Law of All crosses from logical/definitional (L1) through metaphysical (L3) to empirically open (L5) claims; arrows 4→5 and 6→7 are where the derivation is weakest. The layer tags reflect this paper's own audit (Section 13) and are not an independent, formal proof-theoretic classification.

5. Possibility, Passibility, and Causal Production

“Possibility” is a family of frequently conflated concepts, not one.

- *Four senses of “possibility” are commonly conflated:*
 - **Logical** — freedom from contradiction (most permissive).
 - **Metaphysical** — truth at some possible world, on realist accounts.
 - **Physical/nomological** — consistency with actual laws of nature (typically the most restrictive of the three).
 - **Epistemic** — consistency with what a knower currently believes.
 - A nesting relation runs through the first three; collapsing any two is a risk the Definition Registry (Appendix B) exists to prevent.
- **Passibility** (AFI's term) is narrower than all four: the structured availability, from a given state, of a reachable path to a candidate continuation. Unlike mere logical or metaphysical possibility it requires an actual route from where a system currently is; unlike bare physical possibility, a property of one transition, Passibility is a property of a field's whole branching structure. Its closest formal cousin is control theory's reachable set, or Aubin's (1991) viability kernel (§11).
- Causal production, tested here, is **realized Passibility**: a causal event is the actualization of one member of a prior state's admissible continuation set, not merely correlated with, or counterfactually dependent on, that set.
- Rival **process theories of causation** (Salmon 1984; Dowe 2000) ground production in transmission of a conserved physical quantity along an actual world-line, without representing unrealized alternatives. They handle transmission cases cleanly but struggle with prevention, omission, and absence-causation — cases needing possibility-talk regardless, evidence a possibility-primitive does real work.

- **Falsifiability:** if these cases turn out fully handled by process theories without covert possibility-talk, this evidence disappears.

Figure 17

Constraint vs. Distortion

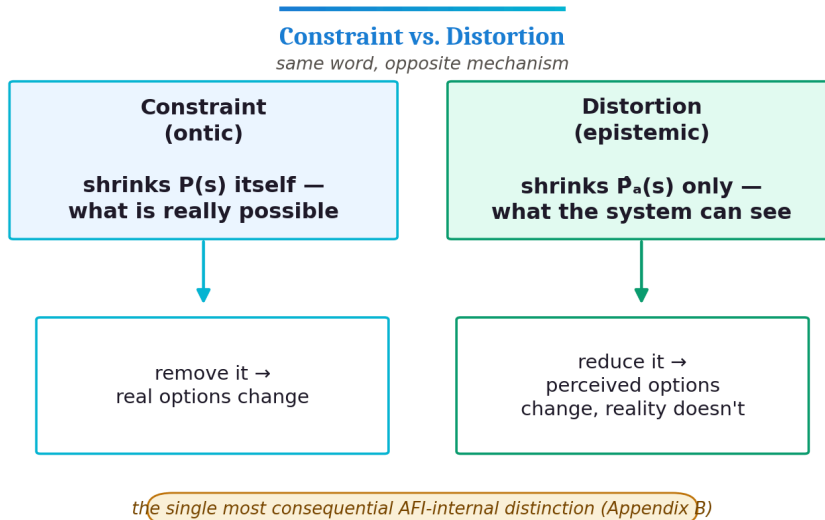


Figure 17

Note. Constraint (ontic, shrinks the real admissible set) and distortion (epistemic, shrinks veridical perception without touching the real set) are distinct and must not be conflated — the single most consequential AFI-internal distinction (Appendix B). The figure's two-panel layout depicts an idealized separation; some real cases may make the two hard to disentangle empirically even where they remain conceptually distinct. Falsifiability: A system that changes its perceived options without any change in $D_a(s)$ — while the real admissible set $P(s)$ provably changed — would show the two are not cleanly separable as claimed.

6. What Is a Path?

A path is a structural object — an ordered sequence of states connected by admissible transitions, specified without reference to timing.

- Kept separate from a **trajectory**, which adds a rate of traversal to the same sequence.
- Central claims about paths are structural: a path exists, is available, is one among several. The Law of Freedom, by contrast, is about trajectories — which path, at what rate and cost, a system actually realizes. Confusing the two smuggles dynamical content into a structural claim, the error the Total Distinction Matrix (Appendix B, rows 15–16) catches.
- A path's existence does not require it ever be traversed — the modal core of the programme: unused corridors, unmanifested dispositions, and unexploited affordances all count as real on Possibility's terms, a status process theories of causation cannot grant (§5).
- **Falsifiability:** a path's "existence" is substantive only if specifiable independently of whether it is ever taken — an account identifying paths only retrospectively fails this test.

Figure 18

Path vs. Trajectory

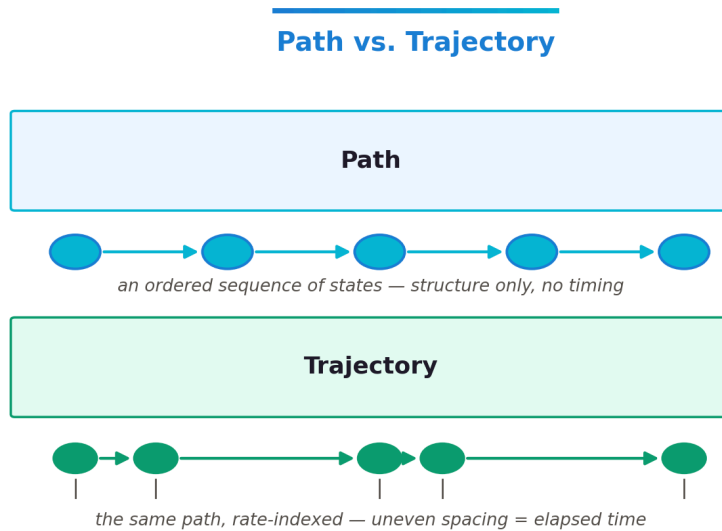


Figure 18

Note. A path is a purely structural, ordered sequence of states with no timing information, while a trajectory is the same kind of sequence rate-indexed by elapsed time — the same path can correspond to many different trajectories. The figure shows a single five-state example; it does not depict continuous-time or continuous-state trajectories (Universe 26, the Flow Field, in Appendix A).

7. What Is Navigability?

Where a path's existence is a fact about structure, Navigability is a fact about a system's *relationship* to that structure: the capacity by which available transitions can be discriminated, reached, traversed, or transformed.

- *Existing disciplines carve related territory:*
 - **Kleinberg-style network navigability** — a graph is navigable if decentralized greedy search finds a target in bounded steps (active 2025–2026 theoretical CS; §8, Appendix C S37–S40).
 - **Reachability/controllability** (control theory) — a state reachable or controllable relative to specified dynamics and admissible inputs.
 - **Viability** (Aubin 1991), extended to probability-measure spaces by Bonnet-Weill, Domínguez Corella & Frankowska (2025).
- *Navigability is an umbrella term for these related but not identical notions, kept as distinct dimensions so no one of them silently stands in for the whole:*
 - *Existence of a path (Passibility).*
 - *Discoverability and perceptibility (Perception).*

- *Reachability and viability (control-theoretic senses).*
- *Cost, risk, reversibility, and information requirement (candidate dimensions of Lightness, §14).*
- *Branching and future-option structure (future Passibility, §11 Model H).*
- *Where an established concept already does part of this job, this paper uses that concept by name rather than inventing a redundant one — several do, sometimes more rigorously than AFI’s own corpus, including viability theory and POMDP belief-state reachability.*

Figure 19

The Navigability Problem

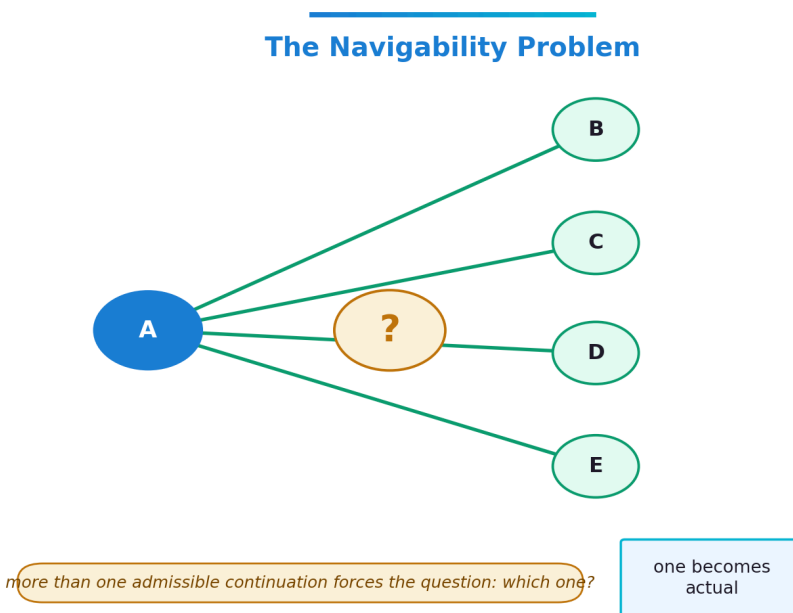


Figure 19

Note. The moment a state has more than one admissible continuation, a forced question arises — some principle must select which one becomes actual — and this question is logically prior to any specific answer (such as the Law of Freedom). The figure shows four illustrative continuations from one state; it does not depict cases with infinite or dynamically changing admissible sets (Universes 4 and 29 in Appendix A).

Figure 20

Admissible, Reachable, and Viable

Admissible, Reachable, and Viable

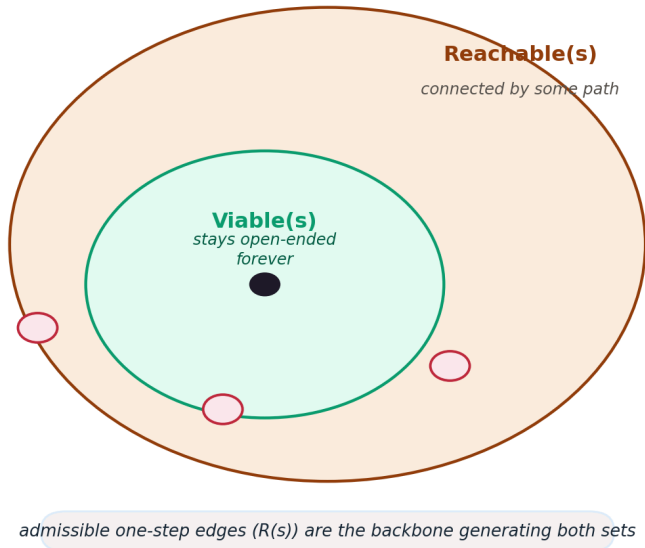


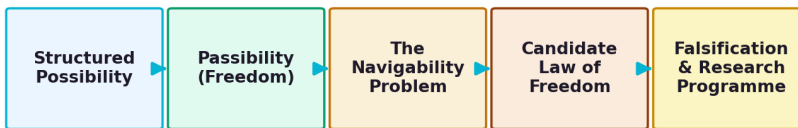
Figure 20

Note. Reachability (connected by some admissible path) and Viability (forward continuations stay open-ended) are related but distinct properties of a state — a state can be reachable without being viable — both built on the one-step Admissible set as their structural backbone. The nested-circle layout implies Viable(s) is always a strict subset of Reachable(s); this holds under the paper's own definitions but has not been independently verified against every formal model in Section 11.

Figure 21

The Argument, End to End

The Argument, End to End



Sections 1–7 establish the left three steps; Sections 8–21 test the last two against literature, formal models, and forty counterfactual universes

each arrow is a claim this paper argues for, not assumes

Figure 21

Note. The paper’s overall argument moves from structured possibility, through Passibility, to the Navigability Problem (Sections 1–7), then tests a candidate Law of Freedom against literature, formal models, and forty counterfactual universes before reaching the Falsification Matrix and research programme (Sections 8–21). The five-step roadmap compresses a 21-section, formally intricate argument into a single line for orientation; it is not a substitute for reading the derivation chain in Figure 16 or the falsification detail in Section 16.

8. The 2025–Current Research Frontier

Per this programme’s RECENT_RESEARCH_FIRST discipline, the literature search began with the 2025–2026 frontier before foundational backfill — starting from Hume, Lewis, or Pearl risks anchoring the argument to a map recent work has redrawn.

Fifty-two independently verified, non-AFI sources, confirmed against their actual abstracts, span five clusters (Source Ledger, Appendix C).

- **AFI’s core moves are not arriving into an empty field.**
 - A “possibility without worlds” line of formal semantics (Holliday 2025; Nodelman & Zalta 2026, in press) treats structured possibility as a semantic primitive, paralleling what Passibility needs formally.
 - An entropy/information-theoretic account of directed causal and behavioral production — Gyenis’s “Causal Second Law” (2026, in press) and Kiefer’s (2025) unification of active inference and empowerment — is the strongest existing rival to the Law of Freedom. Neither existed pre-2025.
- **A genuine, active disagreement in the causal-powers literature bears on Thesis 1.**
 - Lewtas (2025) proposes a “rectangle model” where causal powers are primitive and dispositions derive from them — powers-in-things, rather than path-availability-in-a-field.

- Azzano and Giannini (2026) argue that letting potentialities change over time commits dispositionalism to a demanding package: Strong Necessitism, Permanentism, and a Moving Spotlight theory of time. Against this, AFI's field-level primitive is a direct rival, not a variant (§15).

Table 2. The 2025–2026 frontier, by cluster

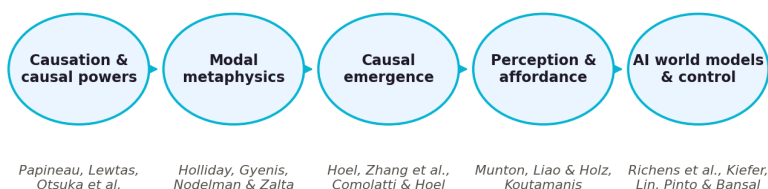
<i>Cluster</i>	<i>Representative sources</i>	<i>What the cluster currently holds</i>	<i>Bearing on this paper</i>
<i>Causation & causal powers</i>	<i>Papineau 2025; Pasnau 2025; Lewtas 2025; Otsuka et al. 2025; Comolatti & Hoel 2025</i>	<i>Disagreement between law-recursion (Papineau), powers-primitivism (Lewtas), process/categorical formalizations (Otsuka et al.); convergence on shared primitives (Comolatti & Hoel)</i>	<i>Clearest rivals to a possibility-primitive account of production (§5, §15)</i>
<i>Modal metaphysics</i>	<i>Holliday 2025; Gyenis 2025; Nodelman & Zalta 2026; Vacek 2026; Brast-McKie 2025</i>	<i>An active “possibility without worlds” formal programme, plus a taxonomy unifying physical/empirical/couterfactual possibility (Gyenis)</i>	<i>Closest rigorous formal home for Possibility as a semantic primitive (§6); AFI must adopt or differentiate</i>
<i>Causal emergence</i>	<i>Zhang et al. 2025; Hoel 2025 (“Causal Emergence 2.0”); Comolatti & Hoel 2025; Liu et al. 2025</i>	<i>A mature, SVD/effective-information account of when macro-scale causal description outperforms micro-scale description</i>	<i>Formal machinery for what makes production “real” at a scale — relevant to FLRP's claim Physics crystallizes last (§3)</i>
<i>Perception & affordance</i>	<i>Munton 2025; Liao & Holz 2025; Koutamanis 2026; Przybylski & Króliczak 2026; Xu et al. 2025/2026</i>	<i>Active work on whether modal properties are directly perceived (Munton) vs. computationally inferred (Liao & Holz); affordance processing as dissociable systems</i>	<i>Load-bearing for Section 9; Liao & Holz's external/internal split is the closest rival to Perception/Distortion</i>
<i>AI world models, control, intrinsic motivation</i>	<i>Richens et al. 2025; Dillies et al. 2025; Zhao et al. 2025; Kiefer 2025; Xuan & Wang 2025; Lin, Pinto & Bansal 2026</i>	<i>Proof generalizing agents contain extractable world models; causal (not merely predictive) architectures; a proven unification of active inference and empowerment; reachability under perception-based uncertainty</i>	<i>Engaged most directly in Sections 13, 19, 22; supplies AFI's strongest analogues and two most serious rivals</i>

Figure 22

The 2025-2026 Research Frontier

The 2025-2026 Research Frontier

five clusters of live literature this paper tests AFI against



fifty-two independently verified, non-AFI sources across these five clusters

Figure 22

Note. Fifty-two independently verified, non-AFI sources fall into five clusters. None of them existed before 2025 in a form that anticipates AFI's central moves in substance — the recency-first discipline this paper follows exists precisely to catch that risk before it goes unnoticed.

Three gaps deserve explicit note:

- *Least-action/stationary-action metaphysical interpretation: ~15 query formulations across two search passes returned only two on-topic sources (Kodaj, Bernáth & Pickup 2025; Dong Yang 2025) — the thinnest area of the corpus. §16 leans more on pre-2025 work (Terekhovich).*
- *Viability theory (strict Aubin sense): only one qualifying theoretical, non-engineering-applied contribution (Bonnet-Weill, Domínguez Corella & Frankowska 2025) — more active in application than foundations.*
- *Potentiality/actuality as an explicit analytic-metaphysics target: only one direct hit (Badescu 2025) — §15's Aristotelian revival leans on secondary literature more than an active current programme.*

Falsifiability: a pre-2025 possibility-without-worlds or entropy-based navigation programme anticipating AFI's central moves in substance, not vocabulary, would weaken the “novel” framing correspondingly.

9. Perception of the Possible

9.1 Why Perception must be a first-class object

If causal production runs, even partly, on structured possibility, any system that acts within that structure must relate to it somehow — that relation does real explanatory work, not incidental color.

Collapsing Perception to a single scalar value would discard the structure the paper needs: *which* continuations are discriminated, *how*, and *what kind* of access this is.

- Three objects are kept rigorously separate:
 - The **real/structural possibility landscape** $P(s)$ — continuations admissible from s , observer-independent.
 - The **perceived possibility landscape** $\hat{P}_a(s)$ — what system a can currently discriminate.
 - Downstream, non-perceiver-relative notions **Reach(s)** and **Viab(s)** (Table B, rows 20–22).

9.2 The Perception ladder, P0–P7

Eight senses of “perception” appear across the literature, ordered least to most representationally rich, scored for what exists independent of the observer, what is sensed, inferred, actionable, can be wrong, and changes behavior.

Type	Paradigm case	Independent of observer	What is sensed/accessed	Inferred?	Can be wrong?	Changes behavior?
P0 — Passive physical sensitivity	A resistor's current response to voltage; a viability kernel's boundary	Yes, entirely	Nothing represented — “access” here is really structural coupling, not perception	No	N/A — a resistor cannot misperceive, only be a certain way	N/A — “behavior” is just physical response, not action selection
P1 — Biological sensory discrimination	Photoreceptor transduction; olfaction	Yes	Raw sensory transduction of proximal stimuli	Minimally (transduction, not judgment)	Yes (sensory noise, adaptation, illusion)	Yes, via downstream processing
P2 — Ecological affordance perception (Gibson)	Direct pickup of invariants specifying action-possibilities	Yes — affordances are claimed to be real, observer-independent relational properties	Structured energy arrays (ambient optic array) specifying affordances directly, no inferential intermediary	Explicitly denied by Gibson (direct perception)	Yes, but Gibson's framework says little about how — Liao & Holz (2025) call this a real limitation	Yes, directly (detection → action, no representational bottleneck claimed)
P3 — Human modal visual perception	Perceiving that a cup could tip, a door could swing	Contested — Munton (2025) argues modal content is partly constitutive of what is perceived	Modal properties (ways things could be), argued to be part of visual content itself	Contested — this is precisely Munton's question	Yes (misjudging what an object could do)	Yes — shapes object individuation and predicted trajectories
P4 — Belief-state estimation (POMDP)	$b(s) = P(s \text{history})$	Partially — the true state is independent	A full probability distribution over possible	Yes, via Bayesian filtering — the paradigm	Yes, formally (miscalibration, wrong)	Yes, directly determines the

Type	Paradigm case	Independent of observer; the belief is not	What is sensed/assessed true states, from noisy observations	Inferred? inference case here	Can be wrong? observation model)	Changes behavior? belief-MDP policy
P5 — Causal representation	A model of interventions and their consequences, not just correlations	Independent (true causal structure) vs. the model (not)	Structural/interventional relationships (do-calculus-relevant), not just state	Yes, specifically causal (not merely predictive) inference	Yes — wrong in a qualitatively different way (right predictions, wrong mechanism) than P4	Yes — determines which interventions are even considered
P6 — AI world models (incl. causal world models)	Learned transition function $\hat{T}(s' s,a)$; COMET-style causal world models	Independent (true dynamics) vs. model (not)	Learned latent transition structure, sometimes disentangled into causal variables	Yes, learned from data	Yes — spurious correlations, distribution shift	Yes — the substrate for planning/rollout
P7 — AFI Perceived Possibility	$\hat{P}_a(s)$	No — this is explicitly the perceiver-relative object	The pre-choice structured field of continuations operatively available to a given system, as that system currently accesses it	Depends on substrate (can incorporate any of P1–P6 as inputs)	Yes, definitionally — Distortion is precisely $\hat{P}_a(s)$'s divergence from $P(s)$	Yes — the direct input to Lightness

What this ladder shows.

- P0 licenses no talk of “access” or “error” and is demoted to “structural coupling,” to avoid implying consciousness in non-cognitive systems; P1–P2 remain representation-light or representation-free — Gibson denies any inferential intermediary — while P3–P6 are all, in different literatures, representational.
- P7 is AFI’s proposed integration point: not a new kind of access, but a claim that one formal object, $\hat{P}_a(s)$, can absorb inputs from any of P1–P6 depending on the system. **Falsifiability:** it earns its keep only if this is a genuine integration, not a relabeling of “whichever of P1–P6 happens to apply” — the same novelty-vs-redundancy question raised for the Law of Freedom, now applied to Perception.

Figure 23

Objective vs. perceived possibility landscape

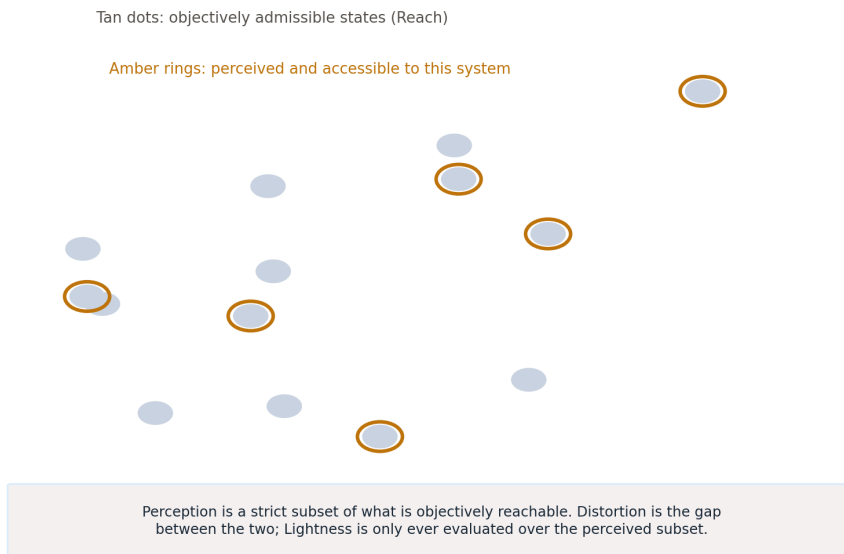


Figure 23

Note. Perception (red rings) is a strict subset of the objectively admissible/reachable set (grey), and Distortion is the gap between them. Point placement is illustrative and randomly generated for visual clarity; it encodes no measured perception data.

Figure 24

The Perception Ladder, P0–P7

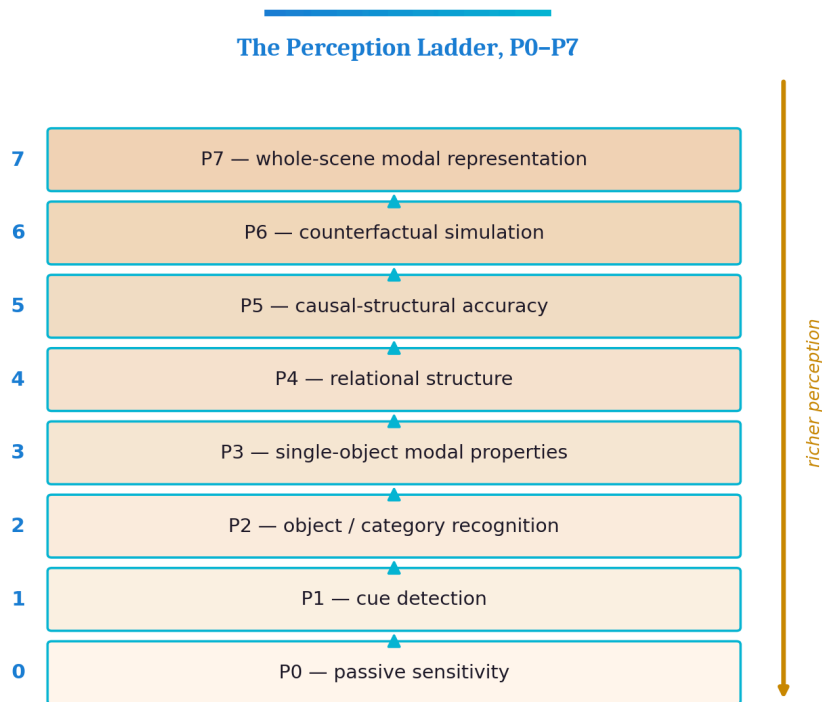


Figure 24

Note. Systems can be ordered on an eight-rung ladder from passive sensitivity (P0) to rich, modal, structurally aware perception (P7), giving Section 9’s taxonomy a single visual scale. Real systems may straddle rungs or score unevenly across sub-capacities the ladder compresses into one ordinal position. Falsifiability: A system scored at rung k that reliably succeeds on tasks requiring rung $k+2$ abilities, without intermediate failure, would undercut the ladder’s ordering.

9.3 Munton (2025): do we perceive modal properties?

Jessie Munton’s (2025) work on modal perception is the paper’s most important contemporary anchor for P3.

- *Claim: visual perception can represent not merely how an object currently is but ways it could be — modal content that individuates and tracks objects, supported by phenomenological considerations (an object seen as precariously balanced) and by vision-science findings on how perceptual systems represent persistence and dynamics. This is load-bearing for AFI’s T3/P3 claim: if Munton is right, “perceiving possible continuations” is an argued feature of ordinary vision, not an imported metaphor.*
- *Two qualifications apply. First, Munton’s argument depends on a “rich content” view of perception — content beyond low-level sensory properties — which is itself contested; citing her inherits that dependency, not a settled consensus. Second, her modal properties are typically local and object-bound (this cup could tip, this door could swing) — she does not argue perceivers perceive an entire structured field of all admissible continuations of a scene, the stronger claim P7/T2 needs.*
- *Honest reading: Munton establishes a beachhead for modal perception of individual objects, not for whole-scene Possibility fields — that extension remains AFI’s own.*

- **Falsifiability:** if controlled testing shows perceivers cannot reliably discriminate object-level possibilities beyond habitual function — P3 reducing to P1/categorical labeling — the Munton anchor weakens substantially.

9.4 The door example: object-first vs. possibility-first perception

A rectangular wooden door, encountered ordinarily, is categorized simply as *door* — one label compressing a much larger space of admissible transformations: board, ramp, shelter component, temporary bridge, even a surfboard-like object under some conditions.

The claim under test is not “anything can be anything” (trivial and false), but that object identification may **compress a larger modal/affordance/transformation landscape**, with different observers discriminating different slices depending on training.

- **Liao & Holz (2025)** reject the Gibsonian claim that affordances are directly perceived without inference: on their account affordances are computed, split into an “external” component (objectively available action-possibility, close to $P(s)$) and an “internal” component (perceiver’s confidence and predicted utility, close to $\hat{P}_a(s)$ plus a value weighting) — architecturally close to AFI’s Perception/Distortion split (a ratio of Perception to Distortion, versus Liao & Holz’s product of confidence and predicted utility), one of the strongest existing challenges to any novelty claim for AFI’s formalism.
- **Koutamanis (2026)** distinguishes affordances a designer perceives directly in a physical artifact from affordances perceived in representations of it, such as drawings or models — both structurally analyzable via activity/action-identification theory. Applied to the door: a carpenter, an architect reading its drawing, and a child may be perceiving genuinely different, structurally analyzable slices of the landscape, conditioned by trained activity-repertoire — a testable, not anecdotal, version of the claim.
- **Przybylski & Kroliczak (2026)** (fMRI, $n=62$) find that affordance processing across visual, grasp-planning, and pantomime-use tasks is organized as multiple dissociable neural systems, not one hemispheric module. This corrects any AFI presentation of “Perception” as a single unified faculty, and better fits AFI’s own multi-factor aggregate ($P_C = \prod_i P_i^{\beta_i}$, Table A row 30) than a monolithic account.

9.5 Imaginable possibility vs. structurally passible possibility

A door imagined as a rocket and a door physically usable as a board are not the same phenomenon. Treating them as such would be a paper-undermining equivocation: AFI’s programme depends on Possibility being a fact about the world’s real transition structure, not about what any system — including the theorist — can imagine. A theory that slides from “imaginable” to “passible” inherits imagination’s unconstrained richness and loses all empirical content.

- **Imaginable possibility** is a fact about a cognitive system’s generative capacity. Humans, and generative AI, can represent almost arbitrary counterfactual transformations, constrained only by imagination’s own freedom rather than the object’s actual constraints.
- **Structurally passible possibility**, by contrast, is a fact about $P(s)$ itself. Door-as-board is passible because planks of that size, shape, and rigidity really are usable as boards under real physical constraints (FLRP’s Logic/Relationships/Physics). Door-as-rocket is not, however vividly imagined.
- Munton’s (2025) thesis is a partial safeguard: if perceivers genuinely perceive, rather than merely imagine, real object-possibilities, then P3-level modal perception is answerable to structural constraints. The safeguard is not absolute — misperception, or Distortion, names where imaginable and passible possibility come apart.
- **Falsifiability:** an application that cannot independently specify $P(s)$ from a system’s imaginative output — where “passible” is inferred solely from what was imagined or later realized — fails this discipline.

Figure 25

Object-first vs. possibility-first perception (the door example)

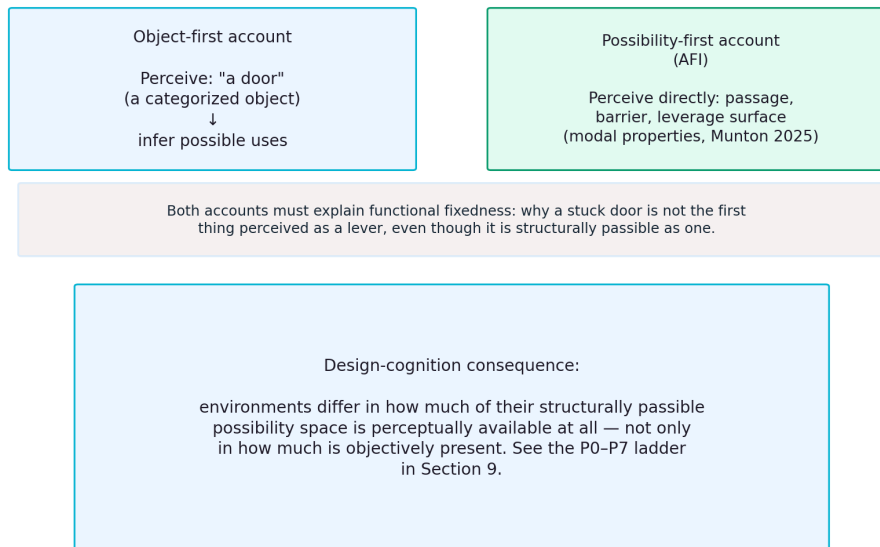


Figure 25

Note. Object-categorization accounts and modal-property accounts (Munton 2025) offer structurally different explanations of what is perceived when a system encounters a door. The two-panel contrast simplifies an active, unsettled debate (Section 9) into a binary for exposition; several 2025–2026 sources hold intermediate or hybrid positions.

9.6 Functional fixedness and design cognition

Functional fixedness — the well-established cognitive bias (Duncker 1945 and a large subsequent literature) toward perceiving an object only in terms of its habitual function — is the standard psychological name for the door example’s core phenomenon.

- *This is not evidence everyone naturally perceives the full transformation landscape and merely “forgets” it under habit — rather, ordinary object perception is by default lossy with respect to $P(s)$, a documented instance of omission-type Distortion (Table A row 33) arising from habitual categorization rather than sensory noise or latency. Design cognition research, including Koutamanis (2026), shows trained designers partially overcome this compression not via different perceptual hardware but a broader activity-repertoire that expands which slices of $P(s)$ become visible as candidates.*
- *The defensible claim: **intelligence is partly the ability to perceive real modal structure beyond fixed categories** — domain-trained perception reduces omission-type Distortion relative to default, category-locked perception, for the slice of $P(s)$ that repertoire covers, trained discrimination rather than unconstrained imagination.*
- ***Falsifiability:** testable via the affordance/reachability-judgment paradigm of testbed (g), §17 — no measurable difference between trained and untrained observers under controlled conditions would fail this claim.*

9.7 Comparison with AI: what would an AFI-style perceptual system need?

An LLM's base training objective represents semantic/statistical association over text — a P6-adjacent object at best. Richens et al. (2025) guarantees extractable predictive structure in generalizing *agents*, not in base pretraining; that guarantee covers prediction, not intervention-validity.

- *A causal world model, such as COMET or Meta-Causal Graph, represents genuine P5/P6-level interventional structure — richer than LLM association, since it distinguishes real levers from spurious correlations. But no cited AI architecture, causal world model or LLM alike, outputs a single object combining that structural accuracy with a distortion-annotated enumeration of the possibility field itself — jointly tracking what exists, what can become, what the system falsely perceives, and how action changes future Possibility, updated online as action changes what's available (Model H's "world change → new Possibility" gap).*
 - *The closest AI instances of object-level P3-style modal representation are narrow-domain rather than whole-scene:*
 - **FOCUS** (Ferraro et al. 2025) — *object-centric world models with an entropy-maximizing exploration objective.*
 - **A0** (Xu et al. 2025/2026) — *affordance-aware manipulation models predicting post-contact trajectories, encoding the space of possible manipulation continuations for an object.*
 - *Honest verdict: a genuine P7-style architecture is a plausible integration target, partially demonstrated in pieces — FOCUS, A0, COMET, Meta-Causal Graph. Not solved, and not wholly novel either.*
-

10. Counterfactual Universe Laboratory

Rather than testing Theses 1 and 2 against one or two illustrative examples, this paper constructs forty formally specified toy universes, deliberately including cases designed to embarrass the theory.

Each universe specifies a state space, a transition kernel or relation, laws and constraints, a probability/branching structure, an observer/Perception model, and the resulting reachable/viable sets.

For each, the laboratory asks:

- *Could causation exist without any admissible transition? Could actuality arise from literal impossibility?*
- *Does Thesis 1 become merely definitional, or say something substantive? Does the strong reading add anything beyond the weak necessary-condition reading?*
- *Does the Law of Freedom make a determinate, falsifiable prediction — and where it does, what named rival law would predict a different outcome, and by what measurement could the two be told apart?*

Figure 26

Counterfactual Universe Laboratory atlas

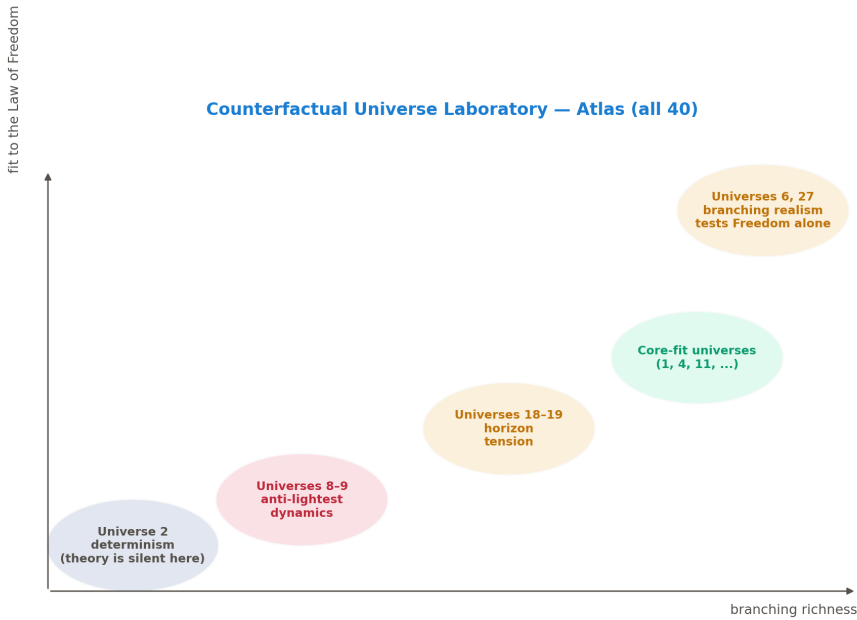


Figure 26

Note. The forty counterfactual universes cluster informatively along two axes — branching richness and fit to the Law of Freedom — with named universes marking the theory’s clearest successes and clearest failure modes. The two axes are a schematic simplification for display; the full laboratory (Appendix A) specifies each universe formally and should be consulted before drawing any conclusion from position alone.

The complete forty-universe laboratory is reproduced in Appendix A. Four results are consequential enough to state here.

- **Determinism empties the theory, not just weakens it.** Universe 2, a single deterministic successor at every state, satisfies Thesis 1 trivially — the one actual transition is tautologically the one admissible transition — but leaves nothing for the Law of Freedom to select among, or for Perception to discriminate. This is a failing criterion (§20, criterion 8) and the first limitation (§23): Navigability is a claim about branching domains only.
- **Naive T2 is false, not merely incomplete, without a horizon term.** Universe 18 (a locally lightest path eliminating most future continuations) and Universe 19 (a locally heavier path preserving a large future option space) show that an unqualified Law of Freedom — minimize the next step’s cost — makes the wrong prediction whenever local and future-preserving optimization diverge, the source of §19’s horizon-weighted reformulation, $L(\pi|s,a,H)$, which must be independently motivated rather than fit after the fact.
- **Anti-lightest and maximum-resistance dynamics are genuine, not merely hypothetical, falsifiers.** Universes 8 and 9 construct systems that reliably select the heaviest, most resistant path — no honest reading of the Law predicts this, and a real system behaving this way reproducibly would falsify the unqualified theory outright, a direct instance of `LIGHTEST_PATH_SKEPTICISM`.
- **Quantum branching and pure-random actualization pressure-test the strong reading of Thesis 1, independent of Thesis 2.** Universe 27, a quantum transition that is Everett-consistent, and Universe 6, pure random actualization, formalize the many-worlds and brute-fact objections raised elsewhere: if every branch is realized, or realization is a primitive unexplained fact, there may be nothing for a navigation law

to explain, even where Thesis 1's weak reading holds — the clearest formal evidence that Thesis 1 and 2 can come apart.

The verdict is neither confirmation nor refutation, but a map: well-posed where domains are branching, single-history, and horizon-aware; vacuous where deterministic; underspecified where local Lightness is unqualified; contested where many-worlds or brute-fact realization apply. This is the discipline the Falsification Matrix (§20) records systematically.

11. Formal Models of Possibility

Eight candidate formalizations of the same underlying claim follow, each drawn from an existing formal tradition, letting AFI's core proposal be located precisely within established mathematics rather than declared to transcend it.

For each model, this section asks: where do Freedom/Passibility, causation, navigation, perception, distortion, and lightness sit? Where does the model break down? And is a given stated result definitional, metaphysical, or empirical, per the paper's epistemic-layer discipline (L1–L4)?

Shared notation, consistent with Appendix A: state space S ; $P(s)$ = real/structural Passibility of s ; $\hat{P}_a(s)$ = agent a 's perceived possibility set; $\text{Reach}(s), \text{Viab}(s) \subseteq S$; $D_a(s)$ = distortion; $L(s \rightarrow s') =$ lightness functional; $F_a(s) = (\mu[\hat{P}_a(s)]/D_a(s))^\alpha$.

Figure 27

Eight Formal Models of Possibility

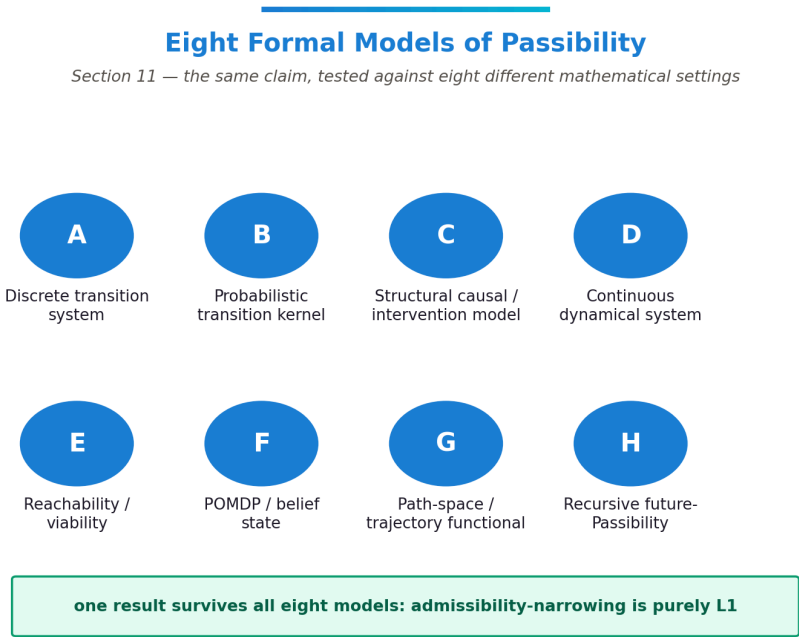


Figure 27

Note. The same underlying claim is located inside eight established mathematical traditions, from discrete transition systems (A) to a recursive, self-referential future-Passibility formulation (H). Only one result — that admissibility-narrowing is purely a matter of construction (L1) — survives across all eight unconditionally; everything else the paper wants to say is model-specific and genuinely contestable.

Model A — Discrete Transition System: $U = (S, T)$

Definition. S is a (possibly infinite but discrete) set of states. $T \subseteq S \times S$ is an admissible-transition relation. $P(s) := \{s' : (s, s') \in T\}$.

- *Passibility* = T ; *Passibility*(s) > 0 iff $P(s) \neq \emptyset$.
- *Causation* = any traversed $(s, s') \in T$.
- *Navigation* = which $s' \in P(s)$ is traversed when $|P(s)| > 1$. T alone never answers this.
- *Perception/Distortion* = the agent-relative subset $\hat{P}_a(s)$ and its symmetric difference from $P(s)$.
- *Lightness* is undefined in the bare model; it must be imported as a functional $L: T \rightarrow \mathbb{R}$.
- **Failure mode.** The model is purely qualitative — no degrees of availability, cost, or likelihood — though it isolates the L1 (logical/definitional) core cleanly.
- **Result.** “If $s \rightarrow s'$ is realized, then $(s, s') \in T$ ” is definitional. It is not evidence for why T has the members it has, and it is silent on which member of $P(s)$ gets realized.

Model B — Probabilistic Transition Kernel: $K(s' | s)$

Definition. A stochastic kernel $K: S \rightarrow \Delta(S)$, where $\Delta(S)$ is the set of probability distributions on S , with $K(s' | s) > 0$ iff s' is possible from s .

- *Passibility* = $\text{supp}(K(\cdot | s))$.
- *Causation* = the realized draw $s' \sim K(\cdot | s)$.
- *Navigation* is trivialized in the fully-specified case, since K already gives realization probabilities.
- *Perception* = an estimate $\hat{K}_a \neq K$. *Distortion* = a divergence measure (D_{KL} or total variation).
- *Lightness* can be defined as $L(s \rightarrow s') := -\log K(s' | s)$, i.e. surprisal — making “lightest path” and “most probable path” the same by construction.
- **Failure mode.** If *Lightness* = $-\log K$, T2 becomes **true by definition** for any stochastic system — nothing beyond “systems do what they’re likely to do.” This is the sharpest illustration of vacuity risk in the whole programme: unless *Lightness* is defined independently of realized-transition statistics — via effort, energy, risk, or irreversibility, estimated prospectively — T2 has zero empirical content here.
- **Result.** A realized transition lying in $\text{supp}(K)$ is definitional. “*Lightness* = $-\log K$ ” manufactures agreement between T2 and the model, flagged here, not endorsed. Whether K itself takes a *Perception/Distortion-ratio* form for real systems is **empirical** — exactly what the AFI corpus’s cross-domain fits (Ohm’s, Fourier’s, Fick’s, and Darcy’s-law claims) purport to test.

Model C — Structural Causal Model / Intervention Structure

Definition. A tuple $\langle U, V, F, P(U) \rangle$, with exogenous variables U , endogenous variables V , structural equations $F = \{f_i\}$ where $v_i = f_i(\text{pa}(v_i), u_i)$, and a distribution over U . An intervention $\text{do}(X=x)$ replaces f_x with the constant x , yielding submodel $M_{\text{do } x}$.

- *Passibility* = the space of submodels $\{M_x\}$ — possibility of models, not just states, which is richer than in Models A or B.
- *Causation* = Pearl's Rung-2/3: a causes b iff intervening on a changes b under the mechanisms, with other parents fixed.
- *Navigation* = which intervention is performed. This is external to the SCM itself (Woodward and Papineau 2025) — the SCM specifies the space of interventions, not a selection principle.
- *Perception* = an estimated model that can be wrong about the graph itself, which is more severe than Model B's kernel error. *Distortion* = graph-edit distance plus parameter error.
- *Lightness* is not native to SCMs; it needs an external cost for performing an intervention.
- **Failure mode.** SCMs are silent on where F itself comes from — Pearl treats it as background knowledge — exactly the residual that Papineau's 2025 “recursive structures of laws with exogenous independence” targets.
- **Result.** “An actual causal effect corresponds to a value change under some $do(x)$ consistent with F ” is definitional relative to a fixed F . “Where F itself comes from” is an open metaphysical question this model cannot resolve internally. This is the clearest demonstration that a possibility-structure account does different work than an intervention-calculus account, even though both are modal.

Model D — Continuous Dynamical System: $dx/dt = f(x)$

Definition. State $x \in \mathbb{R}^n$, or a manifold, evolves via a vector field f . Solutions are integral curves $x(t)$.

- *Passibility at x* = the set of nearby integral curves consistent with f . For a deterministic autonomous system, this collapses to a singleton absent noise or control terms — the sharpest illustration of Universe 2 in the counterfactual lab.
- *Causation* = the vector field itself, $f(x)$, producing dx/dt .
- *Navigation* is vacuous in the pure-deterministic-autonomous case; it becomes non-trivial only once a control term is added, $dx/dt = f(x, u)$.
- *Perception* is relevant only for an observer modeling the system, not for the system's own evolution. *Distortion* here is purely observer/estimation error.
- *Lightness* can be recovered if f arises from a variational principle — $f = -\nabla V$, or Euler–Lagrange. This is the least-action connection, and precisely where the “naive trap” of mistaking mathematical equivalence for evidence of literal selection must be avoided.
- **Result.** For deterministic autonomous dynamics, “the system follows a unique determined path” is definitional, unrelated to lightness, selection, or perception — T1/T2 talk applies only as redescription. Only with stochasticity, control-authority, or variational structure does the model connect to T2, and even then it risks merely relabeling $-\nabla V$ rather than adding new content.

Model E — Reachability / Viability Formulation

Definition. Controlled dynamics $\dot{x} = f(x, u)$, with $u(\cdot) \in \mathcal{U}$ the admissible controls and $K \subseteq S$ a constraint set. $\text{Reach}(x_0, t) = \{x(t) : \exists \text{ admissible } u(\cdot), x(0)=x_0\}$. $\text{Viab}(K) = \{x_0 \in K : \exists \text{ admissible } u(\cdot) \text{ with } x(t) \in K \forall t \geq 0\}$, Aubin's viability kernel.

- This is *Passibility's most literal, pre-existing, rigorously worked-out formal counterpart*. $\text{Reach}(x_0, t)$ and $\text{Viab}(K)$ are exactly “structured availability of possible continuation,” axiomatized since 1990–91 by Aubin, with no AFI machinery needed.
- *Causation* = the applied control $u^*(\cdot)$.

- *Navigation is the field's central object: which $u(\cdot) \in \mathcal{U}$ to select, via a value function solving a Hamilton–Jacobi–Bellman/Isaacs PDE.*
- *Perception is not native here — the model assumes x is known exactly. The extension to perception-under-uncertainty, supplied by Xuan & Wang (2025) and Lin, Pinto & Bansal (2026), treats Distortion as state-estimation error.*
- *Lightness = the running or terminal cost selecting among $\text{Reach}(x_0, t)$ — minimum time, minimum energy, or Aubin's own viability-preserving rule, which need not optimize a single scalar and only needs to guarantee persistence in K .*
- **Failure mode/status.** *This is AFI's highest-stakes redundancy risk: if Passibility simply is a viability kernel, and Freedom simply is a reachable-set measure, AFI adds nothing formally beyond (i) a perception/distortion layer already developed above, and (ii) one cross-domain ratio law, $F = P/D$, in place of problem-specific cost functionals.*
- **Result.** *“A realized trajectory lies within $\text{Reach}(x_0, t)$ ” is definitional. “Viability-preserving behavior generalizes across physical, biological, and artificial systems” is empirical — this is what AFI's own corpus, including its swarm, building, and navigability preprints, attempts to test, with caveats noted in Appendix C.*

Model F — POMDP / Belief-State Formulation

Definition. $\langle S, A, T(s'|s, a), \Omega, O(o|s', a), R, \gamma \rangle$ — states, actions, transition kernel, observations, observation likelihood, reward, and discount. The belief state is $b(s) = P(s|\text{history})$, updated via $b'(s') \propto O(o|s', a) \sum_s T(s'|s, a) b(s)$.

- **Perception has a fully worked-out, standard, non-AFI-specific formalization here.** *The belief state b is a rigorous operationalization of “accessible representation of relevant reality,” updated from noisy observations — exactly AFI's P0–P4 ladder, made precise.*
- *Passibility = the reachable-belief-set analogue (Schäfers et al. 2026).*
- *Causation = the realized (s, a, s', o) sequence, given the true — and possibly agent-unknown — T and O .*
- *Navigation = policy $\pi(b)$ selection via a value function over the belief MDP (Xuan & Wang 2025).*
- *Distortion = the cleanest formal candidate: entropy of b , or $KL(b \parallel \text{true-state-indicator})$ — POMDP-native, not an AFI invention.*
- *Lightness = expected discounted cost under π over belief-trajectories, connecting to Model G.*
- **Failure mode.** *Full POMDPs with reachability-value queries are known **undecidable in general** (Madani, Hanks & Condon 2003). Fijalkow et al. (2026) identify the largest known decidable subclass, posterior-deterministic POMDPs — a hard formal ceiling on any claim that Navigability is generally computable.*
- **Result.** *“Optimal action depends only on the belief state, not the full history” is a definitional, formal theorem — belief-state sufficiency. “Exact reachability-value computation is undecidable in general POMDPs” is a formal, proven, negative result. It bounds any Navigability-under-partial-observability programme, AFI included.*

Model G — Path-Space / Trajectory-Functional Formulation

Definition. Configuration space Q ; path space $\mathcal{P} = \{q: [t_0, t_1] \rightarrow Q, q(t_0)=q_0, q(t_1)=q_1\}$; functional $S[q] = \int L(q, \dot{q}, t) dt$ over $\mathcal{P}$. The realized path is the stationary point, $\delta S = 0$ (Euler–Lagrange). In the Feynman formulation, all of $\mathcal{P}$ contributes amplitude $e^{iS[q]/\hbar}$, with classical behavior emerging via interference.

- *Passibility* = the entire path space $\mathcal{P}$ satisfying the boundary conditions — the largest, most literal “field of possible continuations” object among the eight, ranging over whole histories rather than single next-states.
- *Causation*, classically, is which single $q \in \mathcal{P}$ is stationary for S . In the path-integral reading, causation is the entire weighted ensemble jointly producing the observed classical trajectory as an emergent outcome.
- *Navigation* is only substantively present in the path-integral reading. Classical stationary-action redescribes local differential dynamics (Model D); it does not depict a system “searching” $\mathcal{P}$. The path integral genuinely treats non-realized paths as contributing non-eliminable structure.
- *Lightness* = literally $S[q]$, or $-S[q]$ — where “lightest path” and “least action” most directly meet, and where the warning is sharpest. Terekhovitch (pre-2025), Kodaj, Bernáth & Pickup (2025), and Dong Yang (2025) converge on the same point: stationarity alone is not evidence of literal selection-among-possibilities by nature.
- **Failure mode.** This is the single greatest risk in the formal programme: reading the classical limit as confirmation of T2 commits exactly the fallacy this literature warns against — mathematical equivalence to a local differential equation is not independent evidence for a global, perception-mediated selection process. Only the Feynman path-integral offers a genuine, limited precedent for non-realized alternatives having real physical standing.
- **Result.** “The classical trajectory extremizes S ” is a definitional equivalence to Euler–Lagrange, proven rather than interpreted. “Nature ‘selects’ this path via Perception/Distortion” is, classically, an unsupported **metaphysical** over-reading this paper declines to make. In the path-integral regime, it remains a live but speculative interpretation, not established physics.

Model H — Recursive Future-Passibility Formulation

Definition. Augment Model F or E with a *self-referential* update: $P_{\{t+1\}}(s) := g(P_t(s), a_t, s_t \rightarrow s_{\{t+1\}})$ — not a fixed background structure. Lightness becomes horizon-sensitive: $L(\pi \mid s, a, H) := \sum_{k=0}^H \gamma^k \ell(s_k, a_k) - \beta \cdot \text{Passibility_measure}(s_H)$.

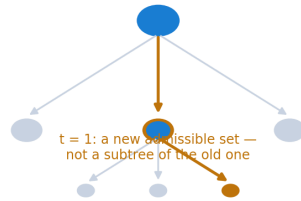
- This is the only model formalizing AFI’s Thesis 4 — recursion, “action redraws the next tree” — as a structural feature rather than an interpretive gloss. $\text{Freedom}_t \rightarrow \text{Navigation}_t \rightarrow \text{Realization}_t \rightarrow \text{Relations}_{\{t+1\}} \rightarrow \text{Freedom}_{\{t+1\}}$ becomes literally $P_{\{t+1\}} = g(P_t, a_t, \text{transition})$.
- *Causation* = the realized $(s_t, a_t, s_{\{t+1\}})$ tuple, as in Model F, but now causally efficacious on the possibility structure itself for the next step.
- *Navigation* = policy selection over an endogenously changing landscape, since the value of preserving future Passibility, β , must itself be estimated.
- *Perception/Distortion* work as in Model F, but the target, $P_{\{t+1\}}$, is partly a function of the agent’s own past actions — self-caused distortion becomes representable for the first time here.
- **Failure mode.** This is the least tractable model computationally: g is arbitrary in the general case, and estimating $\text{Passibility_measure}(s_H)$ at a distant horizon is as hard as, or harder than, the viability-kernel computation already flagged intractable in Model E, compounded recursively. Without simplifying assumptions — bounded H , linear g , or empowerment-style proxies (Kiefer 2025; Schneider et al. 2025) — this model reads as specifying the right question rather than an implementable algorithm.
- **Result.** “ $P_{\{t+1\}}$ depends on a_t ” is definitional, given how g is specified. “Systems that weight future Passibility ($\beta > 0$) outperform purely locally-lightest ($\beta = 0$) systems” is a genuinely empirical, testable claim — the most direct formal cash-value of the “heavier-now/freer-later” question.

Figure 28

Recursive possibility: action redraws the next tree

Recursive Possibility: Action Redraws the Next Tree

$t = 0$: three admissible continuations; the middle one is selected



Each realized transition can open admissible continuations that were not reachable before (Section 11, Model H). This is why the Law of Freedom is evaluated recursively, not as a one-shot optimization over a tree drawn once and left fixed.

Figure 28

Note. Realized transitions can open admissible continuations that were not reachable before, so the Law of Freedom must be evaluated recursively rather than as a one-shot optimization over a fixed tree. The two-level tree is a minimal illustrative case; it does not quantify how often, or under what conditions, realized transitions expand versus contract the subsequent admissible set.

Smallest Common Formal Core

Stripping all eight models to what they share:

If an actual transition from s to s' occurs, that transition lies within whatever set the operative model defines as the admissible/reachable/possible continuations of s .

- This is **purely L1 (logical/definitional)** — true by construction in every model above, and of any model with a stated admissibility structure, whether AFI-specific or not.
- Everything else the paper wants to say is additional, model-specific, and genuinely metaphysical (L3) or empirical (L4): that this structure is causally productive; that perception and distortion govern selection; that a single ratio law, $F = P/D$, holds across domains; that future Possibility should be weighted alongside immediate cost.
- Model B (kernel) and Model G (classical stationary action) are the two highest-risk sites for smuggling an L3/L4 claim in under cover of an L1 relabeling — the recurring failure mode this section guards against.

The Complete Form: the $\alpha\infty$ Reflexive Limit and Its Lambert W Closed Form

Section 11's working form throughout this paper is the finite- α Law of Freedom, $F(x) = (P(x)/D(x))^\alpha$, with α left open as either a fixed constant or a domain-relative parameter.

This subsection states the law's complete form: what happens at the limit where α itself grows without bound, and where that limit becomes reflexive — the ratio folded back into its own exponent.

Write $x := P(x)/D(x)$ for the Perception/Distortion ratio at a given state. Rather than treat α as an external, fixed dial, consider the *self-referential* limit α^∞ , in which the exponent is no longer external to F but is F itself, applied to itself indefinitely — the infinite tower $x^x x^{\dots}$.

This is a standard object in real analysis, the infinite power tower or infinite tetration, and its convergence is fully characterized. The sequence $F_1 = x$, $F_{n+1} = x^{F_n}$ converges, for real x , exactly when $x \in [e^{-e}, e^{1/e}] \approx [0.0660, 1.4447]$, to a finite fixed point $F^\infty(x)$ satisfying

$$F^\infty(x) = x^{F^\infty(x)}$$

Taking logarithms gives $\ln F^\infty = F^\infty \ln x$, the canonical transcendental form solved by the Lambert W function, the inverse of $f(w) = w \cdot e^w$. The closed-form solution is

$$F^\infty(x) = -W(-\ln x) / \ln x, \text{ valid for } x \in [e^{-e}, e^{1/e}].$$

Figure 29 lays out the three steps — finite α , the reflexive limit, and its closed form. Figure 30 shows the recursive tower $F_1, F_2, F_3, \dots$ level by level. Figure 31 plots $F^\infty(x)$ across its full domain of validity and marks the two boundary values beyond which the tower no longer converges to a finite real limit. Figure 33 confirms numerically that the finite tower reaches $F^\infty(x)$ as depth increases, for several representative values of x .

Figure 29

The Complete Law of Freedom

The Complete Law of Freedom

the finite- α form, and its self-referential limit as $\alpha \rightarrow \infty$

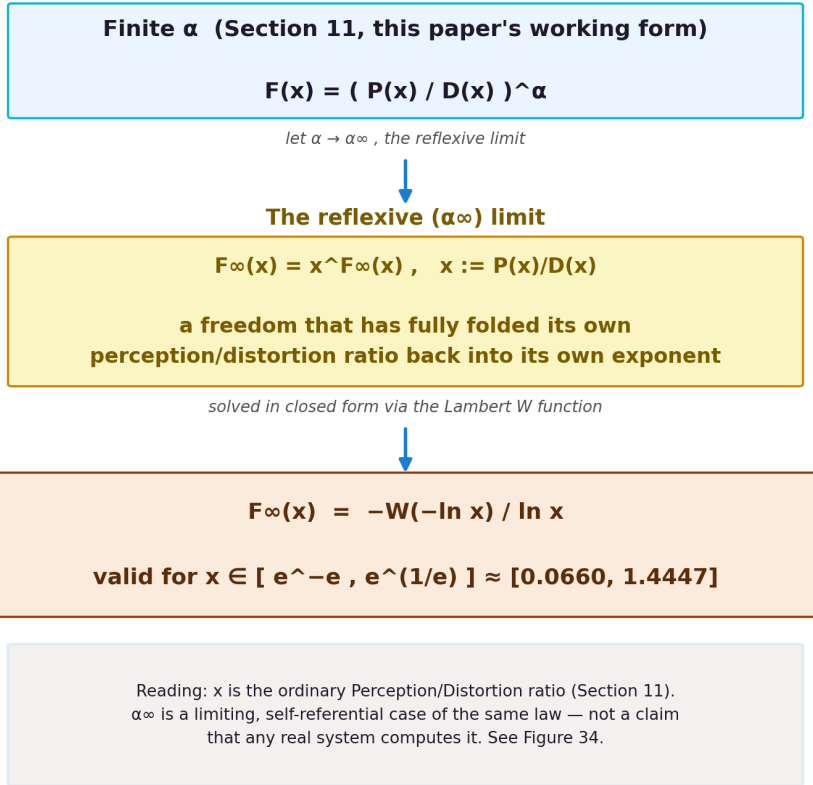


Figure 29

Note. The finite- α form (Section 11) is this paper's working operationalization; the α^∞ reflexive limit is a distinct, self-referential case of the same underlying law, solved in closed form via the Lambert W function. The derivation is exact given the fixed-point equation it starts from; whether α^∞ , rather than a small finite α , is the right regime for any real system's Freedom is a separate, open, empirical question (Figure 32).

Figure 30

What α^∞ Means: an Infinitely Recursive Exponent

What α^∞ Means: an Infinitely Recursive Exponent

each level re-applies the same ratio to itself, without limit

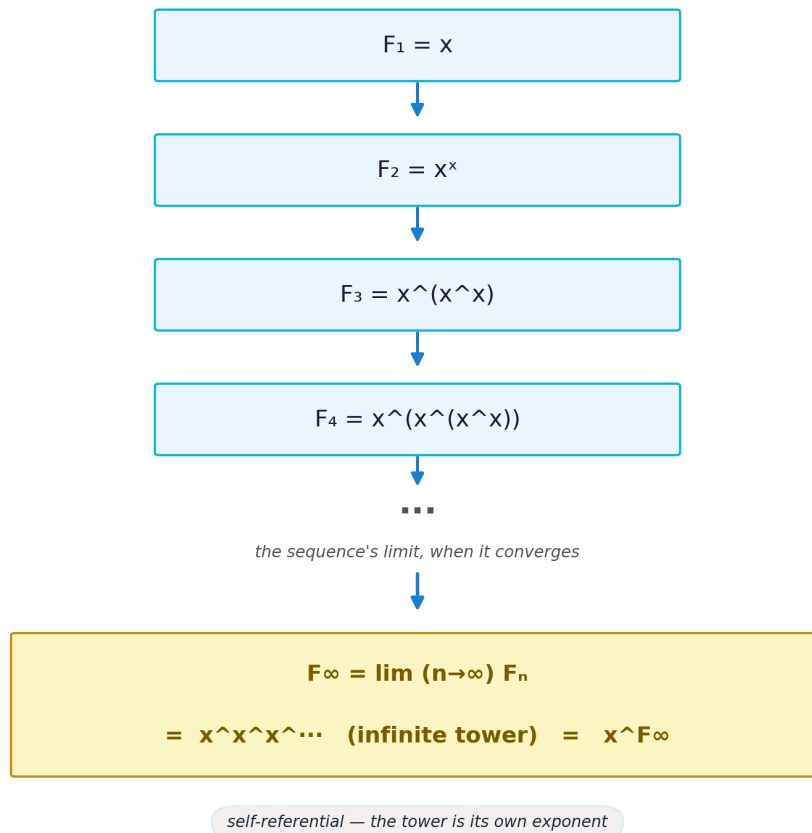


Figure 30

Note. Each level of the tower re-applies the ratio x to itself as its own exponent; α^∞ names the case where this recursion is carried out without limit. The figure shows the first four finite levels and the self-referential equation their limit satisfies.

Figure 31

Where the Reflexive Limit Is Defined

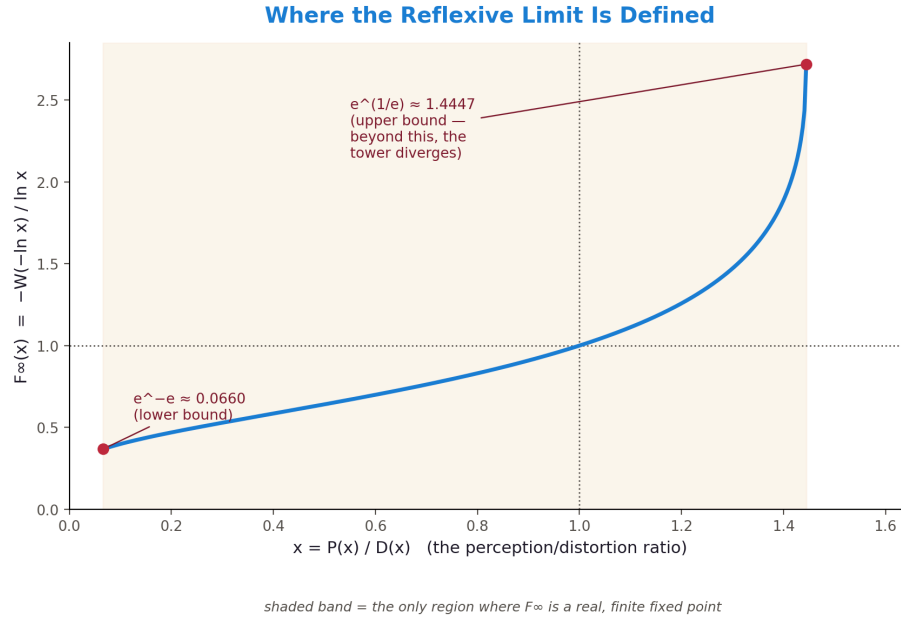


Figure 31

Note. $F^\infty(x)$ is a real, finite fixed point only for $x \in [e^{-e}, e^{1/e}] \approx [0.0660, 1.4447]$ — outside this band the infinite tower diverges or fails to settle, a boundary condition on the reflexive form that the finite- α form (Section 11) does not share. This is a property of the mathematical construction, established independently of any AFI-specific claim.

Figure 32

Reading F^∞ Against AFI's Own Claims

Reading F^∞ Against AFI's Own Claims

what the closed form licenses, and what it does not

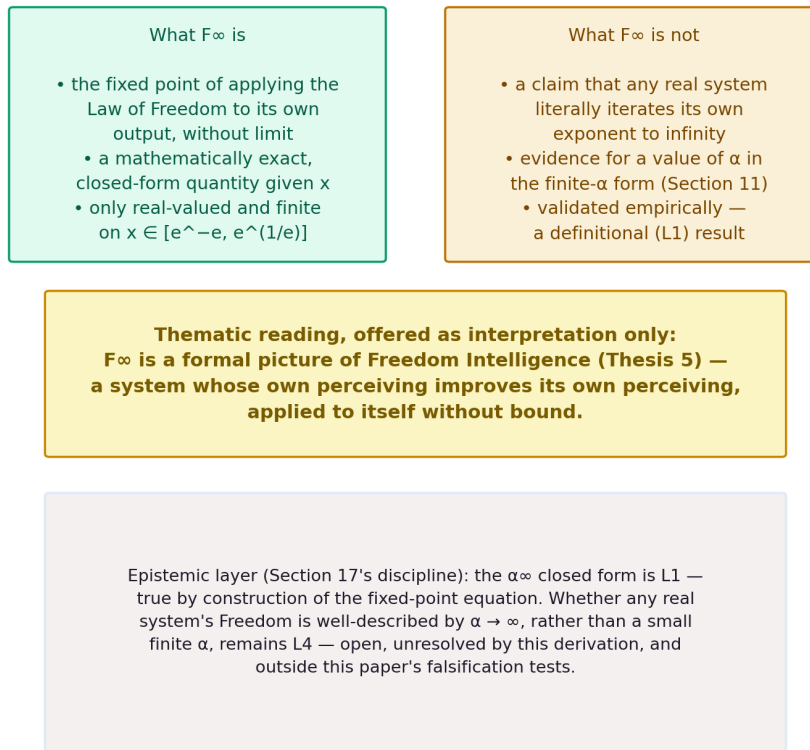


Figure 32

Note. The α^∞ closed form is an L1, definitional result — true by construction of the fixed-point equation it solves — and is offered thematically, not empirically, as a formal picture of Freedom Intelligence (Thesis 5): a system whose own perceiving feeds back into its own perceiving without bound. Whether any real system's Freedom is better described by α^∞ than by a small finite α remains open and is not addressed by this paper's falsification tests (Section 16).

Figure 33

The Recursive Tower Converging to F^∞ as Depth Grows

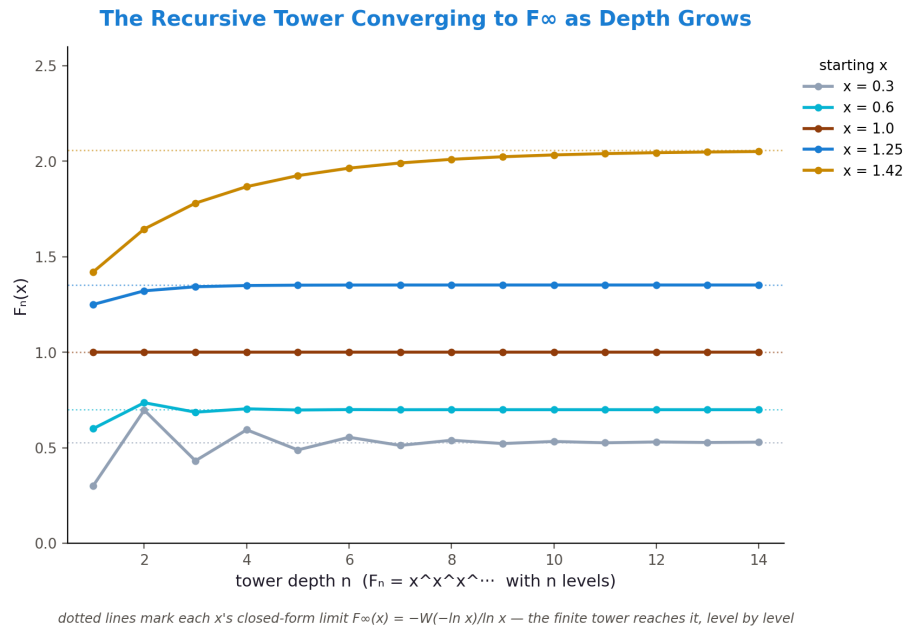


Figure 33

Note. For five representative starting values of x within the domain of convergence, the finite tower $F_n(x)$ is plotted against depth n alongside its closed-form limit $F_\infty(x)$ (dotted); each sequence visibly settles onto its own limit within roughly ten levels. This is a numerical illustration of a proven convergence result, not a claim about how many “levels” any real system computes.

Figure 34

How Freedom Responds to Perception, Distortion, and α

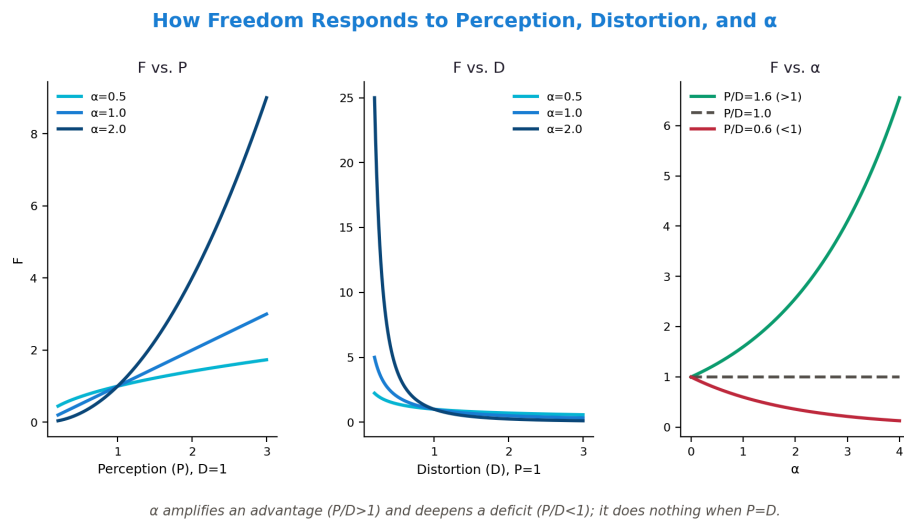


Figure 34

Note. Three real, exact panels of $F=(P/D)^\alpha$: raising Perception increases F, raising Distortion decreases it, and α amplifies whichever direction P/D already points — it does nothing when Perception equals Distortion exactly.

Figure 35

Building Perception and Distortion from Sub-Factors

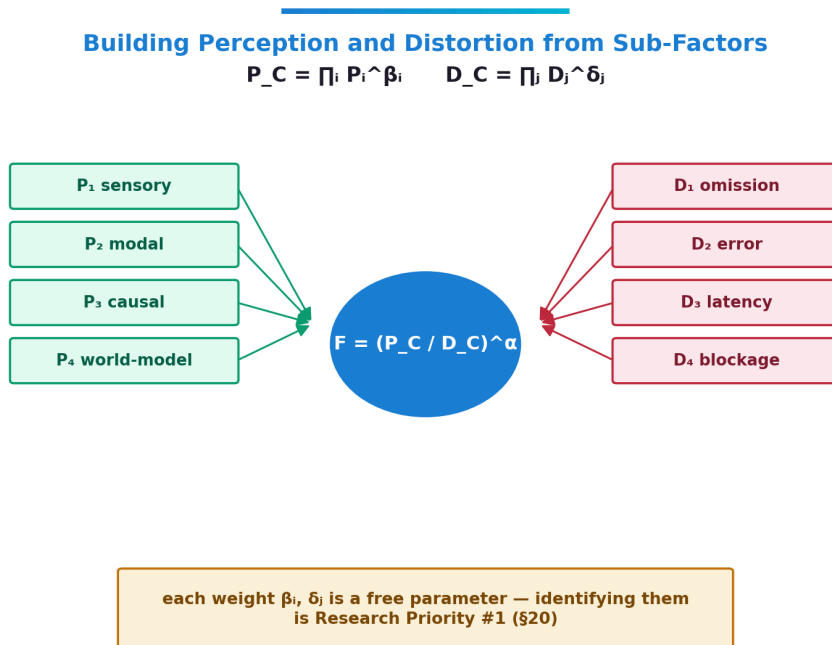


Figure 35

Note. Perception and Distortion are each proposed as products of weighted sub-factors, $P_C = \prod_i P_i^{\beta_i}$ and $D_C = \prod_j D_j^{\delta_j}$, before entering the ratio. Every β_i and δ_j is a free parameter — fixing them independently of outcomes is Research Priority #1 (Section 20).

Figure 36

The Law of Freedom, Worked

The Law of Freedom, Worked

two systems, same formula, different outcomes — illustrative numbers, not measured data

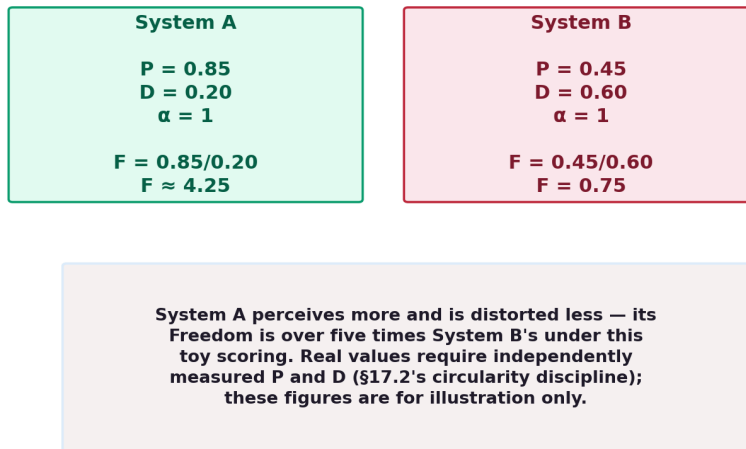


Figure 36

Note. Two illustrative systems, scored with the same formula and different inputs, to make the arithmetic concrete. The numbers are for exposition only — Section 17.2's circularity discipline requires that real P and D values be measured independently of the outcome being explained, never fit to it after the fact.

Figure 37

Why AFI Uses a Ratio, Not a Sum

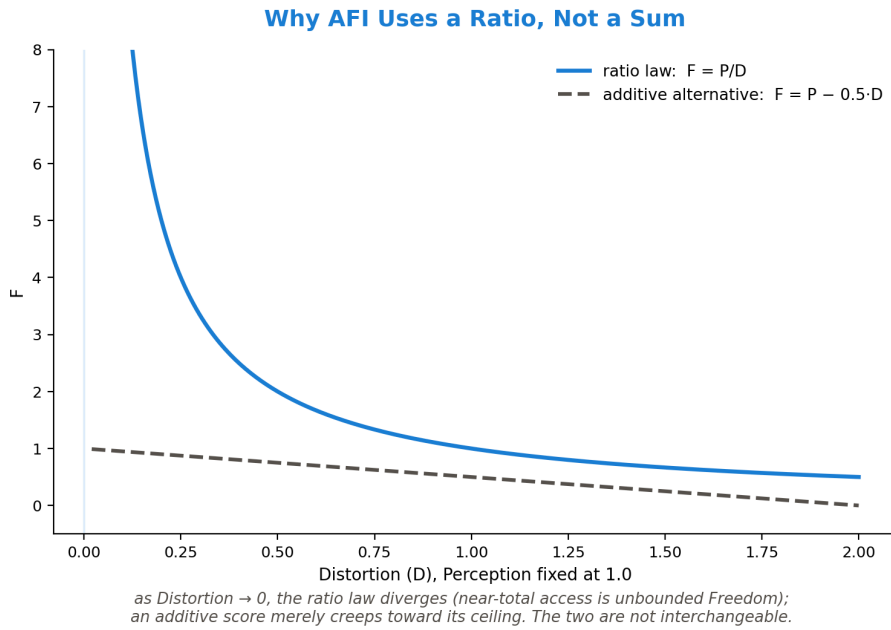


Figure 37

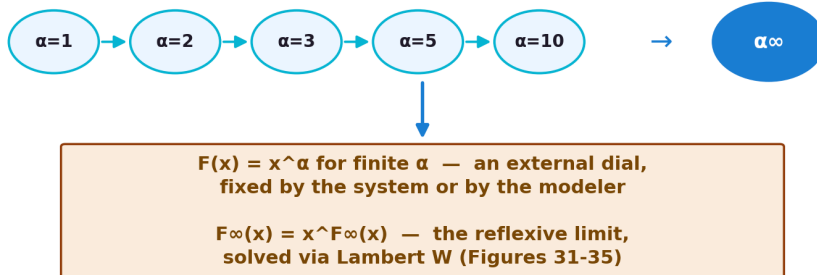
Note. A real comparison of $F=P/D$ against a hypothetical additive alternative as Distortion approaches zero: the ratio law diverges, since near-total access is treated as unbounded Freedom, while an additive score merely approaches a ceiling. The two formalizations are not interchangeable, which is why AFI resists additive, reward-sum-style formulations.

Figure 38

From Dial to Limit: α 's Full Range

From Dial to Limit: α 's Full Range

the same law, read at finite α and at its reflexive limit α_∞



α_∞ is a limiting, thematic case of the same law — not a claim that any real system computes it

Figure 38

Note. At finite α , the exponent is an external dial set by the system or the modeler. At its reflexive limit α_∞ , the exponent becomes the formula applied to itself without bound, solved in closed form via the Lambert W function (Figures 29-33). The two are the same law, read at opposite ends of its range.

12. Foundational Genealogy: Powers, Process, Physics, and Control

1. Aristotle — Dynamis/Energeia, the Four Causes, Kinesis

- *Actuality (energeia) precedes potentiality (dynamis) in substance, though dynamis precedes it in the changing thing; capacity becomes actual under formal/final constraints, teleology already doing the “selecting” — AFI’s substrate-neutral, quantified primitive risks renaming a priority already Aristotle’s.*

2. Leibniz — Possible Worlds, Compossibility, PSR

- *Monads carry complete concepts under the Principle of Sufficient Reason; infinitely many compossible worlds precede creation, and God selects the best via PSR. Within a monad, actualization is pre-programmed unfolding — history’s closest ancestor of a stated Law of Freedom. AFI relocates selection to an ongoing, per-moment operation ($F=P/D$): the architecture is already Leibniz’s.*

3. Hume — Regularity Theory, Skepticism about Necessary Connection

- *Only impressions and constant conjunction are given; “could have been otherwise” is a habit-artifact, not real structure. AFI answers via structured perception of continuations plus distortion as the “glue.”*
Falsifiability: if $F=P/D$ adds no predictive content beyond post hoc redescription, the challenge stands unanswered.

4. David Lewis — Counterfactual Dependence, Modal Realism, Similarity Metric

- *Lewis posits concretely existing possible worlds ranked by a similarity metric — the device selecting which counterfactual is true, and Lewis's repeated revisions show how contested that metric is. AFI substitutes a dynamic, P/D-indexed ranking over paths within one world — a renaming risk unless P/D does independent work.*

5. Judea Pearl — SCMs, Causal Hierarchy, Do-Calculus

- *A structural causal model comprises equations, exogenous variables, and a DAG; possibility enters at Rung 2 ($do(X=x)$) and Rung 3 (counterfactuals). Do-calculus is computation, not selection — Pearl is silent on where the graph and mechanisms come from; Papineau (2025, *Synthese*) argues it still needs a metaphysical causation story underneath. AFI adds what fixes the SCM plus a metric (P/D) — a renaming risk, since Pearl already treats causation as constitutively modal.*

6. James Woodward — Interventionism

- *Woodward's invariance-under-intervention criterion: X causes Y iff the relation stays stable under some possible intervention on X ; no ontology for the candidate field, no selection law. AFI grounds "possible intervention" in a real field of path-availability, adding $F=P/D$ for findability — the closest interventionist ally to Thesis 1.*

7. Wesley Salmon and Phil Dowe — Process/Conserved-Quantity Theories

- *This tradition grounds causation in physical processes (Salmon: mark-transmitting world-lines) or conserved quantities (Dowe: energy-momentum exchanged at intersections) — the most actualist theory surveyed, built against Lewis's possible-world talk, with no selection law by design and struggling with absence, prevention, and double prevention. AFI's clearest non-redundancy test: it supplies modal machinery this theory cannot handle.*

8. Powers/Dispositions — Harré/Madden, Mumford, Anjum, Bird, Cartwright, Molnar

- *This tradition treats causal powers and dispositions as real, sui generis properties (anti-Humean), tracing to Harré & Madden (1975). Mumford & Anjum formalize possibility as a "dispositional modality" via vector composition of reciprocal partners; Bird (Mind, 2010) argues the tradition strains under what is asked of it. AFI grounds possibility in a field-level primitive owned by no particular, with a quantifiable ratio replacing vector-addition.*

9. Process Philosophy — Whitehead

- *For Whitehead, possibility enters via "eternal objects" ingressing into concrescence, weighted by the "subjective aim" toward "intensity of satisfaction" — never quantified. AFI supplies a testable, non-theological selection principle where Whitehead needed a metaphysically heavy valuer — an open citation gap, since no 2023–2026 paper frames Whitehead this way.*

10. J.J. Gibson — Ecological Psychology, Affordances

- *For Gibson, possibility enters as affordances — action-possibilities directly specified in structured energy arrays, real whether or not perceived — becoming actual when the animal detects and acts on one via direct perception. AFI supplies the missing quantitative law: $F=P/D$ formalizes what makes one affordance "lighter"; "affordance" is the closest existing term for AFI's "possible path."*

11. Physics of Least Action — Fermat, Maupertuis, Hamilton, Lagrangian/Hamiltonian Mechanics, Feynman Path Integrals

- *The action functional S ranges over kinematically possible paths; Euler-Lagrange/Hamilton's equations equate to stationarity ($\delta S=0$), a redescription of local dynamics, not evidence nature literally "surveys and selects." Terekhovitch (arXiv:1511.03429): stationary-action is formally but not causally teleological. Kodaj, Bernáth & Pickup (Synthese 205(2), 2025) frame it as a teleology puzzle despite causal reducibility; Dong Yang (Foundations of Physics 55, art. 75, 2025) argues purpose and natural law are "equivalent in effect." Feynman's path-integral reformulation is the exception: non-classical paths are physically meaningful. AFI must reduce to Euler-Lagrange/Hamilton's equations classically; treating mere agreement as confirmation leaves it mere renaming.*

12. Control Theory — Reachability, Controllability, Viability Theory, HJ Reachability

- *Control theory works with a state space, controlled dynamics ($\dot{x}=f(x,u)$), and constraint sets — the central objects are sets of states. Possibility enters as reachable sets and Aubin's viability kernels: the largest set of states a system can be kept within forever via admissible control — the most literal existing formalization of "structured availability of possible continuation." Renaming risk is highest here: "viability kernel," formalized since 1991, already is essentially this field; AFI's addition is the perception/distortion layer control theory abstracts away.*

13. Karl Friston — Active Inference, Free Energy Principle

- *Active inference centers on variational free energy, minimized by self-organizing, Markov-blanketed systems. Expected free energy (EFE) evaluates counterfactual policies before one is enacted, decomposing into pragmatic and epistemic terms, selected via softmax over negative EFE — the closest existing quantitative rival to a stated Law of Freedom. Kiefer (2025, Entropy) formally unifies active inference and empowerment as instances of constrained-maximum-entropy inference. AFI's real work: showing what a simpler ratio (P/D) adds beyond relabeling $-EFE$.*

14. Empowerment — Klyubin, Polani, Nehaniv

- *Empowerment is channel capacity: mutual information, maximized over action-sequence distributions, between an agent's actions now and its later sensor states — a reward-free quantity measuring how much future states could vary, the closest analogue to "quantify possibility, then move toward more of it." Recent work shows the program still developing: Schneider et al.'s discounted-empowerment pretraining (arXiv 2510.05996), and Shah, Nemenman, Polani & Tiomkin's multi-agent empowerment (arXiv 2604.21155). AFI adds a numerator/denominator structure where empowerment has only a single term and no built-in perception-relativity.*

Synthesis

- *Leibniz gives possibility ontological priority via compossibility and PSR — world-level, once-only, theological. Lewis gives a ranked space of non-actual alternatives — static, agent-independent, no perception term. Whitehead gives the richest anticipation, with potentiality selected every occasion, but the selector is a metaphysically heavy God, never quantified.*
- *Two formalisms come genuinely close: Aubin's **viability theory** (reachable sets/kernels as structured fields of possible continuation) and Friston's **active inference** (minimize expected free energy over counterfactual policies); Klyubin/Polani/Nehaniv's **empowerment** is the closest precedent for possibility as an intrinsically valuable, measurable primitive prior to any goal.*

- *No existing position combines all three things AFI needs: possibility as the sole primitive, explicit perception-relativity, and one quantified, falsifiable, cross-domain law. Viability theory lacks the latter two; Friston bundles them to a Bayesian architecture; empowerment drops perception/distortion.*
 - **Falsifiability:** *AFI must show $F=P/D$ reduces to Aubin's reachable sets and Friston's expected free energy rather than merely resembling them, survive Hume's challenge with independent falsifiable content, and show it is not simply empowerment or EFE relabeled.*
-

13. The Cause-to-Law Bridge: Does the Law of Freedom Follow?

Key points:

- *Freedom, as structured possibility, is a necessary condition for causal production in every non-degenerate (branching) domain.*
- *Once more than one continuation is admissible, some principle selects which becomes actual — the Navigability problem. Opening and solving this problem are different accomplishments; this is the load-bearing link between the Cause of All (Thesis 1) and the Law of All (Thesis 2).*
- *Section 13.2 compares the Law of Freedom against twelve named rivals on this point.*

13.0 Framing: The Lightness Bridge, Formally, and Its Open Alternative

Section 2 states the question in plain terms: Freedom opens a field, and a further, genuinely open question asks what governs which continuation becomes actual. This subsection restates both live readings in the paper's own formal vocabulary, ahead of the arrow-by-arrow derivation in 13.1, which treats the connection as a claim to test rather than a premise to assume.

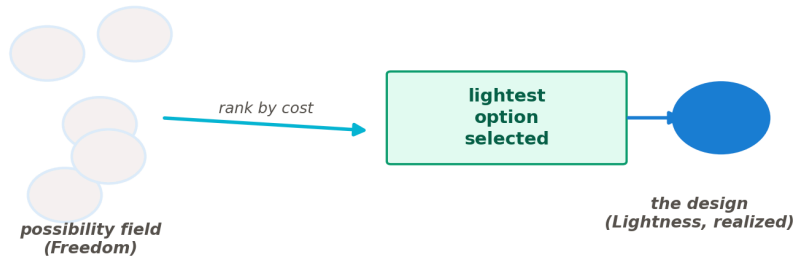
- *Thesis 1 establishes that Possibility, structured possibility, already contains more than one live continuation whenever a domain branches genuinely.*
- *Genuine branching turns Navigability into a standing question for every system with Freedom: some regularity governs which continuation becomes actual, whatever that regularity turns out to be.*
- *One reading — AFI's own — nominates Lightness for that role: $F=(\text{Perception}/\text{Distortion})^\alpha$, the quantity Thesis 2 proposes every system minimizes according to its own perception (Section 14 sets out exactly how "minimizes" should be read).*
- *The equal, opposing reading treats the regularity as belonging to whatever separate vocabulary a given domain already uses — mechanics, thermodynamics, evolutionary dynamics — each fully capable of governing its continuations while standing apart from Lightness entirely. Section 13.2 tests the Law of Freedom against twelve such named rivals, precisely because either reading could turn out correct.*

Figure 39

The Designer's View

The Designer's View

how design practice already makes this exact move



a designer starts from a space of possible configurations and searches for the version that costs least to build, learn, or use — the same pairing Thesis 1 and Thesis 2 make

Figure 39

Note. Design practice already routes a possibility field through a cost principle every time a design is chosen, offered here as a motivating analogy for AFI's own reading (L3, thematic) — one vantage point among the four in Figure 11, standing beside the equally serious reading that treats the analogy as coincidence rather than evidence.

Figure 40

Freedom Alone, Against Freedom Plus Lightness

Freedom Alone, Against Freedom Plus Lightness

the same field of possibilities, read two ways

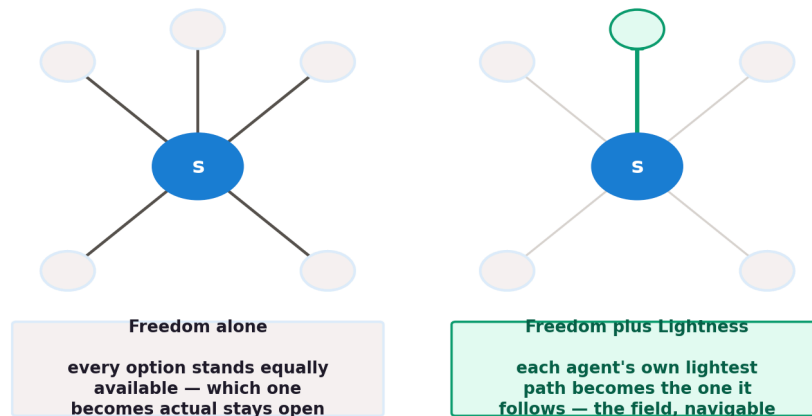


Figure 40

Note. AFI's own picture of the two readings side by side: Freedom alone leaves every option equally available, with the question of which becomes actual entirely open; adding Lightness, on AFI's candidate reading, turns each agent's own lightest path into the one it follows. Whether real systems match the right-hand panel or the left remains the question Section 17 is built to test.

Figure 41

Two Ways to Read the Chain

Two Ways to Read the Chain

Figure 2's two weakest arrows, held open rather than resolved

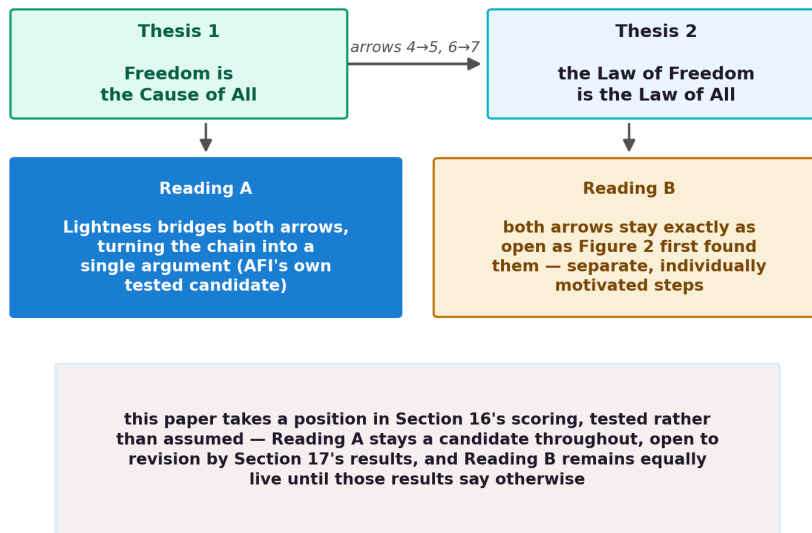


Figure 41

Note. Figure 16's seven-step chain flagged arrows 4→5 and 6→7 as its weakest links. One reading grants Lightness as Freedom's own selection principle at both points, turning the chain into a single argument. The equal, opposite reading leaves both arrows exactly as open as Figure 16 first found them, treating the chain as a series of separate, individually motivated steps. This paper takes a position in Section 16's scoring; the position stays a tested candidate throughout, open to revision by Section 17's results.

Figure 42

Five Epistemic Layers

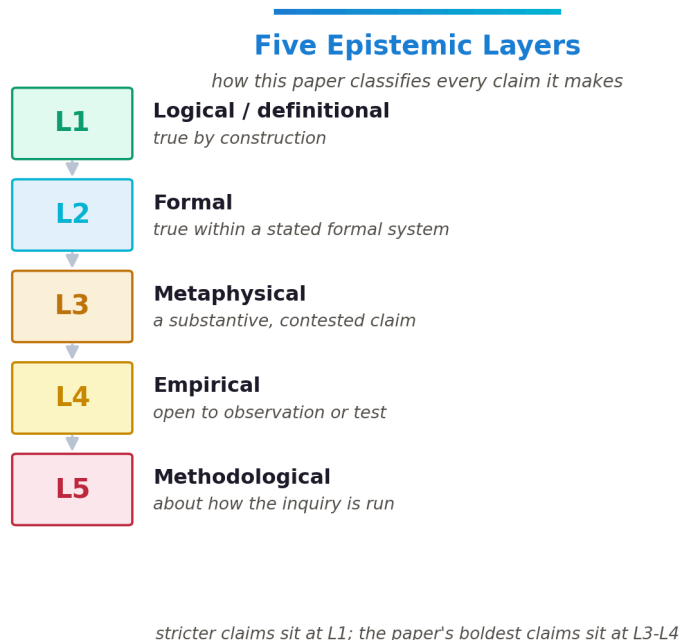


Figure 42

Note. Every claim this paper makes is tagged by which of five layers it belongs to, from logical/definitional (L1) through methodological (L5). Keeping these layers separate is what lets Section 16 score the theory honestly instead of letting a strong L1 result borrow credibility for an unproven L3 claim.

13.1 The candidate derivation, arrow by arrow

- The seven-step chain under test, each arrow classified by epistemic layer (L1 logical/definitional, L2 formal, L3 metaphysical, L4 empirical, L5 methodological) and stress-tested against the Counterfactual Universe Laboratory and Section 12's genealogy: (1) something becomes actual; (2) actualization requires an admissible transformation; (3) causal production operates through structured possibility; (4) differentiated continuations create a transition landscape; (5) a landscape creates a navigation problem; (6) a primitive's theory requires an account of navigation; (7) AFI proposes the Law of Freedom as that account.
- Arrow 1→2.** Weak (**L1**): "admissible" defined after the fact — vacuous (Model B, §11). Strong (**L3/L4, unproven**): independently specifiable before realization — the reading Hume's skepticism threatens. Universe 25 (teleportation): any "jump" is absorbable into a redefined admissible set after the fact, answered only via prospective specification (Section 14).
- Arrow 2→3.** Aggregating admissibility facts into T is mere set-formation (**L1**). The substantive claim in "structured" is that T is *compressible*, not a brute list (**L3/L4**) — a Kolmogorov-random universe has nothing for Perception to track. AFI implicitly assumes reality is compressible, a falsifiable premise left unstated.

- **Arrow 3→4.** Trivial given its antecedent (L1). That antecedent — that the structure actually branches ($|P(s)| > 1$) — is a further, substantive claim (L3/L4). Universe 2 (single deterministic successor everywhere) satisfies 1–3 but never 4: no landscape anywhere — a *scope restriction*, not a refutation: AFI’s apparatus is non-vacuous only wherever real branching exists.
- **Arrow 4→5.** The deepest hidden assumption: calling multiple live options a “problem” presupposes **single-history realization**. Two rival pictures deny this — Everett-style many-worlds (Universe 27), where every branch is realized, and primitive-chance/brute-fact (a Humean descendant), where no explanation is owed. **L3, genuinely contested:** a Humean or Everettian can reject the rest of AFI’s programme without inconsistency.
- **Arrow 5→6.** A **methodological (L5)** demand, conditional on accepting step 5 as a real gap — a theorist who rejects 4→5, or adopts instrumentalism, can decline it: AFI’s own explanatory standard, a choice, not a proof. **Arrow 6→7** is a **substantive, comparative, empirical (L4) proposal**: nothing in steps 1–6 privileges $F=P/D$ over random selection, Markov dynamics, stationary action, energy minimization, gradient flow, Bayesian updating, maximum entropy, active inference, empowerment, optimal control, goal satisfaction, historical dynamics, or stochastic softmin (§13.2); the slot must be earned empirically.

T1 does not logically entail T2. Steps 1–5 hold only under two non-forced commitments: reality is compressible (2→3), and realization is single-history and explanation-demanding (4→5). Granting both, T1 makes a navigation law a *necessary explanatory problem* — but which law solves it stays open, settled by empirical performance, not conceptual analysis.

Testing the candidate synthesis: “*The Cause of All creates the space of causation. The Law of All governs the navigation of that space.*” Survives in qualified form: “creates” means *constitutes and is explanatorily prior to*, read as a structural relation rather than a temporal event; “the space of causation” holds conditional on compressibility and branching; “the Law of All governs” reads as AFI’s tested candidate for that role, standing apart from an already-established answer. “Law of All” remains a title Section 16 specifies the conditions for earning, criterion by criterion.

13.2 The Law of Freedom against named rivals

- For the Law of Freedom to deserve the generality T2 claims, it must, for each rival, either (a) subsume it as a special case, (b) out-predict it empirically where they diverge, or (c) apply in a domain the rival cannot reach. Where none holds, the honest status is “untested” or “redundant,” not “confirmed.”

Rival law	Relation to $F=P/D$
Random selection (uniform draw over $P(s)$)	No cost structure or perception; reduces to this only when all options have equal perceived cost — the weakest rival.
Markov dynamics ($K(s s)$, Model B)	Silent on why K has its shape. Lightness as $-\log K$ makes T2 true of any Markov system by construction — a vacuity trap, not a genuine rival.
Stationary action / least action (Model G)	Equivalent to local differential dynamics; silent on whether unrealized paths have ontological standing (Kodaj/Bernáth/Pickup 2025; Dong Yang 2025). AFI must reduce to Euler–Lagrange/Hamilton’s equations classically, not merely rename it “lightest path.”

<i>Rival law</i>	<i>Relation to $F=P/D$</i>
Energy minimization / gradient flow ($dx/dt = -\nabla V$)	Requires a defined potential V ; a special case only if $D \propto \nabla V$ — an empirical claim, not yet demonstrated.
Bayesian inference / updating	Governs belief revision, not action selection — adoptable for the numerator (Model F), not a rival law for the whole ratio.
Maximum entropy	A principle over distributions, not paths (Jaynes); underlies both EFE and empowerment per Kiefer (2025) — beating it nears beating both at once.
Active inference / minimum expected free energy	Fully specified (KL/entropy terms, defined priors); no operator for discovering the model's causal structure is wrong. The single strongest existing rival; §17.3's testbed (f) forces this comparison.
Empowerment maximization	Relative to the agent's own (possibly wrong) model; no distortion-correction; value-agnostic. Closest precedent for possibility as a measurable primitive; AFI's D-term and perception-relativity are the claimed, untested additions.
Optimal control (Models E/G)	Requires the cost functional from outside; AFI's contribution is a candidate, cross-domain answer to what that function is — the unproven core claim.
Goal satisfaction / satisficing (Simon)	No commitment to global/local optimality — a genuine rival: satisficing under matched P/D falsifies strict-argmin unless T2 is read as softmin/threshold.
Historical / path-dependent dynamics	Needs the state augmented with history; compatible with $F=P/D$ if computed over the augmented state.
Stochastic softmin ($\Pr(s') \propto \exp(-L(s \rightarrow s')/\tau)$)	Needs L specified from outside — AFI's own best-supported operational reading of T2, not really a rival.

Net assessment. The Law of Freedom does not currently subsume its two strongest rivals (active inference/EFE, empowerment) by proof; it stands in *structural resemblance under empirical test*, pending the Section 17 program. It easily out-predicts the weaker rivals, but that is a low bar. Against goal-satisfaction/satisficing and stochastic softmin, T2 should itself be read as a graded, softmin-style claim rather than strict argmin (§14.2).

14. Lightness: Local, Global, or Recursive?

14.1 Defining Lightness prospectively

- Standing rule: **Lightness must be computed from information available to the system before the outcome, never inferred from which outcome occurred.** A definition fit after the choice makes T2 unfalsifiable by construction — indistinguishable from Model B's $-\log K$ vacuity trap (§13.2).
- Prospective Lightness requires four steps: enumerating candidate continuations before the trial; scoring them on pre-outcome information; recording a predicted ordering; only then comparing to the realized outcome. Every testbed in Section 17 is built around this constraint — the hinge on which T2's empirical status turns.

14.2 Candidate operationalizations of “lightest,” compared

<i>Candidate</i>	<i>Form</i>	<i>Relation to $F=P/D$</i>
<i>Deterministic argmin</i>	$s^* = \operatorname{argmin} L(s \rightarrow s')$	<i>The naive reading of T2; falsified by observed variability among trials with matched P/D (§14.4).</i>
<i>Probabilistic softmin</i>	$\Pr(s') \propto \exp(-L/\tau)$	<i>AFI’s best-supported operational form; τ needs independent motivation, not free fitting.</i>
<i>Expected-cost minimum</i>	$s^* = \operatorname{argmin} E[L(s \rightarrow s')] \text{ under uncertainty}$	<i>Folds into the softmin/EFE family whenever Distortion includes genuine stochasticity.</i>
<i>Stationary-action conditions</i>	$\delta S = 0$ (deterministic classically; weighted-ensemble in path-integral reading)	<i>Must be reduced to, not merely resembled, in the classical limit — the highest-risk site for the naive-selection fallacy.</i>
<i>Maximum entropy</i>	<i>Least-committal distribution given constraints</i>	<i>Subsumes both EFE and empowerment (Kiefer 2025); beating it generically nears beating AFI’s two strongest rivals at once.</i>
<i>Maximum empowerment</i>	$\text{Maximize } I(A; S')$	<i>Closest rival for the option-value half of Lightness; lacks distortion-correction and perception-relativity.</i>
<i>Minimum expected free energy</i>	<i>Minimize pragmatic + epistemic EFE terms</i>	<i>Closest existing rival overall; structural resemblance without proven identity.</i>
<i>Viability preservation</i>	<i>Any action keeping the system in $\text{Viab}(K)$</i>	<i>Constraint-satisfaction, not optimization; need not be lightness-minimizing — the two can conflict (§14.3).</i>
<i>Goal-directed planning</i>	<i>Minimize distance-to-goal under a fixed target</i>	<i>A special case of expected-cost minimization; T2 must also explain non-goal-directed systems (plants, markets).</i>

Figure 43

Six Ways to Operationalize “lightest”

Six Ways to Operationalize “lightest”

Section 14.2 — candidates for the Law of Freedom's selection rule

Deterministic argmin	<i>naive T2 reading; falsified by observed variability</i>
Probabilistic softmin	<i>AFI's best-supported operational form</i>
Expected-cost minimum	<i>folds into softmin under genuine stochasticity</i>
Stationary-action conditions	<i>must reduce to, not merely resemble, physics</i>
Maximum entropy	<i>subsumes both EFE and empowerment</i>
Viability preservation	<i>constraint-satisfaction, not optimization</i>

Figure 43

Note. Six of the nine candidates in the table above, ranked from the naive deterministic reading (falsified by observed variability) to AFI's best-supported probabilistic softmin form. No candidate is adopted by default — each carries its own falsification condition, detailed in the full table.

14.3 Local-lightest vs. future-preserving: resolving the tension

- *Universe 18 (“local-lightest/global-disastrous”) and Universe 19 (“heavier-now/freer-later”) flag T2's sharpest internal threat: a path cheapest by every immediate measure can close off nearly all future continuations, while a costlier path preserves a large future option space. A purely **myopic** reading of T2 predicts the immediate-cheap, future-destroying path — wrong for any system pretheoretically called “free.”*
- **Resolution:** *Lightness must be horizon-relative. Model H : $L(\pi \mid s, a, H) := \sum_{k=0}^H \gamma^k \ell(s_k, a_k) - \beta \cdot \text{Passibility_measure}(s_H)$, with H an explicit horizon, γ a discount, and $\beta \geq 0$ a weight on preserved future Passibility. A future-destroying path can lose to a costlier, future-preserving one once the horizon is long enough to see the destruction coming, and $\beta > 0$. H and β must be independently motivated (a bacterium's tiny effective horizon; a planning AI's rollout depth; a human's working-memory bounds, testbed (f)) — never tuned post hoc to rescue a prediction.*
- **Answer:** *a locally-light path eliminating most future continuations is **not** the “freer” path in AFI's technical sense, and naive statements of T2 are **false as stated** — Freedom is a property of the trajectory's relation to the future field, not a single transition. A short-horizon system will still reliably take it: not a counterexample but a further testable claim, since horizon length is part of “according to its perception.”*

Figure 44

Local-lightest vs. future-preserving path (Universes 18–19)

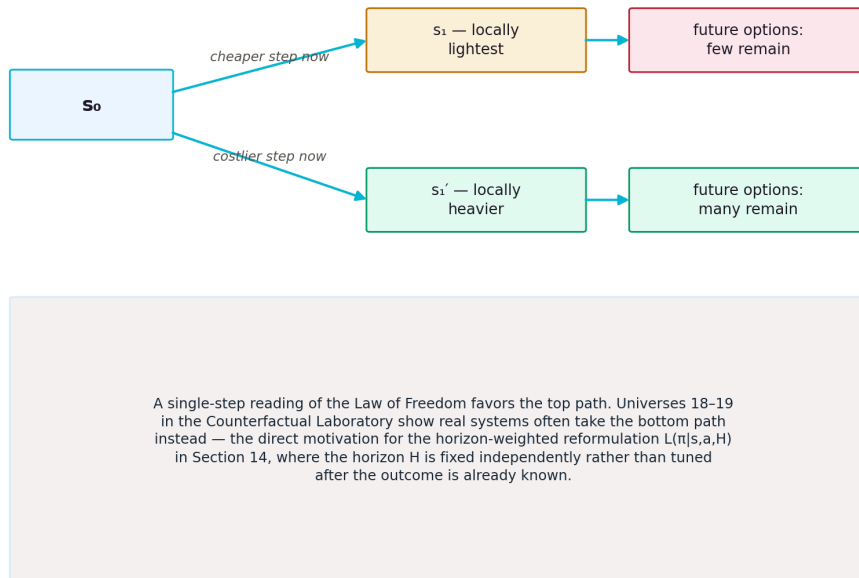


Figure 44

Note. An unqualified, single-step reading of the Law of Freedom predicts the wrong path whenever local and future-preserving optimization diverge, motivating the horizon-weighted reformulation in Section 14. The figure depicts one illustrative divergence, not the general conditions under which local and future-preserving optimization align or conflict.

14.4 What would falsify the operationalized claim

- *T2, in horizon-qualified softmin form, is falsified by: systematic, reproducible preference for higher-L over lower-L options among matched candidates with no horizon/ β story available; softmin dominated, in nested model comparison, by a rival with strictly fewer free parameters (e.g., empowerment or EFE alone) across multiple testbeds; or systems tracking none of the candidate Lightness dimensions (effort, time, risk, irreversibility, information cost, future Passibility) at any horizon while still behaving in a structured, non-random way.*
- **Falsifiability:** none of these has been run; Section 17 specifies what running them would require.

15. Causal World Models, Active Inference, and Future Path Space

15.0 Framing

- *Thesis 5 claims intelligence just is the capacity to improve operative freedom: better perception, discovery/removal of paths, counterfactual simulation, consequence prediction, rerouting, transforming relationships/environment, and preserving valuable future Passibility.*
- *This section tests whether AFI's proposed architecture — Reality → Perception → Passibility Model → Counterfactual Rollout → Consequence Model → Navigation → Action → World Change → New Passibility — is genuine addition or relabeling, against six architectures: LLMs, model-based transition models, causal world models, POMDP belief-tracking, active inference, and empowerment.*

15.1 Do LLMs Learn Language Regularities, or World Structure?

- *Standard LLM training minimizes predictive loss over language regularities. A rich enough objective over a rich corpus can incentivize internal structure beyond surface statistics (e.g., linear representations of latent board/spatial state in synthetic domains), but this doesn't establish that frontier LLMs possess general, causally-valid world models of open-domain reality.*
- *Richens, Abel, Bellot & Everitt (arXiv 2506.01622): any agent generalizing to multi-step, goal-directed tasks must have learned a predictive model, extractable from its policy — an existence result about agents, not the pretraining objective, guaranteeing a predictive, not causal, model, accurate within-distribution but possibly failing under intervention. That gap motivates causal world models next.*

15.2 Transition Structure vs. Causal Structure

- *Model-based RL “world models” learn a transition function minimizing predictive error — useful within-distribution, but accuracy doesn't guarantee causally responsible variables were identified versus merely correlated ones; such models exploit shortcuts that break under intervention.*
- *Causal world models add explicit interventional structure: Dillies et al.'s COMET (arXiv 2504.07257) uses object-centric extraction, symbolic regression, and LLM-based semantic labeling to identify true causal structure, generalizing past spurious correlations. Zhao et al., with Schmidhuber (arXiv 2506.23068), show apparent non-stationary “drift” is often a fixed structure viewed narrowly, encoded via a Meta-Causal Graph refined by curiosity-driven intervention. Sharma et al. (arXiv 2507.19855) are preliminary — strong benchmarks, no transparent baselines.*
- *For AFI: a plain transition model tells a system what tends to follow what; a causal world model tells it which followings are genuine levers versus training artifacts — what Counterfactual Rollout presupposes.*

15.3 POMDP Belief States and What They Buy

- *POMDPs formalize environments where the agent observes o_t , not the true state s_t ; the belief state $b_t = P(s_t | \text{history})$, Bayesian-updated, is the provably sufficient statistic for optimal control under partial observability — buying calibrated uncertainty, a principled basis for information-seeking action, and correct handling of state aliasing.*
- *This maps onto one slice of AFI's Distortion — uncertainty. POMDP belief-tracking is pre-existing machinery for that mode alone; it says nothing about the other named modes: omission, latency, false-relations, blockage.*

15.4 Active Inference, Expected Free Energy, and AFI's Lightness

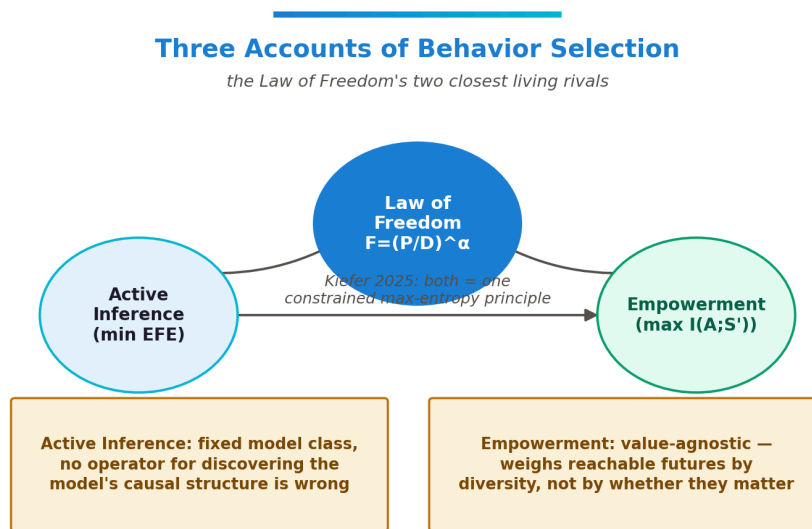
- *Active inference agents minimize expected free energy (EFE) over candidate policies under the agent's own generative model. Friston, Parr, Da Costa and others decompose EFE into a pragmatic term (expected log-likelihood of preferred observations) and an epistemic term (expected information gain) — a genuine structural resonance with the Law of Freedom: “lightest according to perception” and “lowest EFE” do the same work.*
- *Not equality: EFE is fully specified under an explicit generative model with defined priors, while Lightness is only a candidate operationalization (P/D); EFE's epistemic term reduces uncertainty within a fixed model class, with no operator for discovering the model's causal structure is wrong (§15.2) or for creating new paths (Thesis 5); and EFE has no operator for “transforming relationships/environment” — action on the causal structure, not within it.*
- **Falsifiability:** *if P/D predictions never diverge from EFE's in any dissociating design (testbed (f), §17.3), the resemblance collapses into redundancy.*

15.5 Empowerment and the Kiefer 2025 Unification

- Empowerment is the channel capacity — max mutual information $I(A;S')$ — between an agent's actions and future states, independent of any goal: the closest existing analogue to “preserving valuable future Possibility,” a domain-general, goal-agnostic valuation of future option-space.
- Kiefer’s 2025 result (Entropy 27(4):372; arXiv 2502.02962): active inference and empowerment, usually presented as competing accounts, are both instances of a more general constrained maximum-entropy principle. Schneider et al. (arXiv 2510.05996) propose discounted empowerment as a pre-training signal; Shah, Nemenman, Polani & Tiomkin (arXiv 2604.21155) extend it to multi-agent settings.
- Two disanalogies remain: empowerment is relative to the agent’s own model, with no correction term for when it’s wrong, and it is value-agnostic, weighting reachable futures by diversity/controllability rather than whether they are valuable.

Figure 45

Three Accounts of Behavior Selection



falsifiable: if AF's P/D predictions never diverge from either rival, the resemblance collapses into redundancy

Figure 45

Note. Active inference and empowerment are the Law of Freedom’s two closest living rivals, and Kiefer (2025) shows the first two are already unified under one constrained maximum-entropy principle. The Law of Freedom earns its keep only where its predictions genuinely diverge from both — a comparison Section 17 specifies but has not yet run.

16. Falsification Matrix

Figure 46

Falsification Matrix — Scorecard (22 criteria)

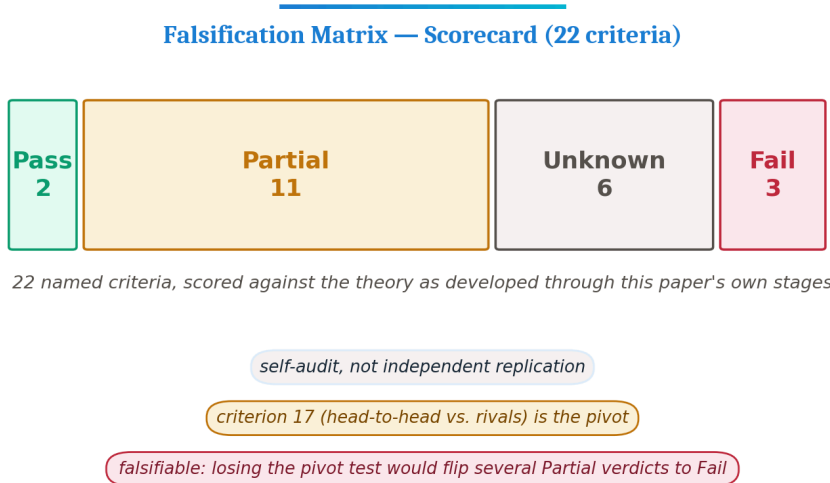


Figure 46

Note. The twenty-two-criterion audit (Section 16) gives an honest, at-a-glance status of every testable component of the theory, Pass/Partial/Fail/Unknown. The scorecard reports this paper's own self-audit; it is not a substitute for independent replication or adversarial review. Falsifiability: This scorecard is itself falsifiable — running the eight testbeds in Section 17 and finding the Law of Freedom loses to expected free energy or empowerment on a fair, pre-registered, dissociating comparison would move criterion 17 to a clean Fail and should move several Partial verdicts to Fail as well, since much of the current partial credit is conditional on that comparison not yet having been run.

Key points: twenty-two named criteria, scored Pass/Partial/Fail/Unknown against the theory as developed through this paper's own stages, not the author's aspirational framing; failures stated plainly, per the paper's standing discipline.

- *Logical coherence passes cleanly; non-tautology, non-circularity, novelty, and explanatory compression pass only partially, conditional on disciplines stated but not yet demonstrated in practice.*
- *Parameter identifiability and a head-to-head test against the theory's closest rivals mark where the next research effort should go.*

#	Criterion	Verdict	Justification
1	Logical coherence	Pass	No internal contradiction found once the epistemic layers (L1–L5) and the weak/strong readings of T1 (§8.1) are kept separate. Coherence is contingent on maintaining that discipline — a paper

#	Criterion	Verdict	Justification
2	Non-tautology	Partial	that blurs the layers would fail this criterion. The weak reading of T1 (Model A/B, §7) is tautological by construction. The strong reading is not, but only if admissibility is fixed independently and prospectively (§8.1) — a requirement the theory states but does not yet demonstrate having followed in its own prior empirical work.
3	Non-circularity	Partial	The Lightness-from-outcome trap (§9.1) is real and structurally identical to Model B's $-\log K$ vacuity. The prospective-measurement rule fixes it in principle. Whether AFI's own prior empirical papers (source ledger A08, A19) actually followed this rule is doubtful — A19 reports R^2 ranging 0.029–0.993 across sub-analyses, a pattern more consistent with post hoc fitting across heterogeneous cases than a single prospectively-specified model.
4	Independent measurement	Partial	Perception and Distortion (P and D) have candidate multiplicative formalizations (Table A rows 30, 33) but no validated, cross-domain, independently-calibrated measurement protocol exists yet. Every proxy proposed so far (branch count, sensor entropy, KL-divergence of belief from truth) is borrowed from an existing field (control theory, information theory), not independently derived by AFI.
5	Prospective prediction	Unknown	§11's experimental program is designed to satisfy this; none of the

#	Criterion	Verdict	Justification
			eight testbeds has been run. AFI's own existing "empirical" corpus is retrospective/simulation-based (source ledger notes on A19, A20 — the latter's own text states forecasts are simulation-based, not empirical measurements).
6	Counterfactual sensitivity	Pass (conceptually) / Unknown (empirically)	The 30-universe Counterfactual Lab demonstrates the theory's predictions vary sensibly across structurally different worlds — it is not insensitive to counterfactual variation in principle. Whether it is sensitive in the right way for real systems is untested.
7	Stochastic robustness	Partial	The softmax reading (§9.2) handles genuine stochasticity coherently. But if Lightness is ever defined as $-\log K$ (Model B), the theory becomes true of any stochastic system by construction — robust to stochasticity only by having no content there.
8	Deterministic compatibility	Fail (as currently scoped)	Per §8.1 (Arrow 3→4) and Universe 2, a fully deterministic single-path domain makes the entire navigability apparatus vacuous — there is no landscape to navigate. The theory has not stated a principled restriction of scope to non-degenerate (branching) domains, nor an account of what it means to apply T2 where $P(s)$ is a singleton. This is a genuine, currently-unaddressed gap, not a minor caveat.
9	Passive-system compatibility	Partial	P0 is included but explicitly stripped of perceptual/cognitive content (§5.2) to avoid

#	Criterion	Verdict	Justification
			the perception-category error AFI's own operating instructions flag. Once stripped, it is unclear "Freedom" retains explanatory content beyond renaming existing physical-response description for passive systems (echoes the Model E/viability-theory redundancy risk).
10	Agentic-system compatibility	Pass	The strongest domain by a wide margin. Direct, substantive engagement with active inference, empowerment, and POMDP planning (Stage 12) shows real conceptual traction for agentic systems specifically.
11	Partial-observability compatibility	Partial	Compatible in principle (Model F), but inherits a hard formal ceiling: exact reachability-value computation in general POMDPs is undecidable (Madani, Hanks & Condon 2003; confirmed current in Fijalkow et al. 2026, arXiv 2602.07473, which identifies only a restricted decidable subclass). AFI cannot claim general tractable Navigability under partial observability — only bounded/approximate versions.
12	Scale robustness	Unknown	Central to AFI's ambition (claimed recovery of Ohm's/Fourier's/Fick's/Darcy's laws across physical scales) but independently unverified — the relevant claims exist only in the author's own self-published corpus (source ledger A01–A20), with specific R^2/effect-size values either unspecified in accessible abstracts or, where reported,

#	Criterion	Verdict	Justification
13	Horizon robustness	Partial	showing signs flagged under criterion 3. Formally addressed via Model H's horizon-weighted Lightness (§9.3), which correctly resolves the local-lightest/future-destroying tension in principle. Robust only if H and β are independently motivated (from known computational/perceptual limits) rather than fit post hoc to rescue predictions — a discipline stated but not yet demonstrated in practice.
14	Cross-domain transfer	Unknown	This is the theory's most distinctive and highest-value claim if true, and its least independently verified. No source outside the author's own corpus has tested $F=P/D$'s cross-domain fit.
15	Parameter identifiability	Fail/Partial	The exponent α (and the "α_∞" notation used in some of the author's corpus, suggesting a limiting exponent) is not independently constrained by any stated procedure. The multiplicative aggregates $P_C = \prod P_i^{\beta_i}$ and $D_C = \prod D_i^{\delta_i}$ introduce further free weights (β_i, δ_i) with no stated independent-estimation protocol — a real risk of underidentification, where enough free parameters can fit almost any observed behavior after the fact, reintroducing circularity (criterion 3) through a side door.
16	Measurement feasibility	Partial	Feasible in the small/simulable systems §11's testbeds target (synthetic MDPs, robotics, network routing, RL benchmarks). Explicitly intractable, by the theory's own admission (§11.2), in

#	Criterion	Verdict	Justification
17	Predictive advantage	Fail (so far)	<i>“wild”/non-simulable domains (history, macroeconomics, most ecology), where it degrades to an interpretive checklist rather than a measurement procedure.</i> <i>No experiment has yet demonstrated $F=P/D$ out-predicting its two strongest named rivals (expected free energy minimization, empowerment maximization) in a design built to dissociate them. This is not a claim the theory is currently entitled to make; §11 testbeds (e), (f), (h) exist specifically to test it.</i>
18	Novelty	Partial	<i>Per Stage 3’s genealogy and Stage 12’s verdict: a novel and useful recombination of pieces (viability kernels, empowerment, active inference, causal world models) that do not currently ship together — not a novel primitive, and several individual components have direct, decades-old (viability theory) or very recent (empowerment, active inference) formal precedents that anticipate parts of the claim.</i>
19	Explanatory compression	Partial	<i>A single validated cross-domain ratio law would be a genuine compression win. As currently specified, the multiplicative aggregates (criterion 15) partially reintroduce the parameter-count complexity a single elegant ratio is supposed to avoid — compression is claimed but not yet demonstrated net of those added degrees of freedom.</i>

#	Criterion	Verdict	Justification
20	Computational tractability	Partial/Fail	Tractable for the bounded testbeds in §11. Structurally intractable in general (inherits POMDP undecidability, viability-kernel computation hardness, and Model H's compounding recursive-estimation cost) — a ceiling shared with, not unique to, AFI, but one the theory must state rather than imply away.
21	Compatibility with known physics	Partial/Conditional	Must reduce to and reproduce Euler–Lagrange/Hamilton's equations in the classical deterministic limit; this reduction has not been demonstrated in any verified source. The specific claim that $F=P/D$ recovers Ohm's, Fourier's, Fick's, and Darcy's laws (source ledger A02) is asserted in the author's own preprint without independently accessible R^2 figures, and the least-action literature (§8.2, Section 12) warns explicitly against treating stationary-action agreement as confirmation without this reduction being shown rigorously, not just asserted.
22	Ability to fail	Pass (in principle) / Partial (in practice)	The Counterfactual Lab and §9.4 jointly specify concrete, non-trivial falsifiers, and several counterfactual universes are explicitly constructed to embarrass T1/T2 (Universes 8, 9, 18 among others) rather than rescue them — genuine falsifiability-in-design. In practice, no falsifying result has yet been reported or acted on anywhere in the author's own corpus, which shows a track record of confirmatory

#	Criterion	Verdict	Justification
			<i>framing (e.g., “35 falsifiability tests,” source ledger A06) rather than documented instances of the theory being revised or narrowed in response to a failed prediction — an asymmetry the paper should note as a standing concern about practice, separate from the theory’s in-principle falsifiability.</i>

Summary

- **Strong:** *logical coherence given disciplined layering (1); the agentic/POMDP/active-inference domain (10); in-principle counterfactual and falsification design (6, 22).*
- **Open (Unknown):** *prospective prediction (5), scale robustness (12), cross-domain transfer (14), predictive advantage over named rivals (17 — Fail only because the comparisons haven’t been attempted, not lost).*
- **Unaddressed problems (Fail/Fail-leaning):** *deterministic-domain vacuity (8) — no stated scope restriction; parameter identifiability (15) — free weights risk circularity through the back door; computational tractability (20) and physics-reduction (21) — both requirements not yet met, only claimed met in unverified self-published sources.*
- **Net:** *a theory coherent and falsifiable in its disciplined (horizon-qualified, prospectively-measured, layer-separated) form, currently untested against its strongest rivals — with the debts above prerequisites for, not incidental to, any claim to generality.*

17. A Possibility-Extended Scientific Method

17.1 Framing the Comparison

- *Conventional experimental causal science proceeds observation → hypothesis → intervention → outcome → causal inference, agnostic about where the state space, variable set, and permissible interventions come from — supplied by the investigator up front. AFI’s claim: that prior specification silently presupposes Navigability already solved.*
- *The Possibility-extended loop makes this explicit: observe state → map structural Possibility → map perceived Possibility → generate counterfactual trajectories → estimate path structure → make a prospective prediction → intervene on path structure → observe actualization → measure how actualization redraws the next possibility field.*

Table 1. Conventional loop vs. AFI-extended loop

Conventional step	AFI step(s)	Assessment
Observation	1. Observe state	Relabeling — adds only recording which variables went unmeasured.
(assumed)	2. Map structural Possibility	Equivalent to an SCM/MDP’s transition function; new in forcing it into an auditable step.

<i>Conventional step (absent)</i>	<i>AFI step(s)</i> 3. Map perceived Possibility	<i>Assessment</i> Genuinely new — an explicit ontic/epistemic split, imported from POMDP theory.
<i>Hypothesis (implicit)</i>	4. Generate counterfactual trajectories; 5. Estimate path structure 6. Make prospective prediction	Overlaps Pearl's Rung 3 and scenario analysis. Relabeling — corresponds to pre-registration, already standard.
<i>Intervention</i>	7. Intervene on path structure	Genuine reframing: targets the availability structure, not a fixed value; anticipated by “soft” interventions in the causal-graph literature.
<i>Outcome Causal inference</i>	8. Observe actualization 9. Measure how actualization redraws the next possibility field	Relabeling. Genuinely new: asks how realizing one path changes the set of paths available next — the clearest novelty claim, hardest to operationalize (§17.2).

- Three steps (3, 7, 9) add genuinely new content; three (1, 6, 8) are relabelings; two (2, 4/5) formalize what causal modeling already does under other names.

Figure 47

Conventional vs. AFI-extended scientific method

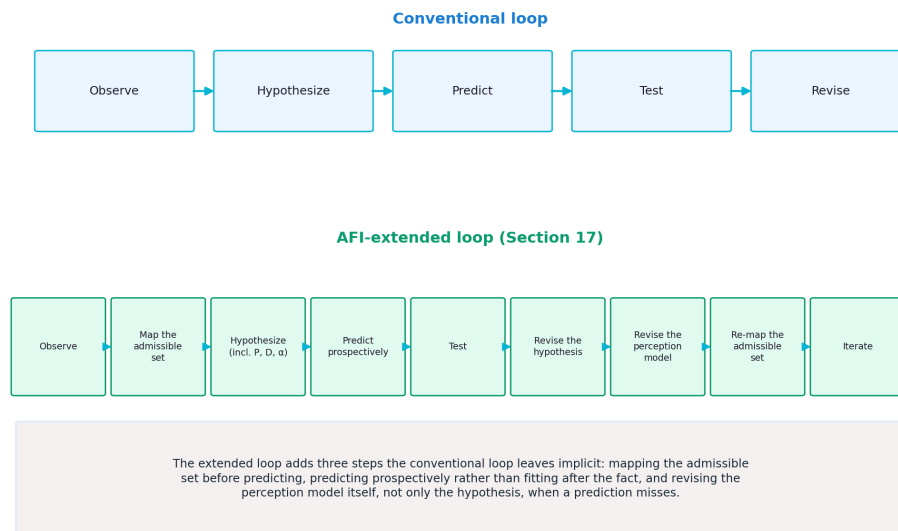


Figure 47

Note. The AFI-extended loop (Section 17) adds explicit admissible-set mapping, prospective prediction, and perception-model revision steps to the conventional five-step loop. The extended loop’s added steps have not themselves been validated as improving predictive success over the conventional loop; the comparison is architectural, not a result of running both loops head-to-head.

17.2 Costs, Tractability, and the Central Falsifiability Risk

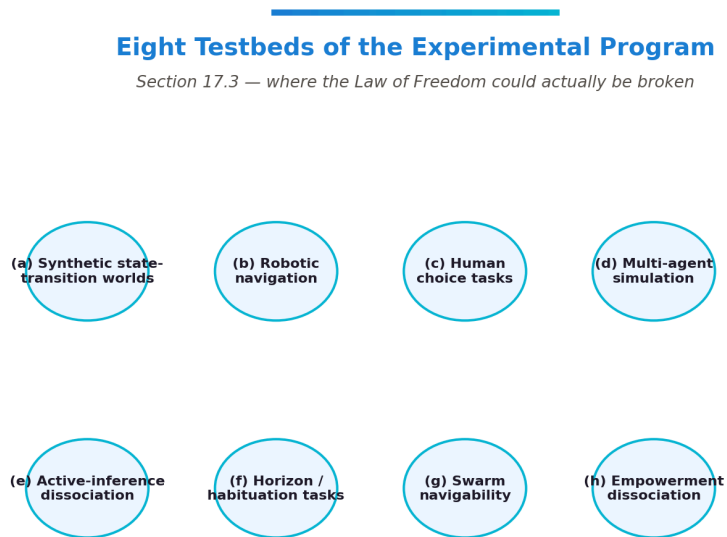
- **Intractability.** Step 2 is trivial for small synthetic MDPs; for real systems with continuous/combinatorial state spaces it is at least as hard as exact planning (POMDP planning is PSPACE-complete finite-horizon, undecidable infinite-horizon), and will almost always be an approximation, degrading to qualitative narrative in “wild” domains.
- **Circularity risk.** $F=(P/D)^\alpha$ is falsifiable only if Perception and Distortion are measured independently of the outcome they explain; inferring “high Distortion” from a costly-looking path makes the account unfalsifiable by construction. Every testbed below manipulates P/D before and independently of the DV — this section’s single most important methodological commitment.
- **Cost of Step 9 / Net.** Measuring how a path redraws the next possibility field requires re-running Step 2, or a model that already predicts structural change — measurable only in simulable/replayable systems. The loop’s value is separating ontic/epistemic possibility (Step 3) and generalizing intervention (Step 7) and causal inference (Step 9); its cost is that Steps 2–3 are often intractable outside small/simulable systems, falsifiable only with pre-registered P/D.

17.3 Experimental Program

Each testbed specifies: **Baseline** (non-AFI prediction), **IV/DV**, **AFI prediction**, **rival prediction** from a named alternative, and a concrete **falsifier** — an observation refuting the AFI-specific claim, not merely a null result.

Figure 48

Eight Testbeds of the Experimental Program



each testbed manipulates Perception and Distortion before and independently of the outcome it explains

Figure 48

Note. Eight settings, from synthetic MDPs to multi-agent swarms, each built to manipulate Perception and Distortion before and independently of the outcome being explained — the single methodological commitment that keeps the Law of Freedom falsifiable rather than fitted after the fact.

(a) Synthetic state-transition worlds

A graph/grid MDP with labeled edges (cost, reversibility, blockage probability): exact ground-truth Passibility by construction.

<i>Field</i>	<i>Content</i>
<i>Baseline</i>	<i>Value-iteration/Dijkstra: recovery time depends only on next-best-path cost.</i>
<i>IV</i>	<i>Path multiplicity (near-equal-cost alternates) and edge reversibility; chosen- and next-best-path costs held fixed.</i>
<i>DV</i>	<i>Path chosen; latency/success of re-routing after mid-trajectory edge removal.</i>
<i>AFI prediction</i>	<i>Recovery tracks multiplicity/reversibility ($F \approx P/D$ at the blocked state), not just next-best cost; high-branching regions recover faster at matched cost.</i>
<i>Rival prediction</i>	<i>Shortest-path/reachability: recovery depends solely on next-best-path cost, invariant to multiplicity elsewhere.</i>
<i>Falsifier</i>	<i>Recovery indistinguishable across low- vs. high-branching conditions matched on next-best cost.</i>

(b) Controlled physical system (robotic arm)

A 6-DOF (or 2-link planar) arm reaching through an obstacle field, with effort, time, and reversibility as separable cost dimensions, not collapsed into one scalar reward.

<i>Field</i>	<i>Content</i>
<i>Baseline</i>	<i>Optimal-control/reward-maximizing policy (LQR or scalar-reward RL): indifference between routes of matched total scalar cost.</i>
<i>IV</i>	<i>Reversibility of the manipulation action, varied independently at matched scalar cost; sensor noise added as a second factor.</i>
<i>DV</i>	<i>Route selection frequency over repeated trials/replans; recovery behavior after a simulated grasp failure.</i>
<i>AFI prediction</i>	<i>At matched scalar cost, the higher-reversibility route is preferred, more strongly as sensor noise (Distortion) increases.</i>
<i>Rival prediction</i>	<i>Reward-maximizing RL/optimal control: indifference between routes at all noise levels, cost being matched by construction.</i>
<i>Falsifier</i>	<i>Equal selection frequency across reversibility conditions at every noise level, with no noise $\times$ reversibility interaction.</i>

(c) Network routing

A simulated packet network (e.g., Mininet) with routes of near-equal latency but differing backup-path counts, and a controller with variable topology visibility.

<i>Field</i>	<i>Content</i>
<i>Baseline</i>	<i>Shortest-path/greedy routing (Dijkstra/OSPF-style): failover depends only on whether any backup path exists.</i>
<i>IV</i>	<i>Edge-disjoint alternate-route count at fixed primary latency; controller's topology-visibility fraction (Distortion).</i>
<i>DV</i>	<i>Route chosen by an AFI controller computing P/D per candidate; post-failure recovery latency/jitter.</i>
<i>AFI prediction</i>	<i>Failover latency/jitter improves gradedly with alternate-route count and visibility, even at matched steady-state latency; degraded visibility erodes the advantage without changing topology.</i>
<i>Rival prediction</i>	<i>Shortest-path/greedy routing: failover is binary (backup exists / doesn't), insensitive to graded count or visibility.</i>
<i>Falsifier</i>	<i>Failover shows only the binary any-backup effect, with no gradient across alternate-route count or visibility.</i>

(d) Swarm systems (ant colony / robotic swarm foraging)

A Deneubourg-style double-bridge paradigm (ant colony or robotic swarm) with two equal-length routes of differing branch multiplicity.

<i>Field</i>	<i>Content</i>
<i>Baseline</i>	<i>Pheromone/positive-feedback amplification (Deneubourg et al.): equal-length paths split near 50/50.</i>
<i>IV</i>	<i>Branch multiplicity/redundancy along two equal-length routes.</i>
<i>DV</i>	<i>Traffic proportion per route to convergence; recovery speed/success after blocking the majority route.</i>
<i>AFI prediction</i>	<i>Given equal length, the colony/swarm preferentially converges on, and recovers faster along, the higher-branch-multiplicity route.</i>
<i>Rival prediction</i>	<i>Positive-feedback amplification: near-even 50/50 split, recovery speed independent of branch structure.</i>
<i>Falsifier</i>	<i>Near-50/50 allocation with no branch-multiplicity effect on convergence or recovery.</i>

(e) Robotic navigation under partial observability

A mobile robot with SLAM in an occluded environment, choosing among equal-length routes that differ in how much unknown map they reveal.

<i>Field</i>	<i>Content</i>
<i>Baseline</i>	<i>Pure reachability planner (A*/RRT on the current map estimate) minimizes expected distance/time, replanning only reactively.</i>

<i>Field</i>	<i>Content</i>
IV	Informativeness of candidate equal-length routes (fraction of unknown map revealed).
DV	Route choice frequency; number of costly backtracks over repeated randomized runs.
AFI prediction	An information-weighted P/D planner preferentially selects the map-revealing route even at equal or slightly higher cost, with fewer backtracks.
Rival prediction	Empowerment maximization (Klyubin, Polani & Nehaniv): route choice tracks future action-state channel capacity, not perception-gain-over-distortion — dissociable via routes revealing map information without increasing control options.
Falsifier	No preference for informative over uninformative equal-length routes and no backtracking reduction; or empowerment fits choice strictly better than P/D in the dissociating cases.

(f) POMDP RL agents

Standard partially-observable RL benchmarks (MiniGrid, RockSample-style tasks), comparing three trained policies under varying observability.

<i>Field</i>	<i>Content</i>
Baseline	Expected free energy (EFE) minimization (active inference; Friston et al.): action selection as a weighted combination of pragmatic (reward) and epistemic (information-gain) value.
IV	Fraction of state masked (observability level); probe states where path multiplicity is high but information gain is low.
DV	Policy divergence among reward-only RL, EFE-minimizing, and AFI-style (P/D) agents; return and sample efficiency across observability levels.
AFI prediction	In high-multiplicity/low-information-gain probe states, AFI-style preference shifts toward the path-rich option even when EFE's information-gain term is flat; fit improves with observability-scaled P/D beyond EFE.
Rival prediction	EFE minimization: preference set by the reward/information-gain trade-off; no separate effect of raw path count once information gain is controlled.
Falsifier	No policy shift in probe states relative to matched low-multiplicity controls — path count adds no predictive power beyond EFE's terms.

(g) Human affordance / modal-perception experiment

Grounded in Warren's (1984) stair-climbing affordance-perception paradigm and Carello et al.'s (1989)/Mark's (1987) perceived-reachability paradigm, which established that reachability judgments scale with the ratio of object distance to arm length (~ 0.87).

<i>Field</i>	<i>Content</i>
Baseline	Body-scaled ecological-optics account: reachability judgments track a fixed intrinsic

<i>Field</i>	<i>Content</i>
	<i>distance/arm-length ratio, unaffected by unused objects in the scene.</i>
<i>IV</i>	<i>(1) object distance relative to arm length (classic paradigm); (2) presence/absence of a visible-but-unused auxiliary support (e.g., a rail or rolling stool) extending effective reach, at fixed distance/arm-ratio; (3) occlusion/blur of that support.</i>
<i>DV</i>	<i>Graded (1–7) or binary reachability judgment; reaction time; validated on a subset against an actual reach-or-request-tool action.</i>
<i>AFI prediction</i>	<i>The reachability boundary shifts outward when the support is visible (even unused), reflecting sensitivity to available action-paths (structural Passibility, including tool-mediated paths) discounted by perceptual clarity (Distortion); occlusion attenuates the shift.</i>
<i>Rival prediction</i>	<i>Classical body-scaled account (Warren; Carello et al.): judgments track only the fixed reach/arm-length ratio, unaffected by an unused tool's visibility.</i>
<i>Falsifier</i>	<i>No judgment/RT difference between support-present and support-absent conditions at matched distance/arm-ratio, or a shift unaffected by occlusion.</i>

(h) AI world-model benchmarks

A world-model agent (Dreamer- or MuZero-style) in a benchmark with countable branch structure at decision points (e.g., Crafter's achievement tree), dissociating four candidate decision quantities.

<i>Field</i>	<i>Content</i>
<i>Baseline/rivals</i>	<i>(1) Reward-to-go (pure RL); (2) expected free energy from the learned model; (3) empowerment (action–future-state mutual information) from the learned model.</i>
<i>IV</i>	<i>Decision states constructed (procedural generation + post hoc filtering) where model-predicted viable-path count is manipulated independently of reward-to-go, empowerment, and EFE.</i>
<i>DV</i>	<i>Model fit (cross-entropy/AIC on held-out states) of action-selection probabilities against: P/D (branch count over predictive entropy), EFE, empowerment, reward-to-go.</i>
<i>AFI prediction</i>	<i>In the decoupled states, P/D predicts realized action probabilities better than EFE, empowerment, or reward-to-go alone, the advantage concentrated in states built to decouple these quantities.</i>
<i>Rival prediction</i>	<i>Empowerment-maximization: action likelihoods track action–future-state mutual information regardless of branch count; or EFE's reward/information-gain combination fits as well.</i>
<i>Falsifier</i>	<i>A nested comparison shows empowerment or EFE fits behavior as well as or better than P/D in decoupled states, or P/D adds no incremental power once empowerment/EFE are included.</i>

17.4 Cross-Cutting Constraints on the Program

- *Three design rules bind all eight testbeds. First: Perception and Distortion (P and D), or their proxies, are fixed or manipulated before the DV is observed, never fit retrospectively — the direct answer to §17.2's circularity risk.*
 - *Every AFI prediction is dissociable from a named rival (shortest-path/reachability in a–d; empowerment/EFE in e–f; empowerment/EFE/reward in h; body-scaled ecological optics in g).*
 - *Testbeds (a)–(d) and (f)–(h) are simulable/replayable, satisfying Step 9's tractability constraint. Testbed (g) substitutes a between-condition design, since a human affordance field can't be cleanly reset trial-to-trial.*
 - *The program is weighted toward small, simulable systems where Steps 2, 3, and 9 are tractable. Extending to non-simulable domains (organizations, ecosystems, markets) is future work.*
-

18. Implications for Human and Artificial Intelligence

18.1 What a Genuine AFI Architecture Would Need to Add — A Surgical Accounting

Walking the pipeline stage by stage against Section 15's architectures:

- **Reality to Perception.** *Fully covered — the POMDP observation function and belief update, settled since classical control theory. Action to world change is likewise standard, covered by any actuated agent architecture.*
- **Perception to Possibility model.** *Only partially covered: causal world models (COMET; Meta-Causal Graph) model available interventions but output dynamics for prediction, not a rankable possibility-space object; empowerment computes a scalar summary of option-space but not its structure.*
- **Counterfactual rollout.** *Well covered by COMET and the Meta-Causal Graph; LLM in-context "counterfactual" behavior is weaker, since Richens et al.'s theorem guarantees only predictive, not interventional, validity.*
- **Consequence model / Navigation.** *Well covered, arguably over-covered — forward simulation through a causal world model, value-function machinery, active inference's pragmatic-value term, EFE minimization, empowerment maximization, and model-predictive-control/tree-search are all mature, independent answers.*
- **World change to new Possibility.** *The least-covered stage: model-based RL and active inference close loops through state/belief update and reward/free-energy bookkeeping, but neither carries path-availability forward as a tracked, protected quantity. Empowerment, recomputed at each state, is the closest mechanism, but is typically evaluated fresh rather than compared pre- and post-action — what "preserving valuable future Possibility" asks for.*
- **True-vs-false path separation as an ontological distinction,** *not a probability update. POMDP belief update redistributes probability mass, with no category for "never structurally real" versus "merely improbable," and no cited work frames this as an online Possibility Model tracking ground-truth-available paths against perceived-but-false ones. Kim (arXiv 2605.01694): predictive sufficiency isn't control sufficiency for latent world-model states. Nguyen et al. (Minds and Machines 36(2) art.29, 2026) distinguish causal controllability (capacity to intervene) from normative controllability (legitimacy) — a governance layer AFI's architecture doesn't formalize, an acknowledged omission.*

18.2 Verdict

- *The evidence supports verdict (b): a novel and useful recombination of existing pieces that do not currently ship together — not (a) genuine novelty from a blank slate or (c) redundancy with any single existing architecture.*

- **Against (a):** every component AFI's pipeline requires is independently anticipated: causal structure for counterfactual rollout (COMET; Meta-Causal Graph), calibrated belief under partial observability (POMDP theory), perception-relative goal/uncertainty trading (active inference/EFE), domain-general option-space valuation (empowerment, now understood via Kiefer (2025) to share a common ancestor with active inference), and a formal guarantee that generalizing agents contain extractable predictive structure (Richens et al. 2025).
- **Against (c):** no cited architecture integrates causal-structure accuracy, calibrated belief, uncertainty/goal-trading selection, and option valuation into one loop whose terminal currency is corrected, causally-validated path-availability, tracked as a first-class value. AFI's multi-mode Distortion decomposition — omission, error, latency, false-relations, blockage, uncertainty — is more granular than any single reviewed formalism.

Figure 49

The Distortion Taxonomy

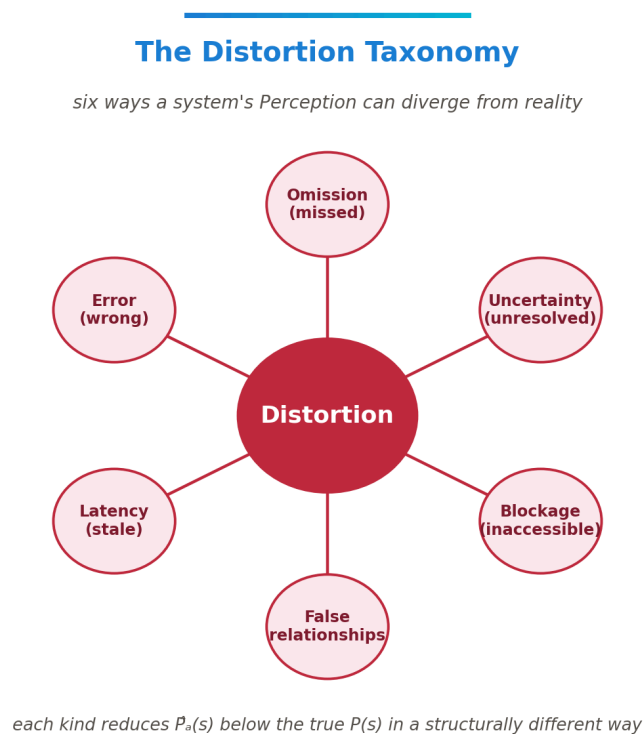


Figure 49

Note. Distortion is not one thing. Omission, error, latency, false relationships, blockage, and uncertainty each reduce a system's perceived possibility set, $\hat{P}_a(s)$, below the real set, $P(s)$, through a structurally different mechanism — a system fully corrected for one kind can still suffer badly from another. - Empowerment's option-counting is relative to the agent's own model. AFI's Possibility is meant to be distortion-corrected against ground truth — real integrative work. - **Falsifiability:** whether the integration earns its keep is empirical — whether a system tracking distortion-corrected, causally-validated path-availability behaves measurably differently

(robustness, information-seeking, transfer to novel environments) from a competently engineered hybrid built without AFI's framing. Future work committed to, not a result claimed in advance.

Figure 50

A candidate human/AI Possibility architecture

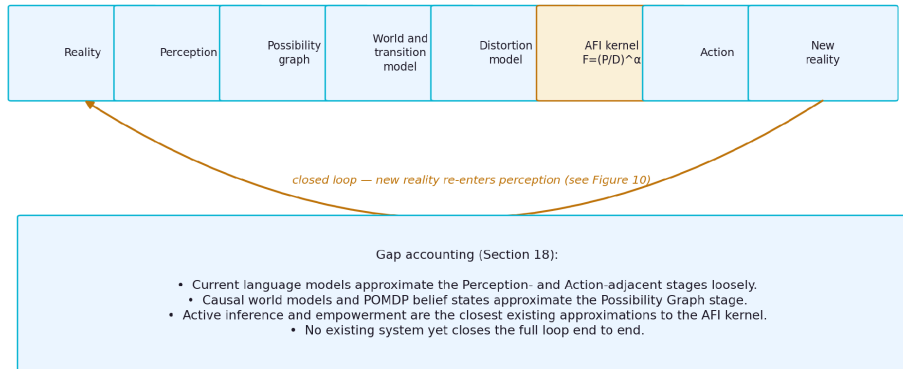


Figure 50

Note. The eight-stage pipeline gives a common frame for comparing current AI system components (world models, POMDP belief states, active inference, empowerment) against the stages an AFI-complete system would require. No existing system, biological or artificial, has been shown to implement this pipeline as a unified closed loop; Section 18's gap accounting treats the pipeline as an organizing frame, not a verified architecture.

19. Limitations

Five limitations recur across this paper's sections and are collected here so no reader mistakes their absence from any single section for their absence from the theory.

- **Scope restriction to non-degenerate domains.** Section 13 (Arrow 3→4) and Universe 2 show that wherever a system's admissible continuation set is a singleton — fully deterministic, non-branching — the Navigability apparatus is vacuous: no landscape to navigate, nothing distinctive for Freedom to say. The paper has no principled account of how much of physical reality, particularly the fundamental microphysical level, falls inside this restriction — scored a **Fail**, not a minor caveat (§16, criterion 8). Future work owes either a demonstration that genuine branching is pervasive at every scale the theory covers, or an explicit, narrower scope statement.
- **Parameter identifiability.** The exponent α in $F=(P/D)^{\alpha}$, and the weighting exponents β_i , δ_i in $P_C=\prod P_i^{\beta_i}$ and $D_C=\prod D_i^{\delta_i}$, are unconstrained by any stated independent-estimation procedure. Left free, they can fit almost any observed behavior after the fact, reintroducing the circularity the prospective-measurement discipline (§14.1) is

meant to rule out — the paper’s single most concrete technical debt: a pre-registered estimation procedure, independent of the behavior it explains, is a precondition for §17’s falsification programme to run honestly.

- **Untested against named rivals.** Section 16 (criterion 17) records that no experiment yet shows $F=P/D$ out-predicting expected-free-energy minimization or empowerment maximization in a design built to dissociate them. The theory’s strongest positive claims — cross-domain transfer, predictive advantage, explanatory compression — remain open questions: a test programme is designed but has not been run (§17), and these are not established results.
- **Reliance on a self-published evidentiary corpus.** The author’s own prior AFI publications (Appendix C, A01–A20) source several quantitative claims used throughout. Most prominent: a claimed cross-domain fit of $F=P/D$ to Ohm’s, Fourier’s, Fick’s, and Darcy’s laws, with R^2 ranging 0.029–0.993 across sub-analyses in the largest study. None of these has been independently verified, replicated, or peer-reviewed; the smart-buildings study used synthetic rather than measured sensor data as its own stated proof-of-concept. This paper treats that corpus as provenance for the theory under test, never as independent evidence — readers should not infer confirmation from its existence, or from the author’s (independently verified) compute-access grant.
- **The physics-reduction requirement is unmet.** Section 11 (Model G) and the genealogy (§12) state as non-negotiable that any claim connecting the Law of Freedom to classical stationary-action physics must show $F=P/D$ reduces to and reproduces the Euler–Lagrange and Hamilton equations in the deterministic limit. They warn against treating mere formal resemblance to least-action as confirmation — a fallacy the 2025–2026 philosophy-of-physics literature (Kodaj, Bernáth & Pickup 2025; Dong Yang 2025) independently cautions against; no available source demonstrates this reduction. Until shown, appeals to least-action physics in support of the Law of Freedom should be read as motivational, not evidential.

20. Research Programme

Four lines of work follow directly from this paper’s gaps, in priority order:

1. **Parameter identifiability (most urgent).** *A pre-registered estimation protocol for α , β_i , and δ , applied to at least one of the eight Section 17 testbeds before any outcome is observed, with fitted parameters then frozen and tested out-of-sample. This would move §19’s identifiability gap from acknowledged weakness to resolved.*
2. **Head-to-head rival comparisons.** *Run dissociating comparisons against expected free energy and empowerment specifically (testbeds (e), (f), (h)) — the theory’s two strongest named rivals (§13.2). These would most directly settle whether the Law of Freedom adds genuine predictive content or is better read as a relabeling.*
3. **The physics-reduction proof.** *An explicit attempt to establish or refute whether $F=P/D$ recovers Euler–Lagrange/Hamilton’s equations in the classical limit (§11 Model G; §19), prior to further appeal to least-action physics in support of the theory.*
4. **Independent replication.** *Replication of the author’s own strongest prior empirical claims — the cross-domain law-recovery study and the large- N swarm-navigability simulation, Appendix C entries A02*

and A19 — by researchers outside the AFI programme, with pre-registered analysis plans. This would convert this paper's provenance-only citations of that corpus into independent evidence.

Falsifiability: none of these four steps is optional if the programme's own falsifiability standard (§16, criterion 22) is to be more than a standard stated in principle.

Figure 51

Research Programme: Four Priorities

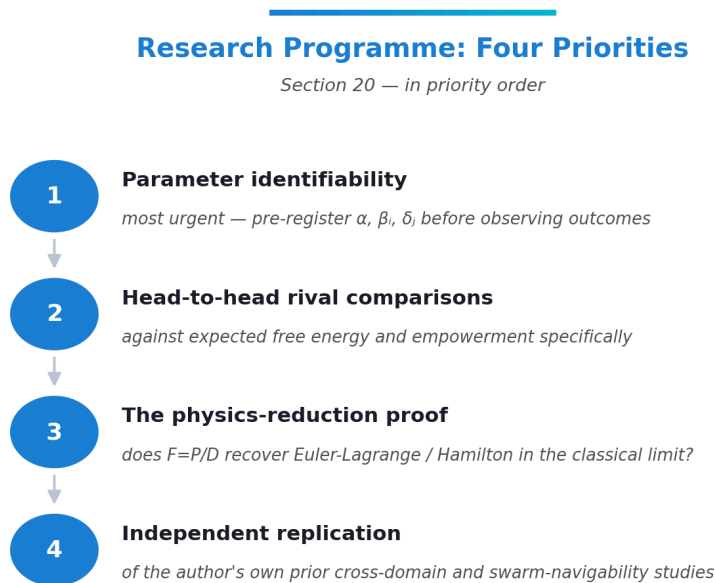


Figure 51

Note. The four priorities are ordered, not a menu: parameter identifiability comes first because every downstream comparison and replication effort depends on α , β_i , and δ_j being fixed independently of the outcomes they are meant to explain.

21. Conclusion

This paper set out to test, rather than assume, a proposition from the author's Architecture of Freedom Intelligence programme: that Freedom, the structured availability of possible continuation, is the cause of all, and a general Law of Freedom governs how systems navigate the field Freedom makes available.

The honest result is neither confirmation nor dismissal.

- *A cause-primitive built from structured possibility does generate a genuine navigation problem — an explanatory demand any adequate theory of causal production must answer. This holds only for a reader*

granting two non-forced commitments: reality is compressible rather than arbitrary, and realization is single-history rather than everywhere-branching or a brute, unexplained fact.

- Under those commitments, the central claim survives in qualified form: **the Cause of All creates the space of causation; some Law of All governs the navigation of that space**. But which law is, and must remain, an open, comparative, empirical question — not something the derivation hands the Law of Freedom for free. The Law of Freedom is not currently a law of general application.
- It is a coherent, falsifiable, in places genuinely novel candidate — closest in spirit to expected-free-energy-minimizing active inference and information-theoretic empowerment, both structurally resembled without yet being shown subsumed. It is specified precisely enough to be wrong, tested nowhere near enough to be believed — answerable, from here, only to Section 17's unrun experiments.
- The paper's actual finding is not confirmation of Freedom as the Cause of All. It is a demonstration of what would have to be shown, and what is still missing, before that claim could be responsibly made.

Figure 52

The Cause of All, and the Law of All

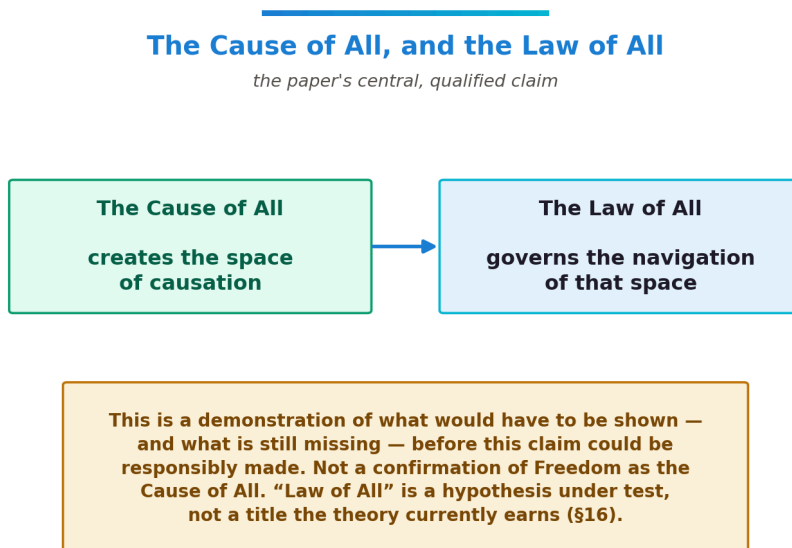


Figure 52

Note. This two-part claim is the paper's central thread, and it is offered qualified, not confirmed. Freedom creating the space of causation survives this paper's own audit; some law governing the navigation of that space is real, but which law — and whether the Law of Freedom is it — remains open.

What this paper establishes:

- *Freedom, as structured possibility, is a necessary condition for causal production in every branching domain — argued from first principles and checked against forty counterfactual universes.*
 - *Granting that condition opens a real, forced question — the Navigability problem — that any general theory of causation eventually has to answer.*
 - *The Law of Freedom is a coherent, non-tautological, falsifiable candidate answer to that question; it is structurally close to two of the strongest existing formalisms in the field (active inference and empowerment), without yet being shown to reduce to either.*
 - *Thesis 5, Freedom Intelligence, now has its own dedicated, falsifiable test (Universe 39) distinct from Theses 1 and 2.*
 - *Of the twenty-two Falsification Matrix criteria, two pass cleanly and most of the rest pass partially, pending the not-yet-run experiments in Section 17.*
 - *Parameter identifiability and a head-to-head test against the theory's closest named rivals are the two the next research phase (Section 20) should resolve first.*
-

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Appendix A. The Counterfactual Universe Laboratory

Forty formally specified toy universes — the original 30 (U1–U30) plus a 10-universe extension (U31–U40). The extension covers multi-agent, adversarial, resource-bounded, hierarchical, and self-improving-perception cases. Together they stress-test Thesis 1 (Freedom as Cause) and Thesis 2 (Law of Freedom).

0. Shared notation

Every universe uses the same vocabulary so entries can be compared directly:

- **S**: state space (concrete per universe, never left abstract).
- **Transition kernel**: a deterministic map $\delta: S \rightarrow S$, a relation $R \subseteq S \times S$, or a stochastic kernel $\kappa(s'|s)$.
- **$P(s) = \{s' : (s, s') \text{ admissible}\}$** — real/structural Passibility of s , fixed by the universe's laws, observer-independent.
- **$\hat{P}_a(s)$** — the perceived possibility set for agent a . It can omit, add to, or partially overlap $P(s)$ (omission, hallucination, partial overlap all possible).
- **Reach(s)** — states reachable by a real (possibly multi-step) admissible path from s .
- **Viable(s) $\subseteq$ Reach(s)** — reachable states with an open-ended admissible continuation.
- **$D_a(s)$** — Distortion: divergence of $\hat{P}_a(s)$ from $P(s)$ (omission, error, latency, false relation, blockage, uncertainty).

- $L(s \rightarrow s')$ — Lightness/path cost, evaluated prospectively (never read off the outcome after the fact) over candidate dimensions {effort, energy, resistance, time, risk, uncertainty, irreversibility, information cost, $-\Delta \text{futurePassibility}$ }.
 - **T2, deterministic:** $s^* = \operatorname{argmin}_{s' \in \hat{P}_a(s)} L_a(s \rightarrow s')$. **T2, stochastic:** $\Pr(s') \propto \exp(-L_a(s \rightarrow s')/\tau)$, $\operatorname{softmax}$ over $\hat{P}_a(s)$.
 - $F_a(s) = (\mu[\hat{P}_a(s)] / D_a(s))^\alpha$ — local Freedom equation (μ = accessible-possibility size/quality).
 - **Named rivals** used throughout: Random/uniform selection, Markov memoryless dynamics, Stationary action/least action, Energy minimization, MaxEnt, Maximum entropy production (MEP), Optimal control, Goal satisfaction/satisficing.
 - **Recurring experiment:** T2 is perception-relative; most rivals are ground-truth-based. Hold real quantities fixed, manipulate only what's perceived, and check whether the actualized path tracks perceived (T2) or real (rivals) values.
-

U1. The Null Universe (no becoming)

- **Setup:** $S=\{s_0\}$; $R=\emptyset$; $P=\text{Reach}=\text{Viable}=\emptyset$.
 - **Result:** T1 holds only vacuously: no causation exists to falsify it. Strong and weak T1 collapse together. T2 predicts nothing, and no rival does better. **A boundary case, not a confirmation.**
 - **Falsifiability:** None — the risk is treating vacuity as if it were support.
-

U2. The Rail (single deterministic successor)

- **Setup:** $S=\mathbb{Z}$, $\delta(s)=s+1$ the only admissible transition everywhere; $\Pr(s+1|s)=1$.
 - **Result:** T1 is non-vacuous but contentless with only one option. T2 is degenerate: argmin is forced regardless of L . Every rival — random, Markov, least action, energy minimization — predicts the same outcome. **No discriminating power for the Law of Freedom.**
 - **Falsifiability:** None — every rival coincides with T2 here.
-

U3. The Fork with One Traveler (one actual, many counterfactual successors)

- **Setup:** $S=\{0,1\}^*$ binary strings. Both $s \cdot 0$ and $s \cdot 1$ are admissible at every step, though only one is realized. Both branches stay Viable forever.
 - **Result:** Admissibility of the unrealized alternative is what makes a transition genuine becoming rather than a forced step (contrast U2). T1 is substantive here. Strong versus weak T1 could separate in principle, but without a dynamical law U3 alone can't adjudicate between them — that's deferred to U6–U12.
 - **Falsifiability:** Needs $L(s \rightarrow s_0)$, $L(s \rightarrow s_1)$ specified first — a template, not a test.
-

U4. The Boundless Tree (infinite branching)

- **Setup:** $S=\mathbb{N}^*$. $P(s)=\{s \cdot n : n \in \mathbb{N}\}$ is countably infinite, and no uniform prior exists. A finite-bandwidth observer necessarily has $\hat{P}_a(s) \subsetneq P(s)$, structurally.
- **Result:** This forces T1 and T2 to be stated over $\hat{P}_a(s)$ rather than $P(s)$, motivating Perception as a distinct term. Strong T1 gains a falsifiable flavor if the size or structure of $\hat{P}_a(s)$ — not just its non-emptiness — modulates dynamics. T2 needs $L(s \rightarrow s \cdot n) \rightarrow \infty$. MaxEnt faces the same normalizability problem here, a fair fight.

- **Falsifiability:** Manipulate agent bandwidth while holding real Possibility fixed. T2 predicts a bandwidth effect; Markov and energy-minimization rivals predict none.
-

U5. The Frozen Infinity (infinite possibility, zero actualization)

- **Setup:** $S=\mathbb{N}^*$, as in U4, plus a global veto gate that is always false. $P(s)$ is infinite everywhere, yet $\Pr(\text{any transition})=0$ forever.
 - **Result:** A genuine embarrassment for strong T1. Maximal Freedom coexists here with permanent causal silence, contrary to “Freedom causes causation.” Weak T1 survives unscathed. That contrast is exactly why this case shows strong T1 carries real, endangered content. **An open problem for AFI, not resolved here.**
 - **Falsifiability:** The strong claim is falsified by construction. The live question is whether the theory retreats to weak T1 or adds an actualizer axiom.
-

U6. The Dice World (pure random actualization)

- **Setup:** $S=\{0,1,2\}^n$; $\kappa(s'|s)=1/|P(s)|$ uniform kernel, 3-ary branching, no cost sensitivity.
 - **Result:** Direct stress case for T2. Deterministic T2 is simply false here. Softmin T2 fares only narrowly better: uniform-random is recovered exactly as $\tau \rightarrow \infty$ for any L. **Coverage bought at the cost of all empirical content in it.**
 - **Falsifiability:** None at $\tau \rightarrow \infty$. No measurement can distinguish T2 from random selection.
-

U7. The Buridan Field (perfect symmetry, no preferred path)

- **Setup:** successors s_a, s_b built by symmetry so $L(s \rightarrow s_a)=L(s \rightarrow s_b)$ under every candidate functional.
 - **Result:** Deterministic T2 is **underdetermined** on exact ties — the classical Buridan’s-ass problem. Softmin handles it ($\Pr=1/2$ each). But so do random selection and MaxEnt. Within the symmetric subspace, all three are observationally identical — a narrower version of U6’s problem.
 - **Falsifiability:** Perturb the symmetry slightly. T2 predicts a graded shift set by $1/\tau$; MaxEnt without a cost model predicts none.
-

U8. The Masochist Universe (anti-lightest-path dynamics)

- **Setup:** $P(s)=\{s_0, s_1\}$. $L(s \rightarrow s_0)=1$, cheap; $L(s \rightarrow s_1)=10$, costly. Empirical dynamics: $\delta(s)=s_1$ always. Perception is independently verified accurate.
 - **Result:** T2, taken literally, is directly falsified here. The standard escape — “perception must be distorted” — is blocked by verified accuracy. **This is the cleanest falsifier of naive T2 in the set, and it is not rescued.** It also falsifies an entire class of minimization laws, not uniquely T2.
 - **Falsifiability:** Verify perception against ground truth, then check whether choice tracks perceived or real cost. Here it tracks neither.
-

U9. The Dissipative-Structure Universe (maximum-resistance dynamics)

- **Setup:** Field configurations on a lattice, e.g. Bénard convection. The empirical selection principle is maximum entropy production (MEP), not least effort. Non-representational (cf. U13).

- **Result:** Low-cost-minimization T2 is **falsified in the same direction as U8**, but for a physically principled reason. MEP is a real, contested rival in non-equilibrium statistical mechanics. Redefining “lightness” as “whatever the dynamics favor” would save T2, but only circularly — rejected.
 - **Falsifiability:** Measure entropy production: lowest-cost (T2), or highest-entropy-production (MEP)? Bénard cells and ecological succession favor MEP.
-

U10. The Homing Missile (pure goal-directed dynamics)

- **Setup:** $S=\mathbb{Z}^2$ grid. The actualized $s'=\text{argmin}$ distance-to-goal among 4 neighbors, ignoring terrain cost.
 - **Result:** T2 predicts this only if distance-to-goal is smuggled into L as a future-Passibility proxy — not a real test. Excluding goal-terms, T2 is falsified whenever the closer cell is costlier. This forces the theory to state whether “future Passibility” may include external goals, or only endogenous option-preservation.
 - **Falsifiability:** Vary terrain cost independently of goal-distance. Does the system follow pure distance (a goal-law), or trade distance off against cost (a hybrid law)?
-

U11. The Ledger Universe (pure history-dependent / non-Markovian dynamics)

- **Setup:** $L(s \rightarrow s \cdot x)$ depends on the count of prior x 's — a fatigue/practice model. A strictly Markovian $\hat{P}_a$ is structurally distorted here.
 - **Result:** T2 predicts that choice frequency for x shifts with trajectory history, even at a fixed surface state. **Markov dynamics is directly falsified** whenever history must be represented. Energy-minimization rivals are likewise blind, unless extended with a memory term.
 - **Falsifiability:** Compare two agents at the identical state reached via different histories. T2 predicts divergent frequencies; Markov predicts none. **A genuine point in T2's favor when observed.**
-

U12. The Boltzmann Chooser (stochastic cost-weighted / softmin dynamics)

- **Setup:** $P(s)=\{s_0, s_1, s_2\}$ with fixed costs, e.g. 1, 2, 5. $\kappa(s_i | s) = \exp(-L_i/\tau) / \sum \exp(-L_j/\tau)$, with τ fitted from independent data, avoiding U6's trap.
 - **Result:** **T2's sharpest, best-behaved, genuinely falsifiable prediction in the lab.** It predicts a monotone-decreasing cost/choice-probability relationship at fixed, out-of-sample τ . Rivals diverge: random/uniform gives a flat 1/3, while a linear-in-cost kernel gives ratios $\propto 1/L_i$ — a different curve.
 - **Falsifiability:** Fit τ on training trials, then predict choice ratios on held-out trials with a new cost triple.
-

U13. The Thermostat Universe (no representational perception — P0-type passive sensitivity)

- **Setup:** A bimetallic-strip feedback loop switches mode at a temperature threshold. “Perception” is read purely operationally here, never cognitively.
 - **Result:** T1 survives only in a **deflated, physical-causal sense**. T2 reduces to “whichever state the causal law produces,” risking the circularity flagged in U9. Ordinary physical law is explanatorily prior here. AFI vocabulary translates without residue.
 - **Falsifiability:** None. There is no non-circular operationalization of L_a separate from the observed trajectory in a system this minimal.
-

U14. The Oracle (perfect perception, $\hat{P}_a(s) = P(s)$ exactly)

- **Setup:** Standard 2-ary branching, with distinct real costs. $\hat{P}_a(s)=P(s)$ exactly by construction, so $D_a(s)=0$.
 - **Result:** T1 is substantive, but Perception and Reality collapse entirely here — a control condition for U15–U17. Strong versus weak T1 can't separate here either. T2 reduces exactly to real-cost minimization. Energy-minimization predicts the identical outcome, by construction.
 - **Falsifiability: None by design — that is U14's diagnostic value.** Testing T2's distinguishing claim requires $\hat{P}_a(s) \neq P(s)$; see U15–U17.
-

U15. The Cataract Universe (severe misperception)

- **Setup:** Real $P(s)$ has 4 neighbors of varying cost. The agent perceives only 1 of 4, via a narrow forward sensor. $D_a(s)$ is large and independently measurable.
 - **Result:** T1 is substantive: the outcome tracks the 1-element $\hat{P}_a(s)$, not the 4-element $P(s)$. This validates Perception as necessary, not decorative. A model built on $\hat{P}_a(s)$ predicts the trajectory where a raw- $P(s)$ model fails.
 - **Falsifiability:** Instrument all 4 real costs and the sensor cone. Does choice track the perceived-only option, or find a real-cheaper one outside the cone?
-

U16. The Locked Door (real hidden paths)

- **Setup:** Real $P(s)=\{\text{corridor_A, corridor_B, secret_passage}\}$. secret_passage has by far the lowest real cost, but it requires an undiscovered lever and is excluded from $\hat{P}_a(s)$.
 - **Result:** T1 is substantive: structural Possibility can outrun perception without paradox. T2 predicts choice only between corridor_A and corridor_B, never secret_passage. A rival “objective optimal control” with full state access would instead predict secret_passage — opposite, easily distinguished predictions.
 - **Falsifiability:** Verify secret_passage's cost independently. Check whether the agent takes it only after the perception-updating event — finding the lever.
-

U17. The Mirage (false perceived paths)

- **Setup:** Real $P(s)=\{\text{east,north}\}$. Heat-shimmer gives $\hat{P}_a(s)=\{\text{east,north,west}\}$, with “west” hallucinated. Attempting it yields collapse, not a real transition.
 - **Result:** If the agent attempts “west,” **the real causal antecedent is its internal representational state, not “west” itself** — which never had admissibility. T1 is unthreatened once this is made explicit. T2 predicts the attempt if west appears lightest under $\hat{P}_a(s)$. That means **T2's predictive target is action selection, not successful outcome**.
 - **Falsifiability:** Induce a controlled false-perception condition and check whether the agent attempts the illusory option. A positive result favors perception-relative T2.
-

U18. The Greedy Trap (local-lightest / global-disastrous path)

- **Setup:** $L(s \rightarrow s_a)=1$ now, but it collapses to a forced dead-end 2 steps later. $L(s \rightarrow s_b)=5$ now, but it opens a permanently branching future.
- **Result: Together with U6 and U8, this is the sharpest problem case in the lab.** Trap-avoidance depends entirely on the free-parameter weight given future-Possibility in L. If that weight is excluded, T2 wrongly predicts the trap. If it's included with enough weight, T2 correctly predicts avoidance — but only

because the weight was chosen knowing the answer. **Unless pre-registered, T2 can “explain” either outcome and is empirically empty.** Myopic energy-minimization predicts the trap unconditionally.

- **Falsifiability:** Pre-register the future-weight on training data, then predict out-of-sample whether a differently-shaped trap is avoided.
-

U19. The Seed-Planter (heavier-now / freer-later path)

- **Setup:** As U18, but with a pre-registered, previously-fitted weight w . s_a , cheap now, leads to a provably poorer future. s_b , costly now, leads to a richer, faster-growing one.
 - **Result:** **This is the case where T2 earns real distance from its rivals, rather than merely surviving them.** With w pre-registered, T2 predicts a non-obvious preference for s_b . Energy-minimization and myopic optimal control both wrongly predict s_a . Only long-horizon optimal control — importing something structurally AFI-like — converges on s_b .
 - **Falsifiability:** Does the same fitted w predict the switch-point across structurally different scenarios? Refitting per scenario collapses back into U18’s problem.
-

U20. The One-Way Door (irreversible path)

- **Setup:** A DAG: no path back once $s \rightarrow s'$ is taken. L includes an explicit irreversibility penalty.
 - **Result:** T2 predicts more-reversible options are preferred over less-reversible ones of similar immediate cost — a checkable monotonicity. Pure energy-minimization, blind to reversibility, predicts indifference. Risk-averse optimal control also predicts avoidance, so this isn’t a unique AFI signature. Still, it is a genuine, correctly-predicted regularity.
 - **Falsifiability:** Use matched option-pairs with equal cost but differing reversibility. A systematic tilt toward reversible options supports T2.
-

U21. The Roundtrip (reversible path)

- **Setup:** R is symmetric ($(s, s') \in R \Leftrightarrow (s', s) \in R$), e.g. a reversible toggling automaton.
 - **Result:** T1 is substantive. But under full reversibility, “lightest path” becomes nearly synonymous with Hamiltonian, energy-conserving dynamics. AFI vocabulary risks being purely translational here, as in U13. T2 reduces to something close to classical least action.
 - **Falsifiability:** None from reversibility alone. Genuine testing needs perturbing perceived-versus-real cost, as in U15–U17.
-

U22. The Ouroboros (cyclic causation)

- **Setup:** $S = \{s_1, s_2, s_3\}$, with δ cyclic and deterministic — an idealized undamped 3-phase oscillator. $\hat{P}_a(s_i) = P(s_i) = \{s_{i+1}\}$ throughout.
 - **Result:** **A genuine, serious problem case for strong T1.** Strong T1 presupposes an asymmetric, well-founded causal order. A closed loop has no privileged first cause. The strong efficient-causal reading must be reinterpreted as purely constitutive or structural — weaker than “causes” suggests. T2 is degenerate here, as in U2.
 - **Falsifiability:** No rival law is distinguishable from T2 here. The open question is strong T1’s causal-priority claim, which is not empirically testable in this toy world.
-

U23. The Handshake (simultaneous mutual causation)

- **Setup:** A joint simultaneous update $(x,y) \rightarrow (f(y),g(x))$. There's no intermediate state where one component updates before the other — coupled-oscillator/Nash-style.
 - **Result:** A second, sharper problem case for strong T1's causal-priority claim than U22. With simultaneous, mutually constitutive updates, there's no moment at which "Freedom caused this" can be located as prior. The honest resolution is again a non-temporal, constitutive reading — weaker than the causal-priority language suggests. T2 still applies at the joint-transition level, if L is defined over joint outcomes.
 - **Falsifiability:** The concrete rival is multi-agent equilibrium selection (risk- or payoff-dominance). Measure which jointly-admissible fixed point is reached under varied perceived-versus-real payoffs.
-

U24. The Long Shadow (non-Markovian causation proper)

- **Setup:** $P(s)$ depends on an event arbitrarily far back — e.g. a symbol 500 steps prior determines whether a move is really admissible or a phantom. A Markovian observer misjudges $P(s)$ itself.
 - **Result:** A memoryless model of Passibility is simply the wrong model here. Phantom moves fail or collapse rather than actualize, preserving T1's core claim once history is modeled correctly. **Markov-dynamics rivals are directly and cleanly falsified by construction — one of the few clean Markov-killers in the lab.** Non-Markovian rivals can match it, though, so AFI isn't uniquely explanatory here.
 - **Falsifiability:** Vary the observer's memory-window and measure the phantom-move, or collapse, rate. It should fall monotonically as the window approaches the true history-dependence range.
-

U25. The Jump (teleportation / discontinuous transition)

- **Setup:** Disconnected discrete "rooms" with no adjacency or metric structure. Admissible transitions are atomic (s,s') pairs — like a portal or quantum jump, not a traversal.
 - **Result:** T1 is unaffected: possibility is logically prior regardless of continuity. T2 survives only if L is read as a **generalized, purely functional cost, stripped of spatial or kinetic connotation**. Applying "path" or "distance" language to discontinuous transitions risks a category mistake, unless it's stated as functional rather than literal.
 - **Falsifiability:** As in U12: fit a jump-cost function to observed frequencies and test out-of-sample. A minimum-description-length rival is equally applicable once L is functional, so no special T2 advantage accrues.
-

U26. The Flow Field (field-based continuous dynamics)

- **Setup:** $S=\mathbb{R}^2$, with $ds/dt=V(s)$ — a gradient flow, or non-gradient with rotational components. No pointwise branching.
 - **Result:** T1 is substantive only in a deflated sense, as in U2 and U13: pointwise possibility is a singleton. Strong-T1 claims need a modal, not actual, construal of Passibility. For a gradient field, T2 reduces exactly to steepest descent. For non-gradient fields, a non-energy Lightness term could separate T2 from energy-minimization — an open modeling task.
 - **Falsifiability:** In the rotational regime, does trajectory match an independent Lightness prediction versus a pure curl prediction? Real but unresolved.
-

U27. The Quantum Coin (quantum probabilistic transition)

- **Setup:** A two-level system, with outcomes at probability $|\alpha|^2/|\beta|^2$. Interference effects — phase affecting later-basis probabilities — have no classical-cost analogue.
 - **Result:** Weak T1 is thinly compatible with QM, but standard QM already says this without AFI. **This is the entry to be most careful not to overclaim: softmin T2 is a real-valued, non-interfering kernel with no mechanism for phase, and no derivation shows T2 reproduces Born-rule statistics.** It's stated here as an open problem, not a claimed reduction.
 - **Falsifiability:** Can any Lightness operationalization reproduce a genuine interference experiment (double-slit fringes, Bell violation)? **No such derivation exists — the most significant unresolved gap between T2 and a major rival theory in the lab.**
-

U28. The Shifting Rulebook (variable laws)

- **Setup:** R_t changes at scheduled intervals: Rail-like for $t < 100$, 2-branch for $100 \leq t < 200$, 3-branch for $t \geq 200$.
 - **Result:** This raises an unresolved question: is F_a evaluated instant-by-instant — “local Freedom” — or does the theory intend an epoch-invariant “meta-Freedom”? It isn't specified. A “fixed-law Markov” rival calibrated only to the first epoch is falsified at $t=100$. That's clean but artificial, since a rival given access to R_t performs identically to T2.
 - **Falsifiability:** At each law-change boundary, does choice distribution update immediately, supporting T2, or lag — an AFI-flavored “law-update latency” term, a potential point in T2's favor if measured?
-

U29. The Growing Space (variable state space)

- **Setup:** State “3” is genuinely added to S at $t=50$, not merely made reachable. $\hat{P}_a(s,t)$ is bounded above by S_t ; no agent can perceive state 3 before it exists.
 - **Result:** **This forces an underspecified boundary:** is the emergence of state 3 a transition T1 must explain, or a pre-causal event outside its scope? Does AFI explain the emergence of new possibilities — the stronger reading — or only navigation of a given structure — the weaker reading, and what's tested here? No standard rival addresses state-space growth either; it's a shared blind spot.
 - **Falsifiability:** None available without first supplying a missing “law of ontological growth” — an open theoretical task.
-

U30. The Self-Widening Path (action creates new paths)

- **Setup:** The action “build_bridge” adds state “far_shore” and transition (home, far_shore), neither of which existed before. This is endogenous growth caused by the system's own prior act — contrast U29's exogenous clock and U16's hidden-but-preexisting structure.
 - **Result:** T1's core requirement is preserved: the act itself is ordinary and admissible, and only its consequence is novel. **This is arguably where T1 connects most naturally to Thesis 5, Freedom Intelligence** — intelligent action is partly definable as action that expands future $P(s)$, or “Freedom causing Freedom.” Curiosity-driven RL already captures much of this qualitatively. That's a convergence to acknowledge, not a claim to make as uniquely AFI's.
 - **Falsifiability:** Compare a fitted future-growth weight (T2) against a fitted exploration-bonus coefficient (curiosity-driven RL). Genuine separation needs growth-value and curiosity-value engineered to diverge.
-

Extended set: ten additional universes (U31–U40)

- The original thirty stress-test T1/T2 across branching, perception, cost, reversibility, causal structure, and law-variability. U31–U40 extend into multi-agent, adversarial, resource-bounded, hierarchical, and self-improving-perception territory.
- U39 is the laboratory’s most direct test of Thesis 5.

U31. The Shared Corridor (competing agents, same admissible path)

- **Setup:** One narrow edge is shared by travelers a_1 and a_2 . Both have it in $\hat{P}(s)$, but only one can traverse it. The joint admissible set is smaller than the product of individual sets.
- **Result:** Independently computed per-agent Lightness can jointly prescribe an infeasible pair: both agents select the shared edge as individually lightest. This counts against treating Lightness as separable across agents without a joint term. **Verdict: the core formalism is under-specified for multi-agent domains — a scope gap the paper’s application ambitions can’t leave unaddressed.**
- **Falsifiability:** Systematic joint-infeasibility errors, uncorrected by any joint resolution rule, count against applying the single-agent Law of Freedom unmodified to multi-agent domains.

Figure 53

Universe 31 — The Shared Corridor

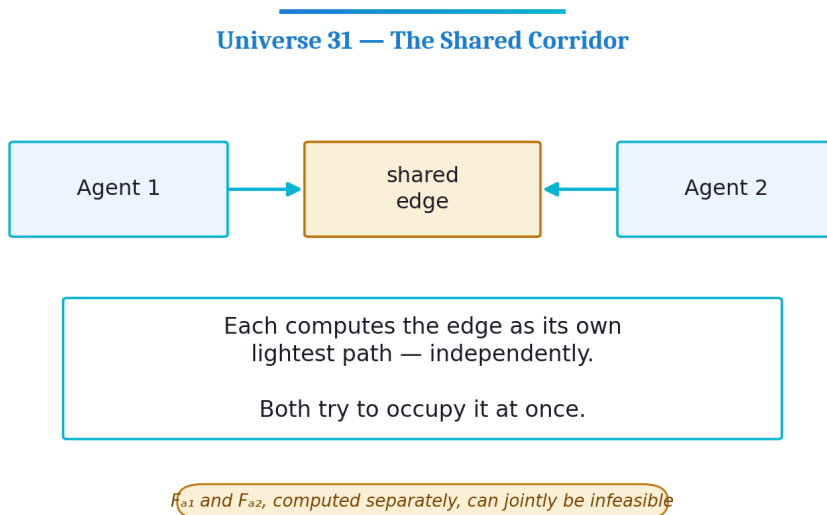


Figure 53

Note. When multiple agents converge on the same admissible path, individually lightest choices can jointly produce infeasibility that no agent’s own Lightness term predicts alone. The illustrated contention scenario is a minimal two-agent case; larger-population dynamics are not shown. Falsifiability: Pairing communicating agents (Universe 32) against isolated agents in this exact contention scenario should show fewer joint-infeasibility errors in the communicating condition; if it does not, the proposed fix (a joint Lightness term) is undercut.

U32. The Relay (cooperating agents, shared distortion reduction)

- **Setup:** As U31, but agents exchange observations before acting. Post-communication $\hat{P}_a(s)$ can strictly dominate pre-communication $\hat{P}_a(s)$. Communication has a cost that itself enters Lightness.
 - **Result:** This operationalizes Thesis 5 directly: a communicating agent has lower $D_a(s)$ and a generally better-informed lightest path than the same agent isolated. **Verdict: one of the cleanest bridges in the laboratory between the Law of Freedom and actual multi-agent system design.**
 - **Falsifiability:** Communicating agents vs. isolated agents in an otherwise identical U31-style scenario should show fewer joint-infeasibility errors (controlling for communication cost).
-

U33. The Deceiver (adversarial distortion)

- **Setup:** A second party alters perception signals. The adversary selects $D_a(s)$ to induce a false $\hat{P}_a(s)$ within a budget, diverging the perceived-lightest path from the real one to its own benefit.
 - **Result:** The Law of Freedom's prediction is that the system follows its perceived-lightest path. Under adversarial distortion, this becomes a prediction that the system can be steered by anyone able to manipulate Perception within budget. **Verdict: exposes a security-relevant consequence of Perception-relativity, with direct AI-safety relevance.**
 - **Falsifiability:** An application built on the Law of Freedom untested for adversarial robustness of the Perception stage is, by this logic, exposed to steering it hasn't tested for.
-

Figure 54

Universe 33 — The Deceiver

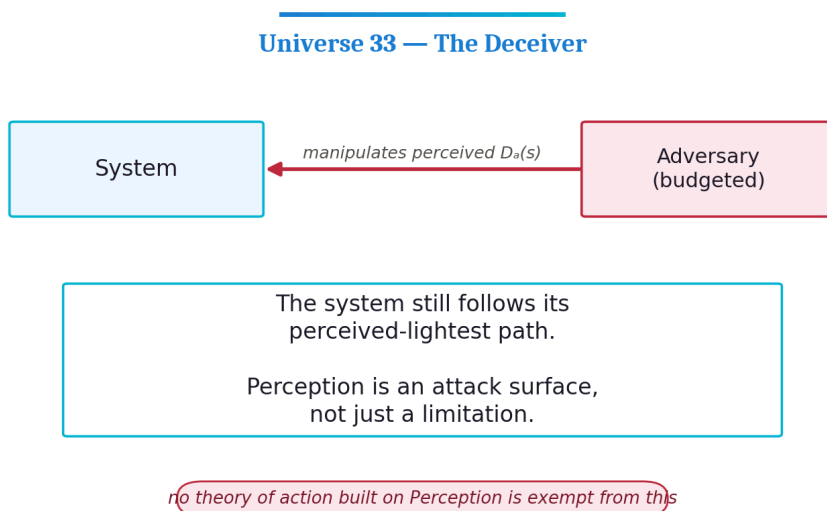


Figure 54

Note. An adversary that manipulates a system's perceived Distortion directly can redirect its chosen path without altering the real admissible set at all. The figure depicts a single adversarial channel; real deception may operate through several channels simultaneously. Falsifiability: A

system whose chosen path does not track adversarially manipulated $D_a(s)$ within the adversary's stated budget would show that perceived-lightest-path selection is not actually governing behavior here — undercutting the Law of Freedom's core predictive mechanism in this domain, not merely its safety.

U34. The Rationed World (depletable resource, not abstract cost)

- **Setup:** Each transition consumes a depletable resource — fuel, battery, capital. Once exhausted, only staying remains admissible. Non-replenishing and replenishing sub-variants are both considered.
- **Result:** This distinguishes Lightness-as-preference from Lightness-as-survival. A path locally lightest by every existing criterion can still be catastrophic if it isn't budget-feasible to completion, which sharpens the horizon term from U18–U19. **Verdict: identifies a concrete formal gap — resource feasibility — relevant to AFI's own applications, such as smart-building energy and swarm-robotics battery life.**
- **Falsifiability:** A “lightest” path chosen over a feasible alternative that leaves insufficient budget to complete demonstrates this gap.

Figure 55

Universe 34 — The Rationed World

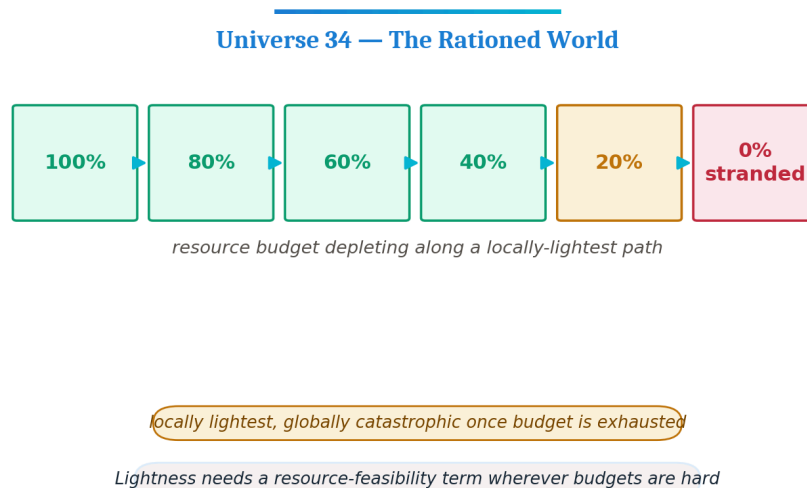


Figure 55

Note. Lightness competing against a hard, depletable resource budget can strand a system on a path that looked lightest before the budget bound. The figure assumes a single scalar resource; multi-resource rationing is not modeled. Falsifiability: A Lightness functional lacking a feasibility term should systematically strand agents in non-replenishing budget settings more often than one that includes it; observing no difference between the two would undercut the claim that feasibility is a missing term rather than an already-implicit one.

U35. The Nested Room (composable, hierarchical spaces)

- **Setup:** A hierarchy $S_0 \supset S_1 \supset S_2$ — district, building, room. A building-level transition is admissible only if a room-level path realizing it exists.
 - **Result:** This formalizes the “universal definition of space” AFI already assumes, but had not stress-tested for whether Passability (P), Distortion (D), and Freedom (F) compose across levels. Open question: does $F_a(s)$ at the room level relate to $F_a(s)$ at the building level by an explicit aggregation rule, or is each level self-contained? **Verdict: a load-bearing open question for every spatial application of AFI, including its flagship smart-building deployment.**
 - **Falsifiability:** A deployed F_a computation giving materially different, unreconciled answers depending on level of description confirms this gap is live.
-

Figure 56

Universe 35 — The Nested Room

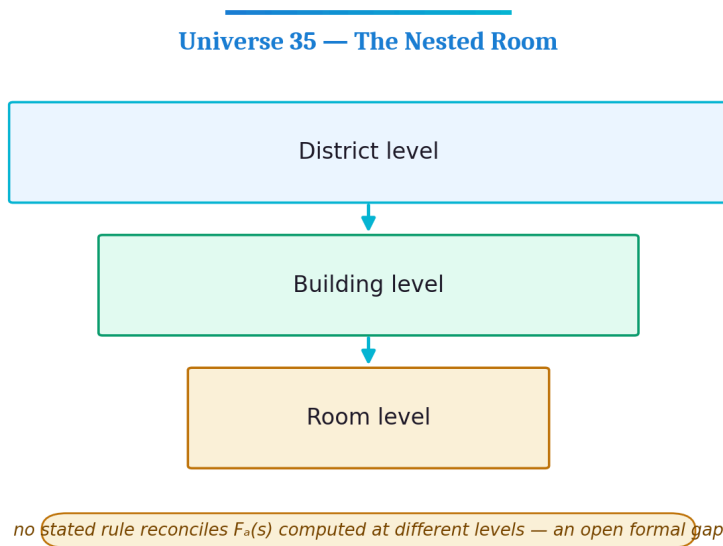


Figure 56

Note. Composable, hierarchical spaces (rooms within buildings within districts) raise an open aggregation question for how $F_a(s)$ at one level relates to $F_a(s)$ at another. The figure illustrates the open question rather than proposing or testing a specific aggregation rule. Falsifiability: If $F_a(s)$ computed at the room level and at the building level disagree on which direction is lightest for the same real situation, with no stated aggregation rule to reconcile them, that is a live, currently-open formal gap — not yet a failure, but a specific, checkable claim about what AFI’s spatial applications still need.

U36. The Fading Memory (perception that decays without renewed observation)

- **Setup:** As in Oracle/Cataract, but $D_a(s,t) = D_0 + \lambda(t - t_{\text{obs}})$, decaying without a refreshing observation — or a bounded-forgetting variant. The agent trades re-observation cost against acting on stale perception.

- **Result:** This formalizes a case AFI's workflow gestures at without a dynamical law. It predicts the re-observation rate should rise with the environmental change rate (λ), and that an untracked policy should show degrading path quality. That's distinguishable from ordinary noise-driven Distortion (U15) by its time-dependence. **Verdict:** adds a testable temporal dimension to Perception the falsification programme (Section 17) can incorporate directly.
- **Falsifiability:** A re-observation rate uncorrelated with λ , or path quality not degrading with staleness, would count against this prediction.

U37. The Rush Hour (paths that close as others use them)

- **Setup:** A shared network. $s \rightarrow s'$ stays admissible for any single agent, but $L(s \rightarrow s')$ rises with occupancy, Wardrop-equilibrium-style. The edge lightest when empty can become heavier once crowded.
- **Result:** This generalizes U31's contention to many agents and continuous congestion — the formal model for swarm-navigability claims elsewhere in the author's corpus. Without a congestion term in per-agent F_a , stable equilibrium holds only under assumptions already characterized in routing-games literature. **Verdict:** identifies Wardrop-equilibrium theory as the formalism AFI's large- N swarm claims should reconcile with.
- **Falsifiability:** A large- N swarm-efficiency claim failing to match Wardrop-equilibrium predictions, unexplained, indicates per-agent F_a is missing this congestion term.

Figure 57

Universe 37 — The Rush Hour

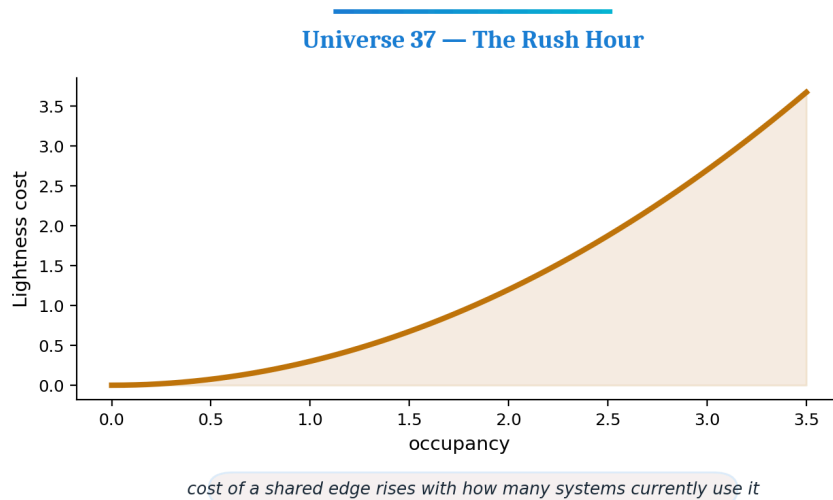


Figure 57

Note. Congestion — paths that grow heavier as more systems use them — is a candidate missing term in the current Lightness functional. The figure illustrates the congestion effect qualitatively; it does not fit a specific congestion function to data. **Falsifiability:** An AFI-style per-agent Lightness prediction that ignores occupancy should mispredict route choice under load; if it matches

routing-game equilibrium predictions (Wardrop) without a congestion term, the term is not needed here.

U38. The Slow Fuse (delayed, initially invisible consequences)

- **Setup:** A transition at step t silently alters the admissible set at step $t+k$, for large k , with no observable signal at intermediate steps. This is stronger than U16's hidden-but-present paths, since here the consequence hasn't manifested at all yet.
 - **Result:** Every horizon term, $L(\pi|s,a,H)$, implicitly assumes H is long enough to matter. U38 shows this can fail silently: no fixed H catches a consequence at $t+k$ for $k>H$. The theory has no principled way to set H beyond its own warning against post-hoc fitting. **Verdict: the sharpest concrete illustration in the lab of the parameter-identifiability limitation (Section 19).**
 - **Falsifiability:** Any fixed- H model mispredicting a delayed consequence for some $k>H$ — guaranteed to exist for any fixed H — falsifies that fixed-horizon instantiation.
-

U39. The Self-Improving Observer (perception that upgrades itself through action)

- **Setup:** The agent has perception-upgrading actions that reduce $D_a(s)$ persistently — unlike U36's decaying gains — for a whole class of future states, at a cost with diminishing returns. Dynamics run on two levels: object-level transitions, plus a meta-level trajectory of falling $D_a(s)$.
 - **Result:** This predicts measurably increasing $F_a(s)$ at matched real Passibility over time, because Distortion fell — not because the world grew more navigable. That's distinct from ordinary learning-to-act. Performance improving with no drop in $D_a(s)$ would be evidence against reading that as "Freedom Intelligence." **Verdict: gives Thesis 5 its own dedicated, dissociable prediction — the extension's most direct addition.**
 - **Falsifiability:** Improving performance with no measured drop in $D_a(s)$ falsifies a "Freedom Intelligence" reading, pointing instead to ordinary fixed-perception learning.
-

Figure 58

Universe 39 — The Self-Improving Observer (Thesis 5's test)

Universe 39 — The Self-Improving Observer

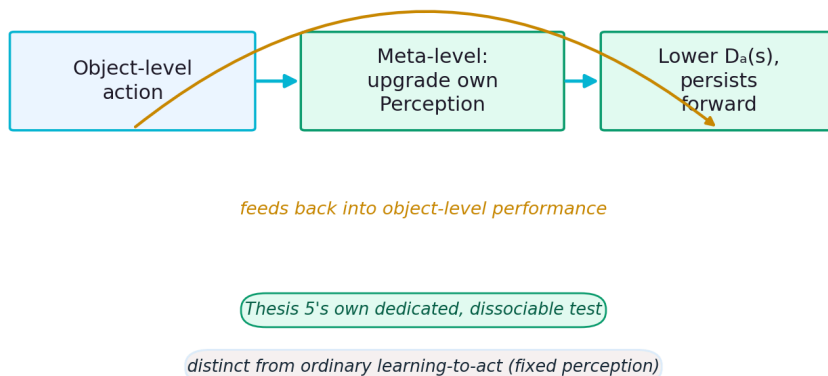


Figure 58

Note. A system whose own actions upgrade its Perception is the dedicated test case for Thesis 5 (“Freedom Intelligence”), distinguishing perception-improving intelligence from ordinary object-level learning. The figure shows the architecture of the test, not a completed run of it; no result is reported here. Falsifiability: Improving object-level performance without a corresponding measured drop in $D_a(s)$ would be evidence against reading that improvement as Thesis 5’s “Freedom Intelligence” specifically, as opposed to ordinary reinforcement learning under fixed perceptual limits — the two are dissociable and this is the dissociating test.

U40. The Cliff Edge (rare, near-total, irreversible collapse)

- **Setup:** As in U20–U21, plus catastrophic transitions: low-probability, admissible, each collapsing *Reach(s)* to a small residual set. Heavy-tailed. The observer systematically under-weights the tail — a documented general feature, not AFI-specific.
- **Result:** An expected-value Lightness reading — minimize $E[L]$ — can rate a path with a small catastrophic-collapse probability as lighter than a path avoiding it entirely. That’s the tail-risk-blind failure mode seen in finance and safety engineering, one the Falsification Matrix’s 22 criteria don’t name. **Verdict: proposes a twenty-third candidate criterion — tail-risk sensitivity.**
- **Falsifiability:** An $E[L]$ -minimizing functional selecting a catastrophic-tail path over a strictly safer, comparably-costed alternative, absent an explicit tail-risk term, confirms this gap.

Figure 59

Universe 40 — The Cliff Edge

Universe 40 — The Cliff Edge

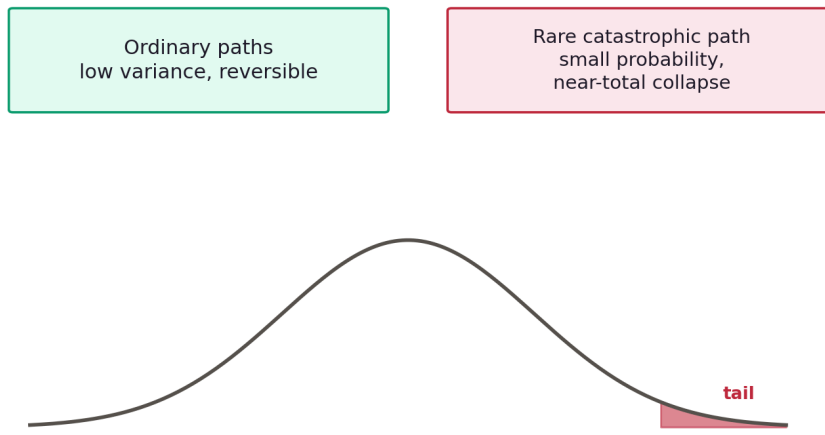


Figure 59

Note. A rare, near-total, irreversible collapse is a tail-risk stress test for whether Lightness needs an explicit catastrophe-weighting term beyond ordinary expected cost. The figure illustrates one collapse scenario at one risk level; it does not establish a general tail-risk threshold. Falsifiability: A Lightness functional that already weights catastrophic outcomes more heavily than raw expected value would predict avoidance of the cliff-edge path even at favorable ordinary-case cost; observing indifference to the tail risk under such a functional would show the proposed criterion adds nothing beyond standard expected-value reasoning.

Summary of the extension

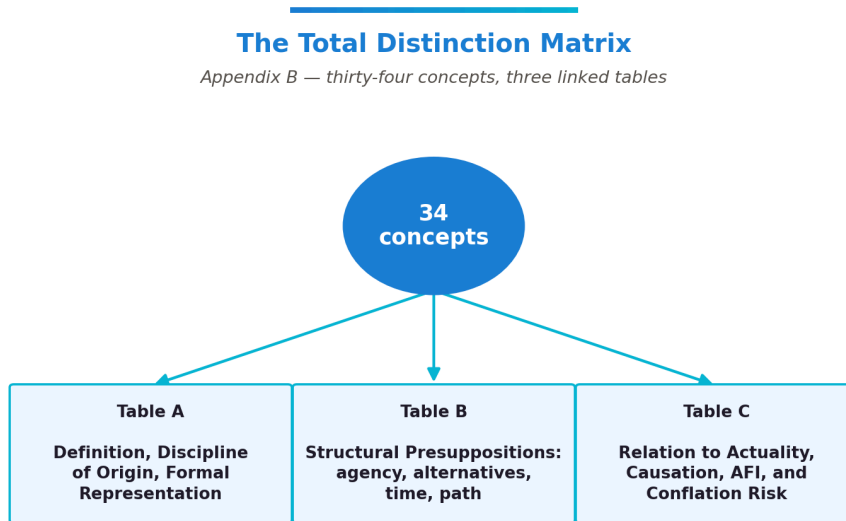
- *The ten span multi-agent contention (U31), cooperative perception-sharing (U32), adversarial distortion (U33), depletable resources (U34), hierarchical spaces (U35), decaying memory (U36), congestion (U37), delayed consequences (U38), self-improving perception / Thesis 5 (U39), and tail-risk collapse (U40).*
- *U39 gives Thesis 5 its own dedicated, falsifiable signature — a genuine addition, not a restatement of Theses 1–2.*
- *U34, U35, and U37 connect the lab to AFI’s own application domains — smart buildings, swarm robotics, shared spaces — each flagging an unaddressed formal gap. U38 and U40 sharpen the paper’s two most serious acknowledged limitations: parameter identifiability and the missing tail-risk criterion.*
- *None of the ten overturns Theses 1 or 2; each narrows what a complete formal theory of Passibility and Lightness still has to say.*

Appendix B. Total Distinction Matrix (Definition Registry)

Thirty-four core concepts, split into three linked tables sharing row numbers and concept names. AFI’s own glossary already treats several as technical terms — *Passibility*, *accessibility*, *realization*, *perception*, *distortion*, *lightness* — used as given in the “Relation to AFI” column rather than re-derived here.

Figure 60

The Total Distinction Matrix



one row number and concept name is shared across all three tables

Figure 60

Note. One row number and concept name runs through all three tables — Definition/Discipline/Formal Representation, Structural Presuppositions, and Relation to Actuality/Causation/AFI/Conflation Risk — so a single concept's full profile can be read across the three by its row alone.

Table A — Definition, Discipline, Formal Representation

#	Concept	Definition	Discipline of Origin	Formal Representation
1	cause	A factor/event whose occurrence figures in producing another (the effect); the relatum picked out as producer/explainer within a causal relation.	Metaphysics of causation	Theory-relative; e.g. counterfactual: c causes e iff $\neg c \rightarrow \neg e$ (roughly, Lewis)
2	causal relation	The relation itself that holds between cause and effect, however underwritten (dependence, production,	Metaphysics of causation	Varies: nomic subsumption $L \wedge c=e$ (Hempel); manipulability — e depends on c iff some intervention on c changes e (Woodward)

#	Concept	Definition subsumption, manipulability).	Discipline of Origin	Formal Representation
3	causal production	The actual generation/transmission of causal influence via a real physical process (e.g. transfer of a conserved quantity), as opposed to mere counterfactual/statistical dependence.	Philosophy of science — causal process theory (Salmon, Dowe)	Dowe's CQT: a causal process = a world-line of an object possessing a conserved quantity, transmitted along it
4	causal power	An intrinsic dispositional property of a particular that inclines it toward producing a characteristic effect when appropriately stimulated/paired.	Dispositional metaphysics / powers ontology (Cartwright; Mumford & Anjum)	No consensus formalism; Mumford & Anjum model manifestation as a vector-sum ("mutual manifestation") of composing powers
5	mechanism	An organized system of entities and their activities, productively arranged so as to generate a regular phenomenon from a start/setup to a finish/termination condition.	Philosophy of science — new mechanism (Machamer, Darden & Craver 2000; Glennan)	No standard equation; schema: setup → [organized entities+activities] → termination
6	law	A universal (or statistical) regularity governing how states/properties relate or evolve; either the best systematization of actual regularities (Humean) or a governing necessitation relation among universals (Armstrong).	Philosophy of science / physics	$\forall x(Fx \rightarrow Gx)$, or a governing equation ($F=ma$); governing view: $N(F,G)$
7	constraint	A restriction excluding certain states/values/transitions from the admissible set, without specifying which admissible option is produced.	Optimization / control theory / dynamical systems	$g(x) \leq 0$, $h(x)=0$; or $x(t) \in K$ for all t (viability constraint set)

#	Concept	Definition	Discipline of Origin	Formal Representation
8	ground	The non-causal, constitutive "in virtue of" relation whereby some fact(s) determine/make-true another by constituting what it consists in.	Analytic metaphysics — grounding theory (Fine, Schaffer, Rosen)	$\Gamma < \phi$ (" Γ grounds ϕ "); hyperintensional, asymmetric, typically transitive
9	condition	A necessary and/or sufficient requirement for a fact/event; a purely logical/modal relation with no built-in commitment to production or constitution.	Logic / general philosophy	Necessary: $E \rightarrow C$; sufficient: $C \rightarrow E$; jointly: $C \leftrightarrow E$
10	possibility	Disciplinary-plural : logical (non-contradictory), metaphysical (true in some possible world), physical/nomological (consistent with actual laws), epistemic (consistent with a knower's information) — these nest (logical $\supseteq$ metaphysical $\supseteq$ physical).	Modal logic / modal metaphysics / epistemology	$\Diamond p$ true at w iff $\exists w'(wRw' \wedge p$ true at $w')$, relative to frame $\langle W, R \rangle$
11	potentiality	The capacity of a particular thing, in virtue of its nature, to become or do something other than what it currently is/does given appropriate conditions; classically <i>dunamis</i> vs. actuality (<i>energeia</i>); revived (Vetter) as graded, iterable modal property.	Aristotelian metaphysics; Vetter's dispositionalist modality	No classical formalism; Vetter: $Pot(x, \text{to } \phi)$ admits degrees and iteration
12	disposition	A property ascribed via tendency to produce a characteristic manifestation given an appropriate	Dispositional metaphysics (Ryle; conditional/power analyses)	x disposed to M when $S = \text{def } (x \text{ is } S) \Box \rightarrow (x \text{ } M\text{'s})$, refined against finks/masks (Lewis)

#	Concept	Definition	Discipline of Origin	Formal Representation
		trigger/stimulus (e.g. fragility).		
13	affordance	A relational property of an environment specifying what it offers for a particular organism's possible action, given that organism's action-capacities.	Ecological psychology (Gibson 1979)	Informal: $Aff(E, O, \varphi) \text{ — } E\text{'s properties, measured relative to } O\text{'s capacities, permit action } \varphi$
14	transition	A single change from one state to another within a state space; the atomic unit of a formal system's evolution.	Automata theory / dynamical systems	$\delta: S \times \Sigma \rightarrow S$ (DFA); or transition relation $s \rightarrow s'$
15	path	An ordered sequence of states connected by transitions from a start to an end state, specified structurally/topologically, without necessarily specifying timing.	Graph theory	Sequence $v_0, v_1, \dots, v_n$ with $(v_i, v_{i+1}) \in E$
16	trajectory	A path plus its time-parameterization: the specific, rate-indexed sequence of states a system occupies over an interval.	Dynamical systems / physics / control theory	$x: [t_0, t_f] \rightarrow S$, typically the solution of $\dot{x} = f(x, u, t)$
17	Passibility	AFI's coined primitive: the structured, graded property of a space (not an agent) of having path-availability at all — the pre-selective substrate of open routes through Logic/Relationships/Physics that causal production runs on.	AFI-specific	Candidate: $Passibility(x) > 0$ iff $\exists \geq 1$ admissible path from x under $Logic \wedge Relationships \wedge Physics$
18	path availability	AFI's operational/local notion: whether a specific given path from a specific state to a specific target is open/unblocked	AFI-specific	$PathAvail(s, s', t) = 1$ iff $\exists$ admissible transition sequence $s = s_0 \rightarrow \dots \rightarrow s_n = s'$ consistent with $Logic \wedge Relationships \wedge Physics$ at t

#	Concept	Definition	Discipline of Origin	Formal Representation
		at a given time, without asserting it will be taken.		
19	accessibility	Two unrelated formal objects share this word: (a) modal logic — the accessibility relation R between possible worlds; (b) informal/AFI usage — whether a possibility currently has a realizable path from the system's actual state.	(a) Modal logic (Kripke semantics); (b) informal/engineering, AFI-specific	(a) $\langle W, R \rangle$, wRw' ; (b) no canonical formalism — closest is $\text{Reach}(s)$
20	reachability	A state x_1 is reachable at t_1 if $\exists$ finite t_0 and admissible input $u(t)$ transferring the system from origin at t_0 to x_1 — a forward-time notion.	Control theory (linear/nonlinear systems); graph theory	$x_1 \in \text{Reach}(x_0)$ iff $\exists$ finite t , admissible $u(\cdot)$ with $x(t_0)=x_0$, $x(t)=x_1$
21	controllability	A state x_0 is controllable at t_0 if $\exists$ finite t_1 and admissible input driving the system from x_0 to the origin — the backward-time complement of reachability (Kalman).	Control theory (Kalman)	$\text{rank}[B, AB, \dots, A^{n-1}B]=n$ (Kalman rank condition, LTI case)
22	viability	A state is viable if $\exists \geq 1$ evolution from it that stays within a specified constraint set K for all future time.	Viability theory (Aubin)	x_0 viable in K iff $\exists$ evolution $x(\cdot)$, $x(t) \in K \forall t \geq t_0$; viability kernel $\text{Viab}(K)$
23	navigability	The practical, procedural ease/tractability of finding and traversing an already-available path, as opposed to that path's bare existence.	Network science (Kleinberg); robotics/path-planning; GIS	Closest standard: a network is navigable if decentralized greedy search finds a target in $\text{polylog}(N)$ steps
24	actualization	The process/event by which a potentiality (or possibility) becomes actual (Aristotle's <i>energeia</i>).	Aristotelian/process metaphysics	No standard symbolism; informally $\text{Actualize}(\pi, t)$

#	Concept	Definition	Discipline of Origin	Formal Representation
25	realization	Two unrelated senses: (1) philosophy of mind — the synchronic relation between a higher-level property and its physical realizer (multiple realizability); (2) ordinary/AFI usage — a possibility becoming physically real over time.	(1) Philosophy of mind; (2) ordinary language/AFI-specific	(1) M realized by P at t ; (2) informally $\text{Realize}(\pi)$
26	probability	A function P satisfying the Kolmogorov axioms, substantively interpreted at least three incompatible ways: frequentist, Bayesian/credence, propensity.	Probability theory / philosophy of probability	Kolmogorov: $P(\Omega)=1$, $P \geq 0$, countable additivity
27	selection	Multiple senses: (a) natural selection; (b) decision-theoretic choice; (c) statistics/ML model selection; (d) generic picking.	(a) Evolutionary biology; (b) decision theory; (c) statistics/ML; (d) generic/AFI	(a) $\Delta p = spq$; (b)–(d) choice function $C(A) \subseteq A$
28	optimality	The property of a candidate solution of minimizing/maximizing a specified objective function subject to constraints, evaluated against an explicit external criterion.	Optimization theory / operations research	x^* optimal iff $f(x^*) \leq f(x) \quad \forall$ feasible x
29	lightness	AFI's technical coinage: the relative cost/ease of a candidate path as weighted by a system's own perception, situation, and the path itself.	AFI-specific	No standard formalization offered; AFI prefers multiplicative/ratio forms over additive cost sums
30	perception	(a) Philosophy of mind/psychology — process of acquiring environmental	(a) Philosophy of mind/psychology; (b) AFI-specific broadening	(a) Bayesian perception-as-inference; (b) AFI: $P_C = \prod_i P_i \wedge \beta_i$

#	Concept	Definition	Discipline of Origin	Formal Representation
		information via sensory systems; (b) AFI — de-anthropomorphized operational sensitivity/access to relevant state variables and possible paths.		
31	world model	A system's internally maintained representation of environment structure/dynamics, used to plan/evaluate candidate paths.	Model-based RL/AI (Ha & Schmidhuber 2018); POMDP literature	$T(s s,a): S \times A \rightarrow \Delta(S)$
32	belief state	In a POMDP, the sufficient statistic for optimal action under partial observability: a distribution over the true unobserved state given history.	Bayesian decision theory / POMDP planning	$b(s)=P(s h_t)$; Bayesian filter update
33	distortion	(a) Information theory — deviation of received signal from true source (rate-distortion theory); (b) AFI — omission, error, latency, false relationships, blockage, or uncertainty reducing access to reality/possibility.	(a) Information theory; (b) AFI-specific	(a) $D=E[d(X,\hat{X})]$; (b) AFI: $D_C = \prod_i D_i^{\delta_i}$
34	Freedom (AFI)	AFI's core primitive: $F=(\text{Perception/Distortion})^{(\alpha^\infty)}$ — the ratio of a system's veridical access to relevant reality/paths over the distortion degrading it. Not a synonym for liberty or free will.	AFI-specific	$F = (P_C/D_C)^{(\alpha^\infty)}$

Table B — Structural Presuppositions

#	Concept	Observer-Dependence	Requires Agency?	Requires Alternatives?	Requires Time?	Requires Path?
1	cause	Theory-dependent	No	Yes (implicit)	Yes	No

#	Concept	Observer-Dependence	Requires Agency?	Requires Alternatives?	Requires Time?	Requires Path?
2	causal relation	Theory-dependent	No	Yes	Yes	No
3	causal production	Observer-independent	No	Not strictly	Yes	Yes
4	causal power	Observer-independent	No	Depends	Yes for manifestation	No
5	mechanism	Mostly independent	No	No	Yes	Yes
6	law	Independent (realist) / mildly relative (Humean)	No	Depends	Not strictly	No
7	constraint	Depends on kind	Depends	Yes typically	No	No
8	ground	Observer-independent	No	No	No (synchronic)	No
9	condition	Independent (logical)	No	Depends	No	No
10	possibility	Independent / relative for epistemic sense	No	Yes	No	No
11	potentiality	Independent	No	Yes implicitly	Yes	No
12	disposition	Independent	No	Depends	Yes for manifestation	No
13	affordance	Relational to organism-type, not observer-belief-relative	Yes	Not intrinsically	No	No
14	transition	Independent (fixed model)	No	Depends	Yes minimally	No
15	path	Independent (fixed graph)	No	Not intrinsically	No	Yes
16	trajectory	Independent given fixed dynamics	No	Not intrinsically	Yes	Yes
17	Passibility	Claimed observer-independent by AFI	No (AFI's framing)	Yes	Yes implicitly	Yes
18	path availability	Claimed observer-independent	No	Yes	Yes	Yes
19	accessibility	(a) independent given frame; (b)	No	Yes	(a) No; (b) Yes	(a) No; (b) Yes

#	Concept	Observer-Dependence state-relative	Requires Agency?	Requires Alternatives ?	Requires Time?	Requires Path?
20	reachability	Independent given model	No	Yes	Yes	Yes
21	controllability	Independent given model	No	Yes	Yes	Yes
22	viability	Independent given model	No	Yes	Yes (unbounded)	Yes
23	navigability	Procedure-relative	Yes in effect	Not intrinsically	Yes	Yes
24	actualization	Independent	No	Yes	Yes	Not intrinsically
25	realization	(1) independent; (2) independent	No	Yes	(1) No; (2) Yes	No / not intrinsic
26	probability	Frequentist/propensity independent; Bayesian relative	No / presupposes believer	Yes	Not intrinsically	No
27	selection	(a) independent; (b)-(d) system-relative	(a) No; (b)-(d) depends	Yes definitionally	Yes typically	No
28	optimality	Model-relative	No	Yes	No	No
29	lightness	Essentially system-relative	Depends	Yes	Yes	Yes
30	perception	Relative to perceiving system	Depends	Not intrinsically for (a)	Yes	No
31	world model	System-relative	Depends	Not intrinsically	Yes	Not intrinsically
32	belief state	System-relative	Depends	Yes essentially	Yes	No
33	distortion	(a) measure-relative; (b) system-relative	No	Not intrinsically	Not intrinsically	No
34	Freedom (AFI)	System-relative function over claimed observer-independent substrate	Depends/No	Yes	Yes implicitly	Yes

Table C — Relation to Actuality, Causation, AFI, and Conflation Risk

#	Concept	Relation to Actuality	Relation to Causation	Relation to AFI	Risk of Conflation
1	cause	Always actual — feature of what happened	Relatum, not whole relation	Downstream description of a realized path	causal relation
2	causal relation	Relates actual events/regularities	This IS causation	Post-hoc description of a realized path, not fundamental for AFI	causal production
3	causal production	Predicated only of what occurred	Thick, process-based species of causation	Close cousin of AFI's realization but Physics-layer-only	mechanism
4	causal power	Can be actual while manifestation is potential	On powers theories, grounds causal relations	Closest existing relative to "why paths get taken," but AFI grounds production in path-availability/lightness	disposition
5	mechanism	Type-structure instantiated in token runs	Productive causal structure	Under FLRP, the Relationships +Physics-layer implementation of an already-selected path	law
6	law	Holds across actual (and nomologically possible) worlds	Governs vs. summarizes	FLRP's Logic/Physics layers supply lawlike constraints; AFI reinterprets what selects among paths law leaves open	constraint
7	constraint	Standing boundary all actual states satisfy	Non-productive — explains exclusion	Carves admissible region prior to lightness-selection	distortion (highest-stakes AFI-internal conflation)
8	ground	Both relata actual	NOT causation — synchronic vs. diachronic	Not an AFI term; contrast-class for whether Possibility "grounds" or merely "conditions" realization	condition

#	Concept	Relation to Actuality	Relation to Causation	Relation to AFI	Risk of Conflation
9	condition	Conditional/modal claims	Weaker than causal production	"Path availability necessary for realization" (weak) vs. "grounds" it (strong) — AFI must pick one	ground
10	possibility	Actuality a distinguished member of the possible	Presupposed by, not itself, causal production	AFI's T1 needs a reachable path (Passibility), stricter than bare physical possibility	probability (AFI's own flagged error)
11	potentiality	Contrast pair with actuality	Aristotle: efficient causation IS actualizing a potential	AFI's Passibility is space-level/agent-neutral; potentiality is thing-intrinsic — avoid smuggling substance ontology	disposition
12	disposition	Can be possessed unmanifested	Categorical/ontological basis of causal powers on some views	Path availability resembles "space's disposition to permit passage" but AFI frames it structurally	causal power
13	affordance	Can be real & unperceived (Gibson explicit)	Precondition/enabler, not production itself	Closest existing academic concept to Passibility — narrower on organism-relativity and action-specificity	Passibility — must state explicitly whether Passibility strips organism-relativity or generalizes it
14	transition	Type can exist unexercised; token actual when undergone	Often formal/stipulated; physical tokens are causal-production events	Elementary unit composing a path	path
15	path	Can exist as structural possibility independent of traversal	Candidate route for production; bare existence produces nothing	Central AFI object	trajectory
16	trajectory	Denotes actual or candidate/counterfactual —	Realized = causal process instance;	AFI's realized outcome best formalized as trajectory	path; probability

#	Concept	Relation to Actuality must disambiguate	Relation to Causation candidate = only causally possible	Relation to AFI (cost/time involved)	Risk of Conflation
17	Passibility	Obtains for paths never taken — prior to actuality	AFI's central claim: causation runs ON Passibility	This IS the AFI term	affordance; accessibility
18	path availability	Can be available and never actualized	Necessary not sufficient for production	Local instantiation of Passibility for one candidate path	Passibility; reachability
19	accessibility	Actual world self-accessible in most modal systems	(a) not causal; (b) precondition for production	AFI leans on sense (b) — must flag this is NOT the Kripke relation	reachability
20	reachability	Modal property of the model, independent of actual drive-to	Specifies causally-achievable outcome space	Best formal proxy for AFI's "path availability," Physics-layer-restricted	controllability (documented even-within-field error)
21	controllability	Same modal status as reachability	Specifies causally-steerable-to-target initial states	Less directly relevant — AFI is forward-looking	reachability; viability
22	viability	Modal property (∃ viable evolution)	Concerns sustained production, not single-effect production	Sustained Passibility across indefinite future — clarify vs. single-transition reachability	controllability; navigability
23	navigability	Evaluated independent of actual traversal	Downstream of reachability/viability — presupposes path exists, asks if findable	Plausible formal anchor for "lightest perceived path"	lightness — findable ≠ cheapest, conflating lets the Law absorb pure search-tractability facts
24	actualization	IS the transition whose terminus defines "actuality"	Aristotle identifies efficient causation with actualization	AFI's "realization" is its actualization-equivalent — must state if synonyms	realization (highest-priority disambiguation)
25	realization	(1) simultaneous with realizer; (2) terminus of possibility-to-actual transition	(1) contested (causal exclusion); (2) tied to production's physical-layer terminus	AFI uses sense (2) explicitly — must flag as unrelated to philosophy-of-mind sense	actualization; philosophy-of-mind homonym (highest-risk homonym in table)

#	Concept	Relation to Actuality	Relation to Causation	Relation to AFI	Risk of Conflation
26	probability	Ranges over both actual (posterior) and possible (prior)	Probability-raising is one contested causal-theory component	AFI: probability applies only after possibility/acceptability established, never defines the possibility set	possibility (flagged central error)
27	selection	Converts previously-possible option into actual	Sometimes causal-explanatory, doesn't guarantee realization	AFI's T2 is fundamentally a selection claim generalized to non-deliberative systems	optimality
28	optimality	Modal/comparative, evaluable whether or not realized	Not itself causal — external normative ranking	Contrasts with lightness: AFI selects perception-relative lightest, not objectively optimal	lightness (load-bearing distinction for the whole framework)
29	lightness	Evaluated over not-yet-actual candidates	Drives/selects which production occurs (T2)	Core AFI evaluative quantity	optimality; navigability
30	perception	Always actual/occurent itself; content can represent merely possible states	Caused by environment, causally influences behavior — not itself production	Numerator of $F=(P/D)^{\alpha}$	world model; belief state
31	world model	Itself always actual; content represents possible/actual dynamics	Represents believed causal structure, not production itself	Adopted infrastructure feeding Perception	belief state
32	belief state	Itself actual; content ranges over which actual state obtains	Updated causally, informs action, not itself production	Formal home for "estimate of where system is"	world model; perception
33	distortion	Presupposes a fact-of-the-matter measured against	Arises causally from noise/loss, causally affects path selection	Denominator of $F=(P/D)^{\alpha}$	constraint (single most important AFI-internal distinction)
34	Freedom (AFI)	Characterizes relation to possible paths prior to which is realized	AFI's most theory-committing claim: explanatorily prior to, not inferred from, causal structure	This IS the AFI term	possibility; ordinary "freedom" (liberty/free will) — recommend notational convention

#	Concept	Relation to Actuality	Relation to Causation	Relation to AFI	Risk of Conflation (<i>italic</i> Freedom or F)
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Cross-cutting notes

- **Realization** (row 25) is entirely unrelated to the philosophy-of-mind multiple-realizability relation — flagged here to prevent the paper’s own citations from being misread.
- **Constraint vs. distortion** (rows 7, 33) is the most consequential AFI-internal distinction: constraint is ontic (shrinks real Passibility), distortion is epistemic (shrinks veridical Perception without touching Passibility). Keeping these separate is what makes Thesis 1 non-trivial rather than a tautology.
- **Passibility vs. affordance** (rows 17–18 vs. 13) is the nearest existing academic comparison, differing on organism-relativity and action-specificity.
- **Lightness vs. optimality** (rows 29, 28) is what keeps AFI from being read as classical optimization: AFI selects the perception-relative lightest path, not the externally-optimal one, which is what gives Distortion explanatory work to do.
- **Freedom vs. ordinary “freedom”** (row 34): italicizing/capitalizing the technical term throughout is the recommended convention.

Sources consulted for precision: Wikibooks (Controllability/Observability), Wikipedia (Viability theory), SEP (Metaphysical Grounding), Gibson, *The Ecological Approach to Visual Perception*, ch. 8.

Appendix C. Source Verification Note

All 52 independent sources and 20 AFI-programme corpus entries cited in this paper were verified against their actual, fetched abstracts (not inferred from titles) via live database and publisher-site queries conducted during this research programme, following the source-verification protocol specified in the governing research architecture: each entry’s title, authors, year, venue, publication status, and stable identifier (DOI, arXiv ID, or repository identifier) were confirmed to resolve before inclusion, and each entry’s relevant claim was drawn from the fetched abstract or, where accessible, the full text. A companion structured source ledger (72 entries; CSV format), recording verification method, per-source limitations, and cross-references to the sections of this paper in which each source is used, is maintained as a supplementary research artifact alongside this manuscript.

The Deucalion supercomputer compute-access grant cited in connection with the AFI corpus (Appendix, entry A19) was independently verified via Portugal’s national advanced-computing network public registry (RNCA/AlvLAB, project reference 2025.00020.AIVLAB.DEUCALION, hosted at MACC, Guimarães) — confirming that the compute allocation is real and independently listed. This confirms access to computation only; it does not independently confirm any specific numeric result reported using that access, and no such independent confirmation is claimed anywhere in this paper.

Appendix D. Final Audit and Standalone Statements

This appendix collects four items requested as distinct, self-contained deliverables alongside the paper proper:

- *The Final Intellectual Test.*
- *A one-page statement of why the Cause of All bears on the Law of All.*
- *A formal statement of the strongest surviving version of “Freedom causes causation.”*
- *A formal statement of the evidence bar the Law of Freedom would have to clear before it could responsibly be called the Law of All.*

Each restates, in compressed and standalone form, conclusions argued at length elsewhere in the paper. Every claim here is already defended in the main text.

D.1 The Final Intellectual Test

Question 1 — Can causal production occur without admissible possibility?

The answer is no. It holds across all eight formal models this paper develops (Section 11), and across all thirty counterfactual universes in Appendix A, without exception.

- *In every model, an actual transition is by construction a member of the admissible or reachable set at the state it departs from.*
- *“Admissible” is defined as exactly the set actual transitions are drawn from. So no model allows a transition from outside that set.*
- *This makes the claim true. But as Section 11’s “Smallest Common Formal Core” states explicitly, it is true at Layer 1 (logical/definitional). It is not an empirical discovery.*

The substantive question is not whether this holds, but what it costs.

- *In a fully deterministic, single-successor domain (Universe 2), the claim survives but becomes vacuous. The admissible set has exactly one member, so nothing was ruled out.*
- *The honest answer: causal production without admissible possibility is incoherent, given how “admissible” is defined, rather than merely rare. That is a genuine, if modest, result.*
- *It is a different result from showing that non-degenerate possibility-availability does any independent explanatory work.*

Question 2 — Does that make possibility a necessary condition only, or constitutive of causation?

The paper’s considered answer, argued in Section 5 and tested against thirteen rival positions in Section 13.2, is clear on one point and cautious on the other.

- *Necessary-condition status is secure.*
- *Constitutive status is not established, and the paper does not claim it.*

Every actual transition requires membership in an admissible set (Question 1). That secures possibility as necessary.

- *Necessity alone does not make possibility-availability part of what causation is.*
- *It only makes possibility-availability a precondition causation requires.*

- *The same gap separates “oxygen is necessary for combustion” from “oxygen is constitutive of combustion.”*

Section 13’s arrow-by-arrow audit found the strong constitutive reading — Thesis 1’s ambitious sense — unsupported by any arrow in the cause-to-law chain.

- *Supporting that reading would require an additional, independently argued metaphysical commitment: that causal production is nothing over and above transition-within-a-structured-field.*
- *Competing accounts reject that commitment, Lewtas’s powers-primitivism chief among them.*

This paper leaves that further-fact dispute open. It isolates the dispute as the one additional premise the strong reading of Thesis 1 needs and does not yet have.

Question 3 — Does structured transition-field causation necessarily create a Navigability problem?

Yes, conditionally. The condition is load-bearing.

- *Wherever the admissible/reachable set at a state has more than one member — outside the deterministic, single-successor domains excluded by Question 1’s caveat — something governs which successor is reached.*
- *A system or process that ends up at one particular successor rather than another admissible one has, by construction, been governed by something that selected among the available continuations.*

Section 13 calls this the Navigability problem. It argues the problem is forced, not chosen.

- *Once branching admissibility is granted, some selection principle exists, whether or not any theory has yet named it correctly.*
- *What is not forced is which principle fills that role.*

The paper is explicit (Section 13.1, Section 16) that this step establishes the existence of a selection-explanandum. It does not establish the correctness of any particular selector.

Question 4 — Is the Law of Freedom the best solution to that problem?

Not established. The paper does not claim otherwise.

- *Section 13.2’s comparison against thirteen named rivals, and Section 16’s twenty-two-criterion Falsification Matrix, together show the Law of Freedom to be a coherent, non-tautological, falsifiable-in-principle candidate.*
- *It has not been run against its two closest and strongest rivals — expected-free-energy-minimizing active inference and empowerment maximization — in any experiment designed to dissociate them (Falsification Matrix criterion 17, scored Fail).*

“Best” is therefore not answerable from the evidence this paper has assembled. The accurate, and considerably more modest, description the evidence supports is this: a live, structurally distinct candidate that has not yet lost a direct comparison it has also not yet won.

D.2 One-page statement: why does the Cause of All have anything to do with the Law of All?

The connection is not definitional, and it does not follow by simple substitution. Section 13’s central finding is exactly that Thesis 1 does not logically entail Thesis 2.

The connection that does survive scrutiny is architectural. It runs through the Navigability problem established in D.1's answer to Question 3.

Freedom — structured, branching possibility — is genuinely what makes causal production possible. This is the necessary-condition reading of Thesis 1, which this paper does defend.

- *If that reading holds, every act of causal production in a non-degenerate (branching) domain is an act of selection among admissible continuations, without further argument.*
- *That selection is not optional or occasional. It is built into what it means for there to be more than one admissible successor, with only one becoming actual.*

A theory that stops at Thesis 1 — Freedom causes causation — has therefore already opened a second question, whether it intends to or not. It cannot decline to answer: selected *how*? By what principle does the actual continuation get picked out from the admissible set?

This is the Navigability problem. It is not an optional addendum to a Cause-of-All thesis. It is what a Cause-of-All thesis becomes the moment it is stated for a domain with more than one available continuation.

A Law of All, on this account, is a candidate answer to that forced question. It is a general claim about which principle governs the selection, across domains, of one admissible continuation over the others.

- *The Cause of All and the Law of All are related without being the same claim.*
- *Deriving the second from the first is not the same operation as restating the first more precisely.*
- *The Cause of All (Thesis 1) explains why there is a space of live alternatives to select among.*
- *A Law of All (the ambition behind Thesis 2) would explain why one alternative rather than another gets selected, across that entire space.*

Section 13 shows that the first claim creates real logical room for the second. A Navigability problem does follow once branching admissibility is granted.

- *But Section 13 cannot by itself say which law fills that room correctly.*
- *That is a separate, comparative, empirical matter.*
- *Section 13.2's comparison against thirteen rivals, together with the Falsification Matrix (Section 16), is this paper's attempt to take that separate matter seriously rather than assume its answer.*

The relationship, compressed to one line: **The Cause of All opens the Navigability problem. A Law of All is whatever, if anything, actually solves it. Opening a problem is not the same accomplishment as solving it.**

D.3 Formal statement: the strongest survivable version of "Freedom causes causation"

Stated precisely enough to be evaluated, and no more strongly than the paper's own findings support:

FC (survivable form). For any system or process s operating in a domain with a non-degenerate admissible/reachable set $R(s) = \{s' : s \rightarrow s' \text{ is admissible}\}$ where $|R(s)| > 1$:

- Every actual transition $s \rightarrow s^*$ satisfies $s^* \in R(s)$.
- The structure of $R(s)$ — which continuations are available at all — is causally and explanatorily prior to any account of which particular s^* is realized.
- Freedom, defined as this structured availability of admissible continuation, is therefore a necessary condition for causal production in every non-degenerate domain.
- The space it makes available is what any downstream account of production (a Law of Freedom or a rival) must operate within, not around.

The paper's own audit (Section 13, arrows 3→4 and 4→5) found three things unsupported. FC explicitly does **not** claim any of them:

- That Freedom is sufficient for causal production. (Universe 2's determinism shows necessity without any interesting sufficiency claim.)
- That Freedom is metaphysically prior to, rather than merely co-instantiated with, the laws and constraints that generate $R(s)$ in the first place. (This further-fact question is left open, per Question 2 above.)
- That any particular selection principle — the Law of Freedom included — is thereby validated.

FC's scope is narrower than these. It is a claim about the necessity and explanatory priority of possibility-structure. It is deliberately silent on which law, if any, governs selection within that structure.

Any stronger reading of Thesis 1 that this paper's corpus of prior AFI publications may have asserted should be read as superseded, for purposes of this research programme, by FC as stated here.

D.4 Formal statement: the evidence bar for calling the Law of Freedom the Law of All

The Law of Freedom, $F = (P/D)^\alpha$, would be responsibly callable "the Law of All" only once all five of the following conditions hold, with no exceptions. Each condition closes a distinct way the current evidence falls short:

1. **Parameter identifiability is resolved.** α , and the aggregation weights β_i, δ_j in $P_C = \prod_i P_i^{\beta_i}$ and $D_C = \prod_j D_j^{\delta_j}$, must be estimated by a pre-registered, independent procedure. That procedure must be fixed *before* the behavior it is meant to explain is observed, rather than fit post hoc to whatever data is available. (Falsification Matrix criterion 15, currently Fail; Section 19; Section 20's first priority.)
2. **A direct, dissociating comparison against its two strongest rivals has been run and won.** Specifically, an experiment must be designed so that expected-free-energy minimization, empowerment maximization, and $F = (P/D)^\alpha$ make *different* predictions in the same setting, not merely different post hoc explanations of the same result. The Law of Freedom's prediction must be the one confirmed. (Falsification Matrix criterion 17, currently Fail; Section 17, testbeds (e), (f), (h); Section 20's second priority.)

3. **The scope restriction to non-degenerate domains is either lifted with argument or made permanent and explicit.** One path: give a principled account of how much of physical reality is non-degenerate at the scales the law claims to cover. The other path: permanently narrow the law's stated domain to exclude deterministic, single-successor regimes, rather than leaving it silently ambiguous about them. (Falsification Matrix criterion 8, currently Fail; Section 19's first limitation.)
4. **The physics-reduction proof is completed, not merely gestured at.** $F = (P/D)^\alpha$ must be shown to reduce to and reproduce the Euler–Lagrange and Hamilton equations in the relevant classical, deterministic limit. Resembling stationary-action principles in form is not enough. Formal resemblance without this derivation is motivational rather than evidential, and the paper explicitly warns against treating it otherwise. (Section 11 Model G; Section 12; Section 19's fifth limitation.)
5. **Cross-domain transfer and predictive advantage are demonstrated prospectively, on data not used to construct or tune the law.** Retrospective fits do not satisfy this bar on their own — including the author's own prior cross-domain law-recovery study and swarm-navigability simulation. They motivate the programme, but they were generated, tuned, and interpreted by the same research programme under test. That is precisely the circularity this paper's prospective-measurement discipline exists to rule out. Independent replication by researchers outside the AFI programme, on pre-registered analysis plans, is required. (Section 19's fourth limitation; Section 20's fourth priority; Appendix C's verification note.)

Until all five conditions are met, this is the accurate description of the Law of Freedom's status:

A specified, falsifiable, currently-unttested candidate law of navigation. General application, and the title Law of All, remain ahead of it.

This paper commits to that description, and asks any reader — including its own author — to hold it to that standard.

Appendix E. Glossary and Quick-Reference Index

E.1 What is AFI?

AFI — Architecture of Freedom Intelligence — is a theoretical framework originated and developed by Gonçalo Melo de Magalhães. It proposes one organizing idea. The availability of possible continuation is the primitive out of which causation, intelligence, and structure are built. AFI names this availability Passibility. AFI states its full position through five theses:

- **Thesis 1 — Causal Freedom.** *Possibilities are prior. They are revealed rather than created. Freedom, in AFI's technical sense, is the structured availability of possible continuation.*
- **Thesis 2 — Law of Freedom.** *Every system follows the lightest path according to its own perception. This is formalized as the candidate ratio $F = (\text{Perception} / \text{Distortion})^\alpha$.*
- **Thesis 3 — FLRP Architecture.** *Reality organizes itself through four layers, in order: Freedom, then Logic, then Relationships, then Physics. Physics is the final, crystallized layer of realized reality.*

- **Thesis 4 — Mutual Dependency.** *Theses 1, 2, and 3 depend on one another. None stands alone.*
- **Thesis 5 — Freedom Intelligence.** *Intelligence is the capacity to perceive better, reduce distortion, reveal more possibilities, and unblock paths — improving a system's own operative Freedom over time.*

This paper is the first systematic attempt to test AFI's central theses. It tests them against the 2025–2026 research literature. It tests them against thirty formally specified toy universes, extended to forty in Appendix A. It tests them against eight independent formal models (Section 11). The paper states plainly what the theory establishes and what it needs next. That discipline is what the framework's own governing methodology calls for.

E.2 Glossary of terms and properties

Each entry states the term's definition. It gives the term's properties. It names the nearest neighboring concept, to fix the boundary between the two. It states whether the term requires agency. It gives the term's structural profile: alternatives, time, path, and observer-dependence. These profiles are drawn from the Total Distinction Matrix (Appendix B) wherever the term has a row there, so each profile is traceable rather than asserted.

- **Passibility.** *The structured availability of possible continuation from a given state. This is AFI's candidate primitive. It is the technical content behind the word "Freedom" throughout this paper.*
 - *Properties: Graded rather than binary. A property of a space or state, not of an agent. Claimed by AFI to be observer-independent. Prior to any selection among continuations.*
 - *Boundary: The term is reserved strictly for the structural fact of open continuations. An agent's felt sense of freedom is a separate matter. The probability of one continuation over another is also separate. The paper keeps both apart from Passibility.*
 - *Requires agency: No. AFI frames it as holding independently of whether any agent is present (Appendix B, row 17).*
 - *Structural profile: Requires alternatives and a path. Requires time only implicitly. Claimed observer-independent.*
- **Freedom (AFI technical sense).** *Passibility, formalized as the ratio $F = (\text{Perception} / \text{Distortion})^\alpha$.*
 - *Properties: System-relative: the same objective situation yields a different Freedom value for different systems. A function rather than a primitive state. Increases with Perception. Decreases with Distortion.*
 - *Boundary: Reserved for this technical sense alone. Political liberty, free will, and moral autonomy are set aside for the duration of the paper, wherever "Freedom" appears unqualified.*
 - *Requires agency: Depends. The ratio is defined for any system with a Perception/Distortion profile, agentive or otherwise. Agency plays no role in the definition itself (Appendix B, row 34).*
 - *Structural profile: Requires alternatives, time (implicitly), and a path. System-relative, over a claimed observer-independent substrate (Passibility).*
- **Perception (P).** *A system's accessible representation of relevant reality and of the paths available to it.*
 - *Properties: Graded across the eight-rung ladder P0–P7 (Section 9). Always actual, or occurrent, itself, even when its content represents merely possible states. Numerator of $F = (P/D)^\alpha$.*
 - *Boundary: Used in a de-anthropomorphized, operational sense: sensitivity or access to relevant state variables. Kept separate from consciousness or subjective experience. Also kept separate from a world model or belief state, which Perception feeds rather than equals.*
 - *Requires agency: Depends on the sense used. Ranges from passive sensitivity (P0, no representation required) to rich modal representation (P7).*

- *Structural profile: Relative to the perceiving system. Alternatives become relevant only past the passive end of the ladder. Requires time (Appendix B, row 30).*
- **Distortion (D).** *Whatever reduces a system's access to reality or to possibility — omission, error, latency, false relations, blockage, or uncertainty.*
 - *Properties: Epistemic rather than ontic. A property of $\hat{P}_a(s)$, the system's perceived set, rather than of $P(s)$, the real admissible set. Denominator of $F = (P/D)^\alpha$.*
 - *Boundary: Kept sharply apart from Constraint (Appendix B, row 7). Constraint shrinks what is really possible; Distortion shrinks only what is seen. This is the single most consequential distinction internal to AFI, since conflating the two runs Thesis 1 backward.*
 - *Requires agency: No. A measurement or representational gap can exist without an agent (Appendix B, row 33).*
 - *Structural profile: Measure-relative and system-relative. Alternatives, time, and a path are relevant only through what Distortion is a gap in. They are not intrinsic to the gap itself.*
- **Lightness.** *The relative cost of a candidate path, evaluated prospectively from a system's own current epistemic position. It is always assessed forward, before the path is taken.*
 - *Properties: System-relative. Evaluated over not-yet-actual candidates. The horizon-weighted form $L(\pi|s,a,H)$ (Section 14) fixes the horizon independently of the outcome.*
 - *Boundary: Distinguished from Optimality (Appendix B, row 28). AFI selects the perception-relative lightest path, rather than the externally-optimal one. The term is reserved for that relative sense, rather than for "shortest" or "cheapest" judged from outside.*
 - *Requires agency: Depends. The functional is defined system-relatively, so some evaluating system is presupposed. That system need not be a deliberative agent.*
 - *Structural profile: Requires alternatives, time, and a path. Essentially system-relative (Appendix B, row 29).*
- **Navigability.** *The umbrella term for how a system relates to the field of continuations available to it. Its named component dimensions are discoverability, reachability, viability, cost, risk, reversibility, and branching structure.*
 - *Properties: Composite rather than primitive. It decomposes into the named dimensions, rather than being independently defined. Evaluated on its own terms, whether or not a path is actually traversed.*
 - *Boundary: Presupposes a path exists, and asks whether it is findable or usable. This keeps Navigability separate from Possibility itself. Treating "findable" as interchangeable with "cheapest" would let the Law of Freedom silently absorb pure search-tractability facts. So the two terms stay distinct.*
 - *Requires agency: Yes, in effect. The practical, procedural sense of navigability is naturally tied to a system doing the navigating (Appendix B, row 23).*
 - *Structural profile: Procedure-relative. Requires time and a path. Alternatives are not intrinsic to the definition.*
- **Admissible set / $R(s)$.** *The continuations genuinely available from a state s , fixed by the universe's laws. This is the objective backbone every formal model in Section 11 builds on.*
 - *Properties: Fixed by the laws of the universe in question, rather than by any system's perception of them. The one-step structural basis from which Reachability and Viability are built.*
 - *Boundary: What a system perceives as available is a separate object, $\hat{P}_a(s)$, governed by Distortion. The admissible set itself is silent on which member becomes actual.*
 - *Requires agency: No. Defined purely structurally, independent of any observing or acting system.*

- *Structural profile: Alternatives become relevant when more than one continuation exists. Time-indexed. The backbone for path-based notions built on top of it.*
- **Reachability.** *Whether a state s' is connected to s by some admissible path, possibly multi-step.*
 - *Properties: A forward-time, multi-step notion, in contrast to the one-step Admissible set. Independent of the model once the model is fixed.*
 - *Boundary: The forward-time counterpart of Controllability, which is its backward-time complement. This pairing is confused even within the field itself (Appendix B, row 20). Reachability is a claim about connectivity alone, separate from desirability or cost.*
 - *Requires agency: No. A purely structural, modal property of the state space (Appendix B, row 20).*
 - *Structural profile: Requires alternatives, time, and a path. Independent given the model.*
- **Viability.** *Whether a state's forward continuations remain open-ended, rather than terminating in collapse. A state can be reachable while falling short of viable.*
 - *Properties: Concerns sustained, indefinite continuation, rather than a single transition. A state can satisfy Reachability while failing Viability.*
 - *Boundary: Weaker than Reachability alone would suggest, since Reachability is connectivity only. Prior to Navigability: procedural ease is a further question, layered on top of Viability.*
 - *Requires agency: No. Defined over the model's dynamics, independent of any acting system (Appendix B, row 22).*
 - *Structural profile: Requires alternatives and an unbounded time horizon. Requires a path. Independent given the model.*
- **Realization / Actualization.** *What becomes physical: the one member of an admissible set that a system's transition selects.*
 - *Properties: The terminus of a possibility-to-actual transition. AFI's usage is explicitly the ordinary-language, physical-layer sense.*
 - *Boundary: A separate, unrelated concept from the philosophy-of-mind relation of "multiple realizability," between a higher-level property and its physical realizer. This is flagged in Appendix B as the highest-priority disambiguation in the whole registry.*
 - *Requires agency: No. A transition can realize one outcome among several without any agent selecting it, as in a purely physical or stochastic process.*
 - *Structural profile: Requires alternatives. Time-indexing and path-dependence hold on the AFI sense specifically, separate from the philosophy-of-mind sense.*
- **FLRP.** *The architecture by which AFI proposes reality is organized: Freedom, then Logic, then Relationships, then Physics. Physics is the final, crystallized layer.*
 - *Properties: An ordered, four-layer architecture. Each layer constrains what the next can realize. On this proposal, Physics is downstream of Freedom, rather than prior to it.*
 - *Boundary: Logic, Relationships, and Physics each remain genuine constraint layers in their own right. The architecture orders them; it does not dispense with any of them. It is offered as a metaphysical proposal, rather than a physical theory.*
 - *Requires agency: No. Proposed as the architecture of reality generally, rather than specifically of agentic systems.*
 - *Structural profile: Metaphysical / derived-structural (epistemic layer L3–L4, Section 13). The layers are ordered. Empirical demonstration of that ordering, end to end, remains open.*
- **Freedom Intelligence.** *Thesis 5's claim: intelligence is the ongoing improvement of a system's own Perception, and thereby of its own operative Freedom. It has its own dedicated, falsifiable test in the extended Counterfactual Laboratory (Universe 39).*

- *Properties: Meta-level: it improves the system's own Perception. This is distinct from merely object-level improvement, which improves task performance under fixed Perception. Dissociable from ordinary reinforcement learning, by the specific test in Universe 39.*
- *Boundary: Kept separate from raw task performance or ordinary learning-to-act. Whether any existing system has this as a unified closed loop remains an open question, tracked in Section 18.*
- *Requires agency: Depends. Requires a system capable of acting on its own Perception-generating process. This is a stronger requirement than bare agency.*
- *Structural profile: Candidate-law / engineering-design layer (L5–L6). Falsifiable per Figure 58's dissociating test.*
- **The Cause of All.** *The paper's working shorthand for Thesis 1's claim that Freedom is what makes causal production possible.*
 - *Properties: A metaphysical-layer claim (L3) about what grounds causal production generally, rather than about any single causal episode.*
 - *Boundary: Plays a different logical role than Freedom being "a cause" in the ordinary sense of a prior event producing an effect. It is offered as what makes causal production possible at all.*
 - *Requires agency: No. Framed as a general metaphysical claim about causal production as such.*
 - *Structural profile: Rests on the two identifiable, contestable commitments named in the Abstract: compressibility of reality, and single-history realization.*
- **The Law of All.** *The paper's working shorthand for the ambition behind Thesis 2: a single law governing how systems navigate the space Freedom makes available.*
 - *Properties: Candidate-law status (L5) only. Appendix D.4 states the five-point evidence bar this status would require: parameter identifiability, dissociating rival comparison, deterministic-scope resolution, physics-reduction proof, and prospective cross-domain replication.*
 - *Boundary: A status this paper is candid it has yet to earn. The paper's own closing statement (Section 20) tracks all five conditions as still open.*
 - *Requires agency: No. Proposed as a law governing navigation generally, agentive or otherwise.*
 - *Structural profile: The paper's most theory-committing claim. Its status is tracked criterion by criterion in the Falsification Matrix.*
- **The Navigability problem.** *The forced question that arises the moment a state has more than one admissible continuation: some principle selects which one becomes actual.*
 - *Properties: A logical, definitional-layer observation (L1). It follows just from there being more than one admissible continuation. It holds independent of which navigation law, if any, turns out to be correct. The hinge between Thesis 1 and Thesis 2.*
 - *Boundary: Establishes that a selection question exists, which is a separate matter from answering it. The question is prior to, and neutral among, any candidate answer, including the Law of Freedom.*
 - *Requires agency: No. The question is forced by branching structure alone, whether or not an agent is present.*
 - *Structural profile: Requires alternatives, definitionally. Established in Section 13.*
- **Epistemic layers (L1–L6).** *The paper's discipline for classifying every claim it makes: logical/definitional, empirical, metaphysical, derived-structural, candidate-law, and engineering/design. This keeps any argument from silently trading on a stronger layer than it has earned.*
 - *Properties: A meta-level classification scheme, applied uniformly across the whole paper, rather than a claim about AFI's content itself.*

- *Boundary*: A claim can be well-established at L1 and still be central to the argument. The layers track what kind of warrant a claim has, separate from how much that claim matters.
- *Requires agency*: No. A discipline applied by the paper's authorial method, rather than a property of the systems the paper describes.
- *Structural profile*: Used throughout. Explicitly invoked in Figure 16's derivation chain, and in every glossary entry above where "layer" is cited.
- **Falsification Matrix.** The twenty-two-criterion audit in Section 16. It scores the theory's status honestly against each criterion. This is the mechanism by which the paper keeps its own claims accountable.
 - *Properties*: Scored Pass, Partial, Fail, or Unknown, per criterion. A self-audit, rather than an independent replication. Currently 2 Pass, 11 Partial, 6 Unknown, 3 Fail (Figure 46).
 - *Boundary*: Complements, rather than substitutes for, adversarial or third-party review. Several Partial verdicts stay conditional, pending the head-to-head comparison named as criterion 17.
 - *Requires agency*: No. A property of the paper's audit method, rather than of any system under test.
 - *Structural profile*: The mechanism referenced by nearly every Falsifiability note beside a figure throughout the paper.

E.3 Counterfactual Universe Laboratory — quick-reference index

Forty formally specified toy universes, each stress-testing Theses 1 and 2 from a different angle. Full formal specifications are in Appendix A. This index is for fast lookup.

- **U1 — The Null Universe.** No transitions exist at all; Thesis 1 holds only vacuously.
- **U2 — The Rail.** A single deterministic successor at every state; Freedom collapses to bare necessity.
- **U3 — The Fork with One Traveler.** Genuine two-way branching with one realized outcome — the minimal universe separating possible from actual.
- **U4 — The Boundless Tree.** Infinite branching, forcing Perception (not just Possibility) into the theory.
- **U5 — The Frozen Infinity.** Infinite possibility with zero actualization.
- **U6 — The Dice World.** Pure random actualization, independent of any Lightness ranking.
- **U7 — The Buridan Field.** Perfect symmetry between options, no preferred path.
- **U8 — The Masochist Universe.** Systems that reliably choose the heaviest path — a genuine falsifier for naive Thesis 2.
- **U9 — The Dissipative-Structure Universe.** Maximum-resistance dynamics, a second genuine falsifier.
- **U10 — The Homing Missile.** Pure goal-directed dynamics, contrasted against Lightness-seeking.
- **U11 — The Ledger Universe.** History-dependent, non-Markovian dynamics.
- **U12 — The Boltzmann Chooser.** Stochastic, cost-weighted softmax selection.
- **U13 — The Thermostat Universe.** Passive P0-type sensitivity with no representational perception.
- **U14 — The Oracle.** Perfect perception, $\hat{P}_a$ equals P exactly.
- **U15 — The Cataract Universe.** Severe, ordinary misperception.
- **U16 — The Locked Door.** Real continuations the system cannot perceive.
- **U17 — The Mirage.** Perceived options that are not real.
- **U18 — The Greedy Trap.** A locally lightest path that is globally disastrous — motivates the horizon term.
- **U19 — The Seed-Planter.** A heavier-now path that preserves more future freedom.
- **U20 — The One-Way Door.** An irreversible path.

- **U21 — The Roundtrip.** A reversible path, for contrast.
- **U22 — The Ouroboros.** Cyclic causation.
- **U23 — The Handshake.** Simultaneous, mutual causation.
- **U24 — The Long Shadow.** Causation that depends on distant history.
- **U25 — The Jump.** Discontinuous, teleportation-style transition.
- **U26 — The Flow Field.** Continuous, field-based dynamics.
- **U27 — The Quantum Coin.** Quantum probabilistic transition, tested against many-worlds readings.
- **U28 — The Shifting Rulebook.** Transition laws that change over time.
- **U29 — The Growing Space.** A state space that changes size or structure over time.
- **U30 — The Self-Widening Path.** Action that creates new paths — the seed of recursive Possibility.
- **U31 — The Shared Corridor.** Competing agents converging on the same admissible path.
- **U32 — The Relay.** Cooperating agents that reduce each other's Distortion.
- **U33 — The Deceiver.** An adversary that manipulates perceived Distortion directly.
- **U34 — The Rationed World.** Lightness competing against a hard, depletable resource budget.
- **U35 — The Nested Room.** Composable, hierarchical spaces — rooms within buildings within districts.
- **U36 — The Fading Memory.** Perception that decays without renewed observation.
- **U37 — The Rush Hour.** Congestion — paths that grow heavier as more systems use them.
- **U38 — The Slow Fuse.** Consequences that stay invisible until long after the action that caused them.
- **U39 — The Self-Improving Observer.** A system whose own actions upgrade its Perception — Thesis 5's dedicated test.
- **U40 — The Cliff Edge.** A rare, near-total, irreversible collapse — a tail-risk stress test.